Understanding Child Marriage: Theory and Evidence for Boys and Girls

Abigail Stocker*

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Abstract

This paper examines how child marriage rates for both boys and girls respond to exogenous shocks to rainfall, temperatures, and conflict. Using individual-level data from India, Indonesia, and Nepal, I empirically estimate the effects of shocks on child marriage. Low rainfall and high temperatures, which reduce income, decrease the annual probability of child marriage for boys and girls by 1-8%. Exposure to conflict, which increases the risk of experiencing conflict-related violence, decreases child marriage for boys and increases it for girls by up to 30% and 3%, respectively. Effects are similar regardless of the child's age, spousal age gap, or direction of the marriage transfer. I also develop a theoretical household bargaining model, which predicts that negative shocks to income or to child marriage preferences reduce child marriage rates. These findings suggest a perverse relationship between income and child marriage, which is relevant for policymakers seeking to simultaneously reduce child marriage and poverty.

Keywords: Gender, Child Marriage, Marriage Transfers, Income, Asia, Development
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1 Introduction

Children around the world marry before turning eighteen, despite the well-documented negative effects of child marriage on child well-being.[1] On average in less developed countries, 6% of boys and 36% of girls marry by 17, with 10% of girls marrying by 14 (UNFPA, 2022, UNICEF, 2023). Child marriage is typically the choice of the parents, not of the child, and many factors, such as income, norms, household composition, and demographic characteristics, play a role in this household-level decision (Bates et al., 2007, Bicchieri et al., 2014, Gazi et al., 2022).[2] However, these factors are far from random, making it difficult to establish a causal relationship between any given attribute and child marriage, let alone understanding the full picture of how child marriage decisions are made.

Recent literature has tried to solve this problem by focusing on the impacts of exogenous shocks on child marriage.[3] Exogenous economic and social shocks could impact child marriage for both boys and girls through multiple channels, such as changing income, migration patterns, expectations for the future, and risk. However, the existing causal research on child marriage has two major limitations: first, these papers focus almost solely on child marriage for girls, ignoring male child marriage; second, they focus only on one shock in one or two contexts, providing only a small picture of the causal determinants of child marriage. As a result, little is known about the causal role of various factors in the child marriage decision, particularly for boys.

To improve our understanding of child marriage, we need to identify how marriage behavior responds across a wide variety of situations. This includes comparing both boys and girls, different kinds of shocks, and contexts with bride price and dowry. To address this, I study the impacts of multiple types of shocks on child marriage matches for both boys and girls in three countries: India, Indonesia, and Nepal. I focus on plausibly exogenous shocks that directly impact household agricultural income and the risk to a household of experiencing conflict-related violence, resulting in associated impacts on decisions in the market for male and female child marriage. Including both boys and

[1]The existing literature shows that female child marriage is correlated with negative consequences later in life, including poor health outcomes, malnutrition, isolation, depression, maternal mortality, infant mortality, risk of physical and emotional violence, lower education, lower bargaining power, and poverty (Chakravarty, 2018, Duggal, 2023, Field and Ambrus, 2008, Garcia-Hombrados, 2021, Jensen and Thornton, 2003, Jensen et al., 2023, Parsons et al., 2015, Rasmussen et al., 2021, Vogl, 2013). Child marriage is defined as marriage before age 18.

[2]Specific demographic characteristics of the individual or household that predict child marriage include gender, birth timing, caste, or location of residence (Binu et al., 2022, Kamal, 2010, Sekine and Carter, 2019, Tapsoba, 2022, Yogi, 2020); transitory characteristics of the individual or household, such as changes to income, migration incentives, risk levels, or needs for insurance (Anukriti and Dasgupta, 2017, Jensen and Thornton, 2003); and outside factors related to the surrounding culture, religious traditions, weather, or economic conditions (Anukriti and Dasgupta, 2017, Corno and Voena, 2023, Jensen and Thornton, 2003, Rosenzweig and Stark, 1989).

[3]See, for example, Blanco (2023), Corno et al. (2020), Fiala (2023), and Trinh and Zhang (2020).

girls allows me to identify which types of child marriage matches are most responsive to shocks.

Empirically, I find that low rainfall and high temperatures (which cause transitory changes in household income) reduce the annual probability of child marriage for both boys and girls, while exposure to conflict (which increases the risk of experiencing conflict-related violence) increases the annual probability of child marriage for girls and reduces it for boys. Surprisingly, these results hold across institutional contexts and do not depend on whether dowry or bride price is primarily practiced. These findings suggest that rainfall and temperatures shocks impact child marriage through different channels than exposure to conflict. To further understand the impacts of these exogenous shocks on child marriage, I also develop a theoretical model to explain the mechanisms behind the effects. Finally, I examine heterogeneity across different types of child marriage matches, defined based on the age of the child, the age gap between spouses, and the relative fertility preferences of spouses, and find that the direction of the effect is similar across match types.

For my empirical analysis, I use individual-level data from the Integrated Public Use Microdata Series (IPUMS) International and the Demographic and Health Surveys (DHS) in India, Indonesia, and Nepal between 1967-2010. In these contexts, child marriage is highly correlated with characteristics such as age, gender, and religion. Girls are more likely than boys to marry before 18, and the risk of child marriage increases as individuals approach age 18. Age gaps between husbands and wives are higher for girls who are at a higher risk of child marriage. Spousal age gaps for boys married before 18 are consistently small - less than 2.5 years - for all subgroups.

Using a two-way fixed effects model and variation across time and geography, I estimate the impacts of three types of plausibly exogenous shocks, low rainfall, high temperatures, and exposure to conflict, on child marriage outcomes. Combining the individual-level data from IPUMS and the DHS with climate data from the University of Delaware and conflict data from the Uppsala Conflict Data Program (UCDP), I estimate how shocks impact the probability of child marriage for boys and girls in each country in a given year. I examine heterogeneity based on the type and quality of the match, defined based on the child's age at the time of marriage, the spousal age gap, and the relative fertility preferences of the husband and wife.

In a year with low rainfall or high temperatures, the probability of child marriage for both girls and boys decreases by 1-8% of the mean. I also find evidence that exposure to conflict increases the probability of child marriage for girls by up to 3% and decreases child marriage for boys by up to 30%. In this setting, dowry and bride price practice do not predict the response of child marriage to shocks, since the two cases are symmetric.[4] Relative to the underlying child marriage rates, exposure to shocks has similar

[4]This contrary to Corno et al. (2020), who find that, when comparing Africa and India, bride price and

effects across gender, age of the child, and matches with different spousal age gaps, with stronger effects on matches where women's fertility preferences align with those of their husbands. Results are robust to using alternate shock definitions and samples, controlling for shocks in previous years, and including sample weights in the regressions. The effects are qualitatively similar, though generally weaker, for adult marriages (ages 18-24) and total marriages at the district level. This implies that older individuals are less responsive to negative shocks, perhaps due to the fact that delaying marriage is more costly.

These shocks affect both sides of the marriage market through several channels. Low rainfall and high temperature shocks impact a household's agricultural income, which, in turn, affects that household's ability to afford the transfer payments (dowry or bride price) associated with marriage (Acevedo et al., 2020, Ali et al., 2017, Hoogeveen et al., 2011, Levine and Yang, 2014, Wineman et al., 2017).[5] The direction of the marriage transfer, from the bride's to groom's household (dowry) or groom's to bride's household (bride price), might also play a role. Exposure to conflict likely changes the risk of experiencing physical or sexual violence. In areas with higher risk of conflict-related violence, households might choose to marry their daughters as a way of protecting them from experiencing violence (Sharma et al., 2015). Boys may choose (or be coerced) to actively participate in armed groups, reducing their availability for marriage (Lawoti, 2010). Finally, all three shocks might impact migration, which is closely related to marriage, particularly in primarily patrilocal societies (Kudo, 2015).

To understand which of these mechanisms contributes to the effects of shocks on child marriage outcomes, I develop a theoretical household bargaining model to characterize the household's decision regarding whether their child will marry. Households pay or receive a marriage transfer, and each household has a threshold for the maximum (minimum) amount that they are willing to pay (receive). Transitory shocks to income or parents' child marriage preferences shift these thresholds, impacting both the number of matches taking place in the marriage market and the size of the marriage transfer. If the direction of the marriage transfer adheres to strong norms, negative income shocks reduce the number of matches taking place, since transfer amounts are limited in their ability to adjust and accommodate the drop in income. The number of matches in the market is positively correlated with preferences for child marriage, which also influence households' willingness to pay (accept) for a marriage match. Because markets for child marriage in the case of dowry and the case of bride price are symmetric, the direction of the marriage transfer does not predict the impacts of transitory shocks on child marriage.

The model predicts that the effects of negative shocks could come through different channels. Empirically, I use an instrumental variables approach to provide evidence that

dowry predict the direction of the response of female child marriage to negative transitory income shocks.

[5]Households paying the transfers would be less able to afford a marriage, while households receiving the transfers would be more willing to accept a marriage. Conflict could also affect the financial desirability of marriage, since income and stored wealth or assets could be impacted by conflict.

low rainfall and high temperatures affect child marriage through the channel of income by impacting agricultural yields. Additionally, I identify mitigating effects of irrigation and show that responses to shocks are larger in rural areas. These results support transitory changes in income as a likely channel for the impact of low rainfall and high temperatures on child marriage. Exposure to conflict, on the other hand, does not seem to impact income, but likely has impacts through changes to the risk of violence perceived by the household.

This work contributes to three strands of literature, beginning with the literature on child marriage for boys. This paper is the first to causally identify the effect of exogenous shocks on male child marriage. Across India, Indonesia, and Nepal, 8% of boys experience child marriage. Most of these boys marry girls around their own age, while girls who experience child marriage may marry either young boys or older men.[6] Thus, including boys in the analysis also provides insight into the effects of shocks on female child marriage by allowing for identification of the impacts of shocks on a particular type of match (young girls marrying young boys). In contrast to the large body of research on child marriage among girls, the literature on child marriage among boys is lacking.[7] Papers focused on identifying the causal determinants of child marriage, such as Corno et al. (2020), Corno and Voena (2021), Fiala (2023), and Trinh and Zhang (2020), study only female child marriage. While less common in occurrence than female child marriage, male child marriage still occurs frequently in many countries, has been sorely overlooked, and plays an important role in many marriage markets (United Nations Children's Fund (UNICEF), 2019). This paper captures a fuller picture of the impact of shocks on both sides of the child marriage market through the inclusion of boys in the analysis, showing male child marriage responds to negative shocks.

Second, this paper provides a fuller understanding of the the factors the influence both male and female child marriage by comparing effects across different types of exogenous economic and social shocks in different contexts. This increases our understanding of how big-picture factors, such as transitory income shocks and exposure to conflict, affect outcomes for individuals. Previous literature focuses on a single shock or intervention in one or two contexts. Existing work on the relationship between plausibly exogenous shocks and child marriage studies changes in rainfall (Buchmann et al., 2016, Chort et al., 2022, Corno et al., 2020, Corno and Voena, 2021, Hahn et al., 2018, Maswikwa et al., 2015, McGavock, 2021, Trinh and Zhang, 2020), with some papers on conflict (Fiala, 2023) and earthquakes (Blanco, 2023, Lv, 2023), but none of these papers compare the

[6] Though the occurrence of child marriage among girls is both well-studied and widely condemned, the impacts, drivers, and correlates of child marriage among boys are not well-understood. Few studies even mention child marriage for boys (Jensen et al., 2023, Misunas et al., 2019).

[7] Lv (2023) estimates the impacts of earthquakes in Indonesia on child marriage for both girls and boys but does not find an effect for boys. Figueiredo (2024) estimates the impact of *Progresa* in Mexico on marriage outcomes, and finds that the transfers increase child marriage for girls and do not have a significant effect for boys. Other literature addressing child marriage focuses solely on girls.

magnitudes of different types of shocks.[8] In developing countries such as India, Indonesia, and Nepal, weather conditions and conflict violence play a major role in determining the economic climate and may act as factors in the child marriage decision. I contribute to this strand of the literature by providing a more complete picture of child marriage across contexts, gender, and shocks.

I also show that bride price and dowry traditions do not necessarily predict the impact of shocks on child marriage outcomes when both sides of the market are exposed to the shocks, due to the fact that bride price and dowry settings are symmetric. An especially relevant paper by Corno et al. (2020) estimates the effects of low rainfall on female child marriage in India and sub-Saharan Africa. Their model requires that marriage transfer amounts fully adjust to clear the market and predicts that a negative income shock should decrease child marriage in settings where dowry is practiced and increase child marriage in settings where bride price is practiced. When I test this hypothesis in the settings of India and Nepal (where dowry is primarily practiced) versus Indonesia (where bride price is primarily practiced), I find no differences in the sign of the effect of a negative transitory income shock across settings. This suggests that the direction of the marriage transfer does not always dictate the impacts of income shocks on child marriage. For example, the direction of the marriage transfer will not matter if households on both sides of the marriage market may want to transact but are constrained by social norms.

Third, I examine heterogeneity in the impacts of shocks on different types of marriage matches. The approach taken in the previous literature on female child marriage does not often differentiate between the types of marriage matches that a girl in this setting might make (e.g. marrying an older man or marrying a younger boy). However, match characteristics, such as the spousal age gap, are important for understanding the effects of early marriage on later-life outcomes for girls, such as bargaining power (Tauseef and Sufian, 2024). I contribute by showing that the effects of shocks are similar across matches with children of different ages or with different spousal age gaps, while effects are larger for matches where both spouses have the same fertility preferences.

My findings provide insight into how economic conditions affect child marriage for boys and girls and are relevant for policies targeted at reducing child marriage. Specifically, the findings that transitory reductions in income and exposure to conflict reduce child marriage matter for policymakers whose aim is to reduce child marriage. These findings do not imply that events such as lower rainfall or higher conflict violence should be encouraged. After all, increasing income and reducing conflict have been shown to have many advantages, including improving education and health, reducing mortality,

[8] From a policy perspective, stipend programs have been shown to decrease child marriage and lead to welfare increases for at-risk individuals (Baird et al., 2011, Buchmann et al., 2016, Hahn et al., 2018), and laws restricting child marriage may also have a positive impact, though the evidence is mixed at best (Batyra and Pesando, 2021, Bellés-Obrero and Lombardi, 2023, Maswikwa et al., 2015, McGavock, 2021).

and promoting economic development.[9] However, these results suggest that policies to increase income or reduce conflict can lead to unintended negative consequences, such as higher child marriage rates. Policymakers need to carefully consider all components of the social welfare function when making decisions. This could include targeting locations that experience sudden increases in income or reductions in conflict with programs that are known to reduce child marriage. For example, in settings where policymakers want to reduce poverty through providing cash transfers and simultaneously reduce child marriage, the cash transfers could be made conditional on children in the household under 18 remaining unmarried. Simply put, these findings highlight the complicated relationship between child marriage and the surrounding economic conditions and events.

The rest of the paper is organized as follows. Section 2 includes background information on child marriage, each type of shock, and how shocks are linked with marriage outcomes, Section 3 discusses the data and sample as well as demographic characteristics associated with child marriage, Section 4 presents the empirical strategy, Section 5 provides a discussion of the results, Section 6 introduces the model, Section 7 discusses the mechanisms, Section 8 presents a series of robustness checks, and Section 9 concludes.

2 Transitory Shocks and Child Marriage

This section provides background information on child marriage in India, Indonesia, and Nepal. Many time-varying factors influence child marriage, including income, marriage transfers, and the risk of experiencing conflict-related violence. In my empirical strategy, transitory changes in income and the risk of experiencing conflict-related violence are captured through exogenous changes in rainfall, temperature, and conflict.[10] The remainder of this section describes the types of child marriage matches that occur in India, Indonesia, and Nepal and explains how transitory shocks to rainfall, temperature, and conflict can influence child marriage decisions.

2.1 Types of Child Marriage

For the purposes of this paper and to be consistent across contexts, I define child marriage as marriage before age 18.[11] Child marriage is associated with negative health, education, and well-being outcomes, particularly for girls (Chakravarty, 2018, Parsons et al., 2015, Rasmussen et al., 2021). Certain types of child marriage, such as marriage for girls under age 15, are more likely to be associated with these negative outcomes

[9] See, e.g., Aizer et al. (2016), Dincecco et al. (2021), Gertler (2004), Tsujimoto and Kijima (2020).

[10] I focus on India, Indonesia, and Nepal because child marriage rates are high for both boys and girls and large samples of individual-level data are available, covering similar time periods and including age at first marriage, religion, and location. Additionally, all three countries have traditional marriage transfers, either bride price or dowry, providing a natural setting for income to impact child marriage decisions.

[11] For details on the legal marriage age by country, see the Supplemental Appendix.

(Jensen and Thornton, 2003, Jensen et al., 2023). Spousal age gap, bargaining power, and compatibility also play a role in match type and quality.

Several types of marriage matches can occur: (1) older men with older women, (2) older men with young girls, (3) young boys with young girls, or (4) young boys with older women. The first is not child marriage, and the fourth occurs rarely, so the other two match types (older men with young girls and young boys with young girls) are the relevant cases.[12] The impacts of shocks on child marriage might differ across these categories.

2.2 Shocks to Income, Marriage Transfers, and Risk of Violence

This paper focuses on three factors that impact child marriage decisions and respond to exogenous changes in rainfall, temperature, and exposure to conflict: income, marriage transfers, and risk of violence. Rainfall and temperature primarily impact child marriage through changing household income. The existing literature finds that child marriage responds to financial conditions, showing that conditional cash transfers and scholarships can delay female child marriage (Buchmann et al., 2023, Giacobino et al., 2024). In many developing countries, including India, Indonesia, and Nepal, agriculture is a large source of income, and low rainfall reduces crop outputs (Levine and Yang, 2014), lowering household income. Since certain crops grow best at certain temperatures, high temperatures also impact crop output, potentially reducing agricultural income. Transitory changes to income influence households' ability or desire to pay or receive a marriage transfer, affecting the child marriage decision.

Conflict impacts child marriage by changing the risk of experiencing violence for the child or household. Households use child marriage to protect their daughters from physical and sexual violence, so if violence becomes more prevalent, child marriage might increase in response (Sharma et al., 2015).[13] Conversely, families might not want the expense and commitment of planning a large event during uncertain times. For boys, exposure to conflict directly affects their ability to marry if they are recruited into armed groups (Lawoti, 2010). Additionally, dowry and bride price amounts might respond to conflict (Johnston, 2023, Schlecht et al., 2013), influencing the family's ability to pay the transfers and further impacting child marriage rates.

Finally, rainfall, temperature, conflict, and child marriage are likely linked to changes in migration, which may be correlated with the marriage decision. Marrying off a child may be a way to move that child to a safer location, especially for daughters in patrilocal societies. For example, if rainfall is low, households might want their daughters to marry and move to an area where agricultural output is higher. In this scenario, marriage could be a means of achieving migration, either to improve the outcomes of the daughters or

[12]Only about 1% of marriages occur between a boy who is under 18 and a woman who is 18 or older.

[13]Additionally, attitudes towards risk, beliefs about the future, poverty, broken families, and lack of access to schooling resulting from conflict can cause earlier marriage (Schlecht et al., 2013, Schmidt, 2008).

of the entire household (Corno and Voena, 2023, Rosenzweig and Stark, 1989).

3 Data & Sample

To analyze the impacts of rainfall, temperatures, and conflict on child marriage outcomes, I combine individual-level outcome data with information on district-level shocks. This section describes the data and sample used in my main analysis, which include individuals in India, Indonesia, and Nepal from 1967 to 2010.[14] These three countries are chosen because they have relatively high rates of child marriage for both boys and girls, large samples of individuals over many years, and variation in dowry and bride price practice.

3.1 Data on Child Marriage

Individual-level data come from the Integrated Public Use Microdata Series (IPUMS) International and the Demographic and Health Surveys (DHS) (International Institute for Population Sciences - IIPS/India and ORC Macro, 2000, International Institute for Population Sciences - IIPS/India and ICF, 2017, International Institute for Population Sciences (IIPS) and ICF, 2021, Ruggles et al., 2025). For India, I use the DHS, which includes individuals' district of residence, age, sex, marital status, and age at first marriage.[15] While this data is provided for the full sample of women, age at first marriage is only included for a limited sample of men.[16] To deal with this, I extrapolate age at first marriage for men based on their wives' age at first marriage.

IPUMS provides data for Indonesia and Nepal, including individuals' location of residence (at the district level for Nepal and the kabupaten (regency) level for Indonesia; for the remainder of this paper, both levels will be referred to as districts), age, sex, marital status, and age at first marriage.[17] For Nepal, information on age at first marriage is available for both men and women. For Indonesia, information on age at first marriage is only available for women, so I extrapolate age at first marriage for men.

[14]For details on all data sources used, see Appendix B.

[15]I use DHS surveys conducted in India in 1998-1999, 2015-2016, and 2019-2021. While the DHS is representative for women between ages 15 and 49, it is not representative for men. The men's sample is based on men living in the households chosen for the women's sample. However, if these men are not representative of all men in India, then this sample is not representative for men.

[16]Age at first marriage can refer to marriage, consensual union, and cohabitation.

[17]For Nepal, the Central Bureau of Statistics provides data from the National Population Census 2001 and 2011. Data from Statistics Indonesia is available for 1976, 1980, 1990, 1995, and 2005. The 1976 data is not fully representative, so I run the analysis without 1976 in the Supplemental Appendix.

3.2 Data for Rainfall, Temperature, and Conflict Shocks

Average annual rainfall and temperatures come from the University of Delaware.[18] Using shapefiles for India,[19] Nepal,[20] and Indonesia,[21] I calculate the land-area-weighted average rainfall and temperature for each district, providing the annual average level of rainfall and annual average temperature across the whole district for each year from 1900 to 2017.

I fit annual rainfall to a district-specific gamma distribution using data from 1950 to 2017. A year is considered to have low rainfall if the rainfall for that year is below the 15th percentile of the district-specific gamma distribution.[22] Low rainfall is associated with reduced crop output, lowering household income (Levine and Yang, 2014). To define the temperature shocks, I use temperature data from 1950-2017. A year is considered to have high temperatures if the average temperature for that year is above the 90th percentile of all values for that district between 1950 and 2017.

Conflict data come from the Uppsala Conflict Data Program (UCDP), which includes incidents between 1989 and 2022 (Davies et al., 2023, Sundberg and Melander, 2013). The incidents reported include both organized violence and civil strife. I measure conflict exposure using an indicator for a district having any conflict-related deaths in that year.[23]

Figure 1 shows the number of years with low rainfall, high temperatures, and any conflict-related deaths from 1950-2017 in each of the 466 districts in India. Similarly, Figure 2 shows the distribution of rainfall, temperature, and conflict shocks across the 257 districts in Indonesia, and Figure 3 shows the distribution of rainfall, temperature, and conflict shocks in the 72 districts in Nepal. Not all districts receive the exact same number of shocks, but the distribution of rainfall and temperature shocks is relatively uniform. The distribution of years with any conflict deaths is less uniform, so district fixed effects in my main specification control for time-invariant differences in conflict prevalence across districts.

3.3 Additional Data Sources

To identify the mechanisms behind the main effects of shocks on child marriage, I use several additional datasets. First, the Indonesian Family Life Survey (IFLS) provides information on bride price amounts (including both monetary and non-monetary transfers) in Indonesia (Frankenberg and Karoly, 1995, Frankenberg and Thomas, 2000, Strauss

[18]University of Delaware Terrestrial Precipitation data provided by the NOAA PSL, Boulder, Colorado, USA, from their website at https://psl.noaa.gov. I use total monthly precipitation and average monthly temperature in 0.5x0.5 gridded cells, where 0.5 degrees is equivalent to 50 km at the equator.

[19]https://international.ipums.org/international/gis_yrspecific_2nd.shtml, accessed 16 March 2024.

[20]https://data.humdata.org/dataset/cod-ab-npl?), accessed 16 March 2024.

[21]https://international.ipums.org/international/gis_harmonized_2nd.shtml, accessed 16 March 2024.

[22]This follows Burke et al. (2015) and Corno et al. (2020).

[23]For Nepal, more detailed and comprehensive conflict data are available from the Informal Sector Service Centre (INSEC), listing the universe of known incidents in the conflict between the government and Maoist opposition groups between 1996-2006. I use this for robustness in the Supplemental Appendix.

et al., 2004, 2009, 2016). Second, I use data from the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) from the ICRISAT Data Repository to measure the amount of irrigated land in each district in India and provide information on yields at the crop level for each district. Finally, I use district-level data on Gross Regional Domestic Product (GRDP) for Indonesia from 2010-2017 from BPS - Statistics Indonesia, the Indonesian Bureau of Statistics.

3.4 Age at Marriage Extrapolation & Matched Couples Sample

The data for India and Indonesia do not have information on age at first marriage for all men. To deal with this, I match the married men and women to obtain a sample of matched couples.[24] For the matched couples, the age at first marriage for the wife is used to generate age at first marriage for the husband by comparing their ages at the time of the survey under the assumption that individuals are only married once. This provides an approximate measure of age at first marriage. The matched husbands are then merged back with the single men to create the full dataset of married and unmarried men.

Even when spouses can be matched, there is likely some measurement error, but this should attenuate the estimates and create bias against finding impacts of shocks on male child marriage or differences in effects on different types of matches. For example, measurement error in age at first marriage would occur if a couple was divorced and then both spouses re-married. Additionally, I do not have a way to impute age at first marriage for men whose marital status is reported as divorced or widowed. Despite these drawbacks, this method still provides a viable estimate of age at first marriage for men.

I also create a sample of all matched couples, restricting only to individuals who are married and who have a spouse present in the data. I use this sample of matched couples to examine heterogeneity in the effects of shocks on different types of matches based the age gap between husband and wife and relative fertility preferences.

3.5 Sample Restrictions & Summary

For the main analysis, I use individual-year-level samples for both men and women, as well as for the matched couples. Starting with the data from IPUMS and the DHS, I restrict to individuals born after 1955 who are at least 18 at the time of the survey.[25] I drop individuals who report marriage before age 12 and individuals with missing information

[24]Details for the matching process can be found in Appendix C. I am able to match 57% of married men in India and 93% of married men in Indonesia. The difference in match rates likely comes from the difference in sampling frames, since the Indonesian data comes from a Census that interviews every household member, while the Indian data does not.

[25]This follows Corno et al. (2020), to minimize misreporting for locations where child marriage is illegal. For robustness, I include only individuals 21 and older and only individuals 25 and older in the Supplemental Appendix.

on age or age first married, but I retain individuals who were never married.[26]

To create the individual-year-level sample, I expand the individual-level data into a retrospective panel, with six observations for each individual, one for each of the years when they were between the ages of 12 and 17. Individuals are matched with shocks in each year, and they are included in the dataset until (and including) the year that they marry.[27] For example, if an individual marries at age 15, that individual would appear in the data at ages 12, 13, 14, and 15. The individual would not be in the data for ages 16 or 17, since they are already married at that point and no longer on the marriage market. If an individual never marries, they are included in the data for each year between ages 12-17. Outcomes are also at the individual-year-level, with the main outcome being an indicator for the individual marrying at that specific age.

Panel A of Table 1 describes the sample of individuals, separately for girls and boys. For the sample of matched couples, Panel B shows the average age gap between spouses. In Panel B, the sample size for the matched couples is different for boys and girls. This is because there may be cases where one individual meets the requirements for inclusion in the sample (for example, being born after 1955), while the spouse does not.[28]

3.6 Demographic Characteristics & Child Marriage

This subsection describes the demographic correlates of child marriage for boys and girls, including age, religion, and cultural norms. Table 2 shows that, in my sample, 38% of girls and 8% of boys marry before age 18. While child marriage rates at each age are much higher for girls, child marriage for boys occurs in a non-negligible fraction of the population. Panel A shows the cumulative probability of child marriage by age for girls in column (1) and boys in column (2), starting from age 12.[29] Child marriage is less frequent for younger individuals, with only 9% of girls and 1% of boys married by age 15.

Panel B shows the probability of child marriage overall and by country and religion.[30] Religion is an important predictor of child marriage, likely because of differences in norms regarding age at marriage depending on an individual's religious beliefs. For example, Hindu individuals are more likely to experience child marriage than Christian individuals. Geographical location is also a predictor of child marriage. India and Nepal primarily

[26] I drop individuals who marry before age 12 due to potential misreporting, following Corno et al. (2020).

[27] I drop individuals who are missing shock data for at least some years when the individual was between ages 12 and 17, including only those with full information on shocks experienced.

[28] For example, if a man born in 1950 married a girl born in 1960, the girl would be included in the sample of matched women, but the man would not be included in the sample of matched men.

[29] Though child marriage before age 12 does occur in the data, I only consider marriage beginning at age 12. This is due to data limitations and concerns over misreporting.

[30] Religion is measured at the time of the survey, not at the time of the marriage. However, the religion that an individual is born into is plausibly exogenous relative to that individual, there tends to be intergenerational persistence of religion (Copen and Silverstein, 2008), and individuals do not frequently change religions during their lifetime. Evans (2023) reports that, in Indonesia, 100% of adults surveyed reported remaining in the same religion that they were raised in.

practice dowry, while Indonesia primarily practices bride price. The amount of the marriage transfer is associated with a girl's age at marriage, with older women requiring higher dowries or bringing in lower bride price payments (Srivastava et al., 2021, Corno and Voena, 2021).[31] The means in Table 2 suggest that age, religion, and country of residence are important predictors of child marriage for both girls and boys and follow similar patterns across gender.

There is heterogeneity in the types of matches that occur. Panel C of Table 2 shows the average spousal age gap (calculated as husband's age minus wife's age) for individuals who marry before 18, separately for girls and boys and by group. Spousal age gap is a key outcome, since it predicts relative bargaining power within the marriage (Tauseef and Sufian, 2024, Nasrullah et al., 2014). For girls married before 18, the average spousal age gap is 5.46 years, and the age gap is lower if the girl's age at marriage is higher. For boys married before 18, the average spousal age gap is 1.78 years. The age gap for boys is much lower than for girls because boys who marry before 18 usually marry girls their own age, while girls who marry before 18 either marry boys near their own age or much older men.

Age, gender, and country of residence are all correlated with child marriage for both girls and boys. However, boys are consistently more likely to marry someone closer to their own age, while the spousal age gap for girls varies with demographic characteristics.

4 Empirical Strategy

To analyze the impact of low rainfall, high temperatures, and exposure to conflict on the annual probability of child marriage, I use an individual-year-level analysis to estimate the impact in the year that the shock takes place.[32] This section describes the methodology and limitations.

4.1 Individual-Year Level Analysis

The individual-year-level analysis allows an examination of how a shock in a given year impacts the probability of child marriage in that same year, using the following equation:[33]

$$Y_{idt} = \beta_0 + \beta_1 Shock_{dt} + \gamma_d + \delta_t + X_{idt} + \epsilon_{idt} \quad (1)$$

[31] As shown in Figure A1, bride price increases with age until around age 27, after which it decreases. The relationship between bride price and dowry amounts and later-life outcomes for women has also been documented (Lowes and Nunn, 2018).

[32] In the Supplemental Appendix, I also use an individual-level analysis to estimate the impacts of multiple shocks at the individual level.

[33] For robustness, I also include a version of the equation that allows for shocks in previous years to have an effect on the outcome in the current year by including $\beta_2 Shock_{d,t-1}$. $\beta_3 Shock_{d,t-2}$, and $\beta_4 Shock_{d,t-3}$. These estimates can be found in Table A1 and are similar to those estimated using the main specification.

In this equation, Y_{idt} is either an indicator for an individual between the ages of 12-17 getting married in year t or an indicator for an individual entering a specific type of marriage match (defined based on age gap or fertility preferences) in year t. $Shock_{dt}$ is an indicator for whether the individual's district of residence experienced a shock during the year t. γ_d is a fixed effect for the individual's district of residence, δ_t is a fixed effect for the year, and X_{idt} is a set of individual-level controls, specifically an indicator variable for the age of the individual in year t. These regressions are run for each country and for boys and girls separately.

Several assumptions are necessary for this analysis. The first assumption is that the shocks are exogenous, guaranteeing that the likelihood of experiencing a shock is not correlated with an individual's observable or unobservable characteristics.[34] The second assumption is that people do not migrate in response to or in anticipation of shocks, meaning that individuals' location decisions are not correlated with shocks.[35]

Estimates of β_1 in equation (1) capture how the probability of child marriage in a given year changes when a shock occurs in that year. Essentially, this regression specification predicts the hazard into child marriage in a given year. Thus, I do not capture changes in the overall probability of child marriage, but rather changes in child marriage probability in a given year.

This specification can also be used to estimate how particular types of marriage matches, defined by different spousal age gaps or partners' relative fertility preferences, respond to shocks. I investigate impacts on different types of child marriage matches in three ways. First, I investigate heterogeneity based on the age of the child at the time of the marriage. I run the regressions only for very young girls or boys, below age 15. Child marriage at younger ages is generally associated with worse outcomes, allowing me to identify the effects on the types of matches that tend to leave individuals worse off.

Second, I investigate heterogeneity by the size of the spousal age gap. I define the age gap between a husband and wife as either a high age gap (where the husband's age minus the wife's age is greater than 5 years) or a low age gap (where the husband's age minus the wife's age is less than or equal to 5 years). The outcome variable is an indicator for an individual marrying in a given year ***and*** entering a match with a particular age gap. Thus, this compares the probability of entering a marriage with a high age gap to the probability of entering a marriage with a low age gap or not marrying, for example.[36] If a shock induces girls to marry men who are much older than them, this will impact bargaining power in the marriage relationship (Tauseef and Sufian, 2024).

Third, I investigate heterogeneity by partners' relative fertility preferences; this acts

[34] While it is generally accepted that rainfall and temperature are exogenous, this is less clear with conflict.

[35] In Section 7, I explain in detail how potential violations of this assumption affect the interpretation of the results and show that migration is not a major concern for my analysis.

[36] For this analysis, examining the difference in impact on high age gap and low age gap marriages will only be relevant for women, since most boys who marry before age 18 marry girls close to their own age.

as an additional proxy for match quality and compatibility. I analyze the probability of marriage by whether the husband and wife's fertility preferences are aligned, under the assumption that higher quality matches are those where fertility preferences are similar. The outcome variable in this case is an indicator for an individual entering a marriage in a given year with someone who has the same fertility preferences, measured retrospectively by asking the wife whether her husband wants more, fewer, or the same number of children as her. This analysis compares the probability of entering a marriage where both spouses have the same fertility preferences to the probability of either entering a marriage where the spouses have different fertility preferences or of not marrying.

4.2 Limitations

There are two limitations of the approach described above. First, age heaping could be a concern, since the year of marriage is calculated using the individual's age and age at marriage. If the individual's reported age is biased or somehow correlated with the shocks, age at marriage will be correlated with the shocks mechanically. As long as the misreporting is random, this will only introduce noise and will not bias my estimates.

Second, since not all ever-married individuals in my data can be matched with their spouses, there could be selection into the sample. This would be the case, for example, if individuals' probability of death was correlated with experiencing a shock in the year of their marriage. Thus, outcomes that require information from both individuals in a couple, such the age gap or the distance between spouses' birth locations, will be subject to this concern. To address this, I use survey sample weights as a robustness check to ensure that my sample is representative of the population.

5 Results

This section shows the impacts of low rainfall, high temperatures, and exposure to conflict on child marriage. Since the existing literature emphasizes the importance of institutions (for example, bride price and dowry) in determining the impacts of shocks on child marriage, I present all results separately by country and by gender.[37]

5.1 Individual-Year-Level Outcomes

Experiencing a year with low rainfall, high temperatures, or conflict deaths could change child marriage outcomes in that same year. Table 3 and Figure 4 show the impacts of

[37] While the probability of child marriage is relatively high, the likelihood of child marriage in a given year is relatively low. Most of the point estimates presented in this section are very small because the baseline means are also quite small when taking into consideration only a single year of an individual's life. However, even a small coefficient can result in an economically meaningful effect.

exposure to low rainfall, high temperatures, and conflict on the annual probability of child marriage in India, Indonesia, and Nepal, for girls and boys separately. Panel A of Table 3 shows that a year with low rainfall reduces child marriage for both boys and girls. This is true across all countries, though the effects are not statistically significant in all cases, and the point estimates range from 1-5% of the baseline mean. This suggests that the effects of low rainfall are consistent across contexts and the gender of the child.

The effects of high temperatures in Panel B are similar to the effects of low rainfall and are consistently negative across all countries and for both boys and girls. The point estimates range from 3-8% of the baseline mean for boys and 4-5% of the baseline mean for girls. The fact that both low rainfall and high temperatures affect child marriage in similar ways suggests that transitory shocks to income resulting from changes in agricultural output may be an important mechanism.

These findings are in contrast with the existing literature, which finds that the direction of the effect of low rainfall on child marriage for girls depends on the dominant direction of the marriage transfer (Corno et al., 2020).[38] On the contrary, in my analysis both locations with high dowry prevalence (India and Nepal) and with high bride price prevalence (Indonesia) experience statistically significant negative effects of low rainfall and high temperatures on the annual probability of child marriage. The differences between my estimates and those in the previous literature are likely due to differences in norms in the respective child marriage markets. Corno et al. (2020) study sub-Saharan Africa as a context where bride price is practiced and find that negative income shocks increase female child marriage, while I focus on Indonesia and find that negative income shocks decrease female child marriage. Whether or not the marriage transfer can fully adjust in response to shocks will determine how households react. In Indonesia, transfer amounts seem to be at least partially determined by cultural norms and are not completely flexible. This results in a situation similar to a price floor, leading to less child marriage in the presence of a negative income shock.[39] Additionally, relative levels of utility derived from child marriage may differ greatly across the two settings; if child marriage is more highly valued in sub-Saharan Africa and prices are not limited by social norms, negative income shocks could increase child marriage rates. Finally, the differences in estimated effects could be due to differences in the structure of the marriage markets in the two settings. In Indonesia, child marriage is common for both boys and girls, while in sub-Saharan Africa, child marriage is common only for girls, who mainly marry older men. To further explain the differences between my findings and the existing

[38] Corno et al. (2020) predict that, in markets that practice dowry, negative transitory income shocks will reduce child marriage and, in markets that practice bride price, negative transitory income shocks will increase child marriage. A detailed discussion and replication of the main results in Corno et al. (2020) is found in the Supplemental Appendix.

[39] This is illustrated by my model in Section 6. In the model, this rigidity comes from the fact that the transfer cannot flip between bride price and dowry. The results also hold if a minimum transfer amount is imposed.

literature, I construct the theoretical model presented in the following section.

Exposure to conflict, which affects the risk of experiencing conflict-related violence, also has impacts on child marriage. The effects of exposure to conflict on the annual probability of child marriage in Panel C of Table 3 are more nuanced. In Nepal for both boys and girls and in India for boys, the effects are close to zero.[40] In India, the effects for girls are positive, statistically significant, and 3% of the baseline mean, while in Indonesia the effect for boys is negative, statistically significant, and 30% of the baseline mean.[41] Thus, the impacts of conflict are different than the impacts of low rainfall and high temperatures, suggesting that they operate through a different channel. Additionally, at least in the case of Indonesia, the effects move in opposite directions for girls and boys.

Panel A of Figure 4 shows the effects of rainfall, temperature, and conflict shocks as a percentage of the relevant mean child marriage rates. Relative effects are similar in size for boys and girls and across all three countries. Panel B shows the raw percentage point effects, which have much more variation, due to the fact that the underlying likelihood of child marriage varies significantly by country and gender.[42]

5.2 Heterogeneity by Match Type

While previous research has found that marriage before age 18 is associated with negative impacts, certain types of child marriage are associated with worse outcomes. Determining which types of matches are most impacted by shocks is key for understanding how much delays in marriage timing impact individuals' well-being. To capture differences in effects for different types of matches, I estimate the regressions separately by type of match, defined based on the age of the child, the age gap between the spouses (calculated as husband's age minus wife's age), and the relative fertility preferences of the spouses.

First, to identify heterogeneity by age, I estimate regressions separately for years when individuals are ages 12-14 and years when individuals are ages 15-17. Panels A-C of Table 4 show the impacts for ages 12-14. These results are also presented in Figure 5 and Figure 6. In most cases, these impacts are much smaller than the overall effects, likely because child marriage for ages 12-14 is quite rare. The directions of all estimated effects are consistent with the main results.

Panels D-F of Table 4 show the effects of rainfall, temperature, and conflict on the annual probability of child marriage for children ages 15-17. These point estimates are very similar to those presented in Table 3, suggesting that shocks mainly change child marriage outcomes for individuals who experience shocks between the ages of 15-17. On

[40]For robustness, I also estimate the impacts in Nepal using the INSEC conflict data. These results can be seen in Table A2 and are consistent with the main specification.

[41]The impact of conflict on child marriage for girls in Indonesia is also large and positive, but not statistically significant at conventional levels.

[42]These estimates show the impacts on probability of child marriage in a given year. In the Supplemental Appendix, I show effects on overall probability of child marriage and age at first marriage.

the other hand, the worst types of child marriages, those that happen for very young children, are not impacted as much by the shocks. Panel A of Figure 5 and Figure 6 shows the effects on child marriage scaled by the relevant child marriage rates for girls and boys, respectively. Panel B of each figure shows the percentage point effects. The scaled effects are similar across age groups.

Next, I estimate the effects by spousal age gap, defining the outcome as the probability of entering a marriage with either a high age gap (more than five years) or a low age gap (up to five years) in a given year for girls. [43] These estimates are shown in Table 5 and in Figure 7. Low rainfall and high temperatures in Panels A and B of Table 5 have a stronger impact on low age gap marriages, but the coefficients are negative in almost all cases. Conflict exposure in Panel C increases child marriage in India primarily for girls entering marriages with a low age gap, while in Indonesia the effect is mainly coming from girls entering marriages with a high age gap.

As seen in the means in each panel, a higher fraction of girls enter low age gap marriages than high age gap marriages. This is one explanation for differences in absolute effect sizes for the two types of matches. Panel A of Figure 7 shows the effects relative to the baseline means, while Panel B plots the raw effects. The results in Panel A suggest that the relative effects are similar across match types. Measuring impacts by age gap matters because the age difference between spouses determines bargaining power, which affects a whole host of outcomes related to household decision-making (Nasrullah et al., 2014). In particular, in marriages with higher age gaps, girls are likely to have lower bargaining power. My findings suggest that more low age gap matches are averted as a result of experiencing a shock, but there are still some effects for high age gap matches.

Finally, I measure match quality using differences in fertility preferences across spouses as a proxy for match compatibility. The DHS survey for India asks women whether their husbands want more children, fewer children, or the same number of children as the women do.[44] I estimate effects on the probability of entering a marriage where the spouses have the same preferences versus a marriage where the spouses do not have the same preferences for girls in India in Table A3 and Figure 8. Panel A of Figure 8 shows effects relative to the means, and Panel B shows the percentage point effects. Both the relative and absolute effects are larger for women with the same fertility preferences, suggesting that high compatibility matches are more responsive to shocks.

[43] This probability is estimated relative to both individuals who did not marry in that year and individuals who married but with a different age gap. I do not estimate these regressions for boys, because there is not enough variation in age gap.

[44] There could be misreporting, especially if the woman was not alone when interviewed. The question asked how many children the husband wanted relative to the wife (fewer, the same, more or unknown).

6 Child Marriage Model

In this section, I develop a framework for how households' child marriage decisions can be impacted by changes to income and changes in the risk of experiencing conflict-related violence. Households trade off consumption and child marriage, since they must pay (or receive) a marriage transfer at the time of the match. Some households might prefer child marriage and be willing to pay a transfer, while others might not prefer child marriage and only marry their child if they receive a transfer. The ability and willingness to pay or receive the transfer depends on the household's income and preferences for child marriage.

In this model, pairs of households bargain over a marriage transfer, subject to reservation amounts for each household and to the norm that the sign of the transfer cannot change (relative to the expected sign in that particular market). If households agree on a transfer amount, the match occurs. If the households do not agree on an amount, the match does not occur. There are N pairs of households in the marriage market, household characteristics within each pair are allocated randomly, and there is no assortative matching. In each pair, one household has a daughter and the other has a son.[45]

Marriage transfers, $\tau_i(r)$, move from the bride's family to the groom's family (a positive transfer is dowry, and a negative transfer is bride price). i indicates the pair of households, and r indicates the state of nature (0 if the household does not receive a negative income shock and 1 if the household does receive a shock). The transfer amount is determined through bargaining between the households. I assume that households with daughters in a market with a history of bride price are unwilling to make a dowry payment, and households with sons in a market with a history of dowry are unwilling to make a bride price payment. In other words, this historical norm limits the values that $\tau_i(r)$ can take and implies that the sign of the marriage transfer will not change relative to the expected direction in that marriage market.[46] Households with daughters have an upper bound, $\tau_i^{ub}(r)$, for the transfer that they are willing to pay (this value could be either positive or negative), and households with sons have a lower bound, $\tau_i^{lb}(r)$, for the transfer that they are willing to receive (this value could be positive or negative).

Household consumption is determined by income and marriage transfer payments. For households with girls, consumption is given by $c_{g,i}(r, m_{g,i}) = y_{g,i}(r) - \tau_i(r) \cdot m_{g,i}$. For households with boys, consumption is given by $c_{b,i}(r, m_{b,i}) = y_{b,i}(r) + \tau_i(r) \cdot m_{b,i}$. For both equations, $m_{s,i}$ indicates whether the child marries, where $s \in g, b$ indicates households with girls and boys, and $c_{s,i}(r, m_{s,i})$ represents consumption. The income of each household, $y_{s,i}(r)$, varies across states of nature and across households. I assume that

[45]The model is agnostic to the pairing process. Each household has one child under age 18. However, this is generalizable to other settings, such as cases where girls are under 18 and boys are older.

[46]Empirically, the transfer direction is the same within geographical regions or religious/ethnic groups. Here, I bound the transfer amounts at zero. The predictions of the model still hold if transfer amounts are bounded away from zero.

changes to r affect all households similarly and that the distributions of $y_{g,i}(r)$ and $y_{b,i}(r)$ are identical, meaning that the expected income of the household is the same regardless of whether that household has a son or a daughter.

Parents maximize household utility, given by $u_{s,i}(c_{s,i}(r, m_{s,i}), m_{s,i}) = f(c_{s,i}(r, m_{s,i})) + \gamma_{s,i} \cdot m_{s,i}$, where $f'(c) > 0$, $f''(c) < 0$, and $\gamma_{s,i}$ represents household preferences for child marriage. Preferences vary across households with girls and boys. Assume that $\gamma_{s,i} \sim U(\underline{\gamma_{s,i}}, \overline{\gamma_{s,i}})$ and values can either be positive, indicating a preference for child marriage, or negative, indicating a preference against child marriage.[47] For tractability, I impose $f(\cdot) = log(\cdot)$. I also use a static model to describe the marriage decision in a single period, aligning with the empirical analysis.[48]

6.1 Households with Daughters

Households choose to marry their daughters if the utility of having their daughters married is higher than the utility of having their daughters remain unmarried, which is true if:

$$u_{g,i}(c_{g,i}(r, m_{g,i} = 1), m_{g,i} = 1) > u_{g,i}(c_{g,i}(r, m_{g,i} = 0), m_{g,i} = 0) \tag{2}$$

This provides an upper bound for the values of $\tau_i(r)$ that households with girls are willing to pay, depending on preferences and income. Simplifying and rearranging results in:[49]

$$\tau_i(r) \leq y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}) \equiv \tau_i^{ub}(r) \tag{3}$$

However, in markets where bride price is practiced, the transfer amount that the bride's household is willing to pay must be weakly negative due to the historic norms. Thus, in markets where bride price is practiced, we have the additional restriction that the upper bound of the transfer must be less than or equal to zero:[50]

$$\tau_i^{ub}(r) = min\{y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), 0\} \tag{4}$$

[47] It could be the case that $\underline{\gamma_{s,i}}$ is negative and $\overline{\gamma_{s,i}}$ is positive, or that both endpoints are positive or negative. Households might prefer child marriage as a way to protect or provide for their child (Sharma et al., 2015). Those with a negative preferences could have information about the negative health and education impacts of child marriage or value their child's right to make their own marriage decisions.

[48] Marriage is inherently a dynamic problem. However, if shocks are truly exogenous and transitory, then experiencing a negative shock will not change the expected value of marriage (or the expected value of not marrying) in the following periods. Thus, including a second period in the model changes the magnitudes of the thresholds determining the marriage decision but does not change the testable predictions. For simplicity, I include only the single period.

[49] See Appendix A for detailed derivations.

[50] Here the transfers are negative, since they move from the boy's household to the girl's household. Thus, in absolute value, the bound for girls' households will actually be the lower bound of the transfer amount. Note that when dowry is practiced, it does not matter if the willingness to pay for the girl's household is positive or negative, since the restriction to positive payments comes from the boy's side of the market.

When income changes, as long as the transfer amount is not near zero, $\tau_i^{ub}(r)$ also changes:

$$\frac{\partial \tau_i^{ub}(r)}{\partial y_{g,i}(r)} = 1 - e^{-\gamma_{g,i}} \tag{5}$$

When dowry is practiced and the prevailing marriage transfer is positive, $\gamma_{g,i}$ will be positive (this can be seen from equation (3)). Thus, in this case the upper bound of the transfer and income are positively related.[51] When bride price is practiced and the prevailing marriage transfer is (weakly) negative, the upper bound of the transfer and income are (weakly) negatively related, since $\gamma_{g,i}$ is (weakly) negative. Households also respond to exogenous changes in the distribution of preferences:

$$\frac{\partial \tau_i^{ub}(r)}{\partial \gamma_{g,i}} = y_{g,i}(r) \cdot e^{-\gamma_{g,i}} > 0 \tag{6}$$

As preferences for child marriage for girls increase, the upper bound of the marriage transfer amount increases, regardless of whether dowry or bride price is practiced.

6.2 Households with Sons

Similarly, households marry their sons if the utility of having their sons married is higher than the utility of having their sons remain unmarried, which is true if:

$$u_{b,i}(c_{b,i}(r, m_{b,i} = 1), m_{b,i} = 1) > u_{b,i}(c_{b,i}(r, m_{b,i} = 0), m_{b,i} = 0) \tag{7}$$

This provides a lower bound threshold for the amount that the boy's household is willing to accept:

$$\tau_i(r) \geq y_{b,i}(r) \cdot (e^{-\gamma_{b,i}} - 1) \equiv \tau_i^{lb}(r) \tag{8}$$

However, when dowry is practiced, the transfer amount must be weakly positive because of the historical norms, so the lower bound of the marriage transfer can be re-written as follows:

$$\tau_i^{lb}(r) = max\{y_{b,i}(r) \cdot (e^{-\gamma_{b,i}} - 1), 0\} \tag{9}$$

When transfer amounts are not near zero:

$$\frac{\partial \tau_i^{lb}(r)}{\partial y_{b,i}(r)} = e^{-\gamma_{b,i}} - 1 \tag{10}$$

Thus, when dowry is practiced and transfers are (weakly) positive, the lower bound is (weakly) positively related to income, and when bride price is practiced and transfers are

[51] If the upper bound of the marriage transfer is at/near zero, this relationship will be weak or nonexistent.

negative, the lower bound of the transfer is negatively related to income. Additionally, as preferences for child marriage go up, the lower bound of the transfer falls:

$$\frac{\partial \tau_i^{lb}(r)}{\gamma_{b,i}} = -y_{b,i} \cdot e^{-\gamma_{b,i}} < 0 \tag{11}$$

6.3 Marriage Transfer Bargaining

When the upper and lower bound of the marriage transfer are not equal, households bargain on an amount between the two thresholds. If there is no transfer amount that they can agree on, the match will not occur.[52] This provides a range of transfer values in equilibrium, varying based on the characteristics of the household. For child marriage to occur in a given market, there must be at least some pairs of households that have characteristics such that there are gains from trade, that is, $\tau_i^{ub}(r) > \tau_i^{lb}(r)$.

A key assumption is that the direction of the transfers does not change relative to the historical transfer direction in the geographical region or cultural group that defines a marriage market.[53] Amounts of the transfer can go to zero, but they cannot, for example, switch from dowry to bride price as a result of experiencing a shock. Thus, in the case where the transfer amount is positive (dowry), the thresholds for the households are defined based on the equations above:

$$\tau_i^{ub}(r) = y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}) \tag{3}$$

$$\tau_i^{lb}(r) = max\{y_{b,i}(r) \cdot (e^{-\gamma_{b,i}} - 1), 0\} \tag{9}$$

In the case where the transfer amount is negative (bride price):

$$\tau_i^{ub}(r) = min\{y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), 0\} \tag{4}$$

$$\tau_i^{lb}(r) = y_{b,i}(r) \cdot (e^{-\gamma_{b,i}} - 1) \tag{8}$$

6.4 Comparative Statics

This section explains the impacts on transfer amounts and the number of matches in the marriage market when household preferences change or income changes.

[52] E.g., let households with daughters have bargaining power β_i. Then, the transfer paid will be $\tau_i^*(r) = \tau_i^{lb}(r) + (1-\beta_i) \cdot (\tau_i^{ub}(r) - \tau_i^{lb}(r))$, as long as $\tau_i^{ub}(r) > \tau_i^{lb}(r)$. If $\tau_i^{ub}(r) < \tau_i^{lb}(r)$, the match will not occur.

[53] This generally aligns with the empirical setting. See, for example, Anderson (2007).

6.4.1 Changes in Preferences: Households with Daughters

For households with daughters, if preferences for child marriage increase, all else equal, transfer amounts will increase for some households (those where, previously, $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$) and new matches will occur for some households (those where, previously, $\tau_i^{ub}(r) < \tau_i^{lb}(r)$). If preferences for child marriage decrease, the amount of the transfer will decrease and/or child marriage will not occur for some pairs of households (those where, previously, $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$). For other households, there will be no change (those where, previously, $\tau_i^{ub}(r) < \tau_i^{lb}(r)$). Thus, the number of matches taking place in the marriage market as a whole is weakly positively correlated with levels of utility and preferences for female child marriage.

6.4.2 Changes in Preferences: Households with Sons

For households with sons, if preferences for child marriage increase, all else equal, transfer amounts will decrease for some households (those where, previously, $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$) and new matches will occur for some households (those where, previously, $\tau_i^{ub}(r) < \tau_i^{lb}(r)$). If preferences for child marriage decrease, the amount of the transfer will increase and/or child marriage will not occur for some households (those where, previously, $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$). For other households, there will be no change (those where, previously, $\tau_i^{ub}(r) < \tau_i^{lb}(r)$). Thus, the number of matches taking place in the marriage market is weakly positively correlated with preferences for male child marriage.

6.4.3 Changes in Income

Suppose that income falls due to changes in the state of nature. In the case of dowry, as shown above, $\frac{\partial \tau_i^{lb}(r)}{\partial y_{b,i}} \geq 0$ and $\frac{\partial \tau_i^{ub}(r)}{\partial y_{b,i}} > 0$. So, if income falls, both $\tau_i^{lb}(r)$ and $\tau_i^{ub}(r)$ will fall. A match that counterfactually would have happened, where $\tau_i^{ub}(r) > \tau_i^{lb}(r)$, will no longer happen if $\tau_i^{ub}(r)$ falls enough relative to $\tau_i^{lb}(r)$ such that it is no longer true that $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$. Intuitively, the number of matches decreases if the upper threshold falls enough relative to the lower threshold to close the gap between the thresholds.

For bride price, from above, $\frac{\partial \tau_i^{lb}(r)}{\partial y_{b,i}} < 0$ and $\frac{\partial \tau_i^{ub}(r)}{\partial y_{b,i}} \leq 0$. So, if income falls, both $\tau_i^{lb}(r)$ and $\tau_i^{ub}(r)$ increase. A match that would have happened, where $\tau_i^{ub}(r) > \tau_i^{lb}(r)$, will no longer happen if $\tau_i^{lb}(r)$ increases enough relative to $\tau_i^{ub}(r)$ that it is no longer true that $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$. Intuitively, the number of matches decreases if the lower threshold rises enough relative to the upper threshold to close the gap between the thresholds. In both cases, marriages will be deferred in response to lower income if, on average:

$$\Delta\tau_i^{lb}(r) - \Delta\tau_i^{ub}(r) > \tau_i^{ub}(r) - \tau_i^{lb}(r) \tag{12}$$

Given the assumptions above, this will be true if:

$$E\{e^{-\gamma_{b,i}} + e^{-\gamma_{g,i}}\} > 2 \tag{13}$$

This inequality holds for settings of both dowry and bride price if preferences for child marriage, for either girls and/or boys, are small enough (the distribution is far enough to the left). If this is the case, the number of matches decreases in response to a negative income shock.[54] The impact of income on the number of matches in the market for child marriage does not depend on whether bride price or dowry is practiced. This is because the two cases are symmetric.

6.5 Testable Predictions

There are several testable implications of this model. This subsection explains the predictions of the model when income falls or preferences for child marriage change.

Decreases in income: When income falls due to low rainfall or high temperatures, both the upper and lower bounds of the marriage transfer will approach zero. The model predicts that the number of matches in the marriage market will decrease, as long as preferences for child marriage are weak enough, since the household that makes the payment is more responsive to the shock than the household receiving the payment.[55] The absolute value of the transfer amount will fall for individual matches that still take place, but the average transfer amount in the marriage market could increase, decrease, or stay the same, depending on whether the high- or low-transfer matches are deferred.

Increased Risk of Conflict-Related Violence: If conflict causes the risk of physical or sexual violence to increase, this may increase the desire of parents to marry their daughters as a means of protecting them from potential violence, which would appear as an rightward shift in the distribution of child marriage preferences for girls. In this model, this increases the upper bound of the range of possible marriage transfer values. Thus, the model predicts that the number of matches taking place in the market should increase. Additionally, transfer amounts will increase (in terms of the absolute value of the marriage transfer, dowry amounts will increase and bride price amounts will decrease).

Participation in Conflict: If young boys participate in conflict when conflict violence comes to their district, this would cause the number of boys on the marriage market to decrease, since those boys are likely not looking to get married, decreasing preferences for male child marriage. This would raise the lower bound for the range of marriage transfers. As a result, the model predicts that the total number of matches in the market

[54] A similar condition applies for the case where the transfer amounts are near zero. See Appendix A.

[55] If preferences for child marriage are very high, the opposite could occur. This could explain the findings of Corno et al. (2020) that negative income shocks increase child marriage in sub-Saharan Africa. My result also depends on the fact that the marriage transfer cannot fully adjust to clear the market (in markets with dowry, bride price will never be practiced, and vice versa).

will decrease, and transfer amounts will increase (in terms of the absolute value of the marriage transfer, dowry amounts will increase and bride price amounts will decrease).

7 Mechanisms

Following the model's predictions, this section discusses mechanisms through which rainfall, temperature, and conflict could impact child marriage outcomes: the direction of the marriage transfers, changes in the amount of the marriage transfer, impacts on income, and changes to perceptions of risk. I show that the direction of the marriage transfer is not a key mechanism; however, transitory changes to income and changes to the risk of experiencing conflict-related violence seem to be major channels through which shocks impact child marriage outcomes.

7.1 Marriage Transfer Direction

Existing literature emphasizes the importance of the direction of the marriage transfer in determining the response of child marriage to a covarying income shock (Corno et al., 2020, Trinh and Zhang, 2020). However, I do not find that the direction of the marriage transfer matters in this setting. Since dowry is traditionally practiced in India and Nepal and bride price is primarily practiced in Indonesia, if the direction of the marriage transfer was the main channel through which shocks impact marriage outcomes, the effects should move in opposite directions in these countries. This is not the case, as shown in Table 3.

To provide additional evidence that the differences between bride price and dowry do not drive the effects, I also show within-country estimates by religion. Though there may be imperfect compliance, Hindus are more likely to practice dowry, Muslims are more likely to practice bride price, and Christians do not have a set marriage transfer tradition.

For the case of India in Table A4, the impacts of shocks are consistent across religious groups, except for differences in the impact of rainfall shocks on girls, where Hindus and Christians experience opposite effects. In Indonesia, effects are also similar across religious groups, as shown in Table A5. If anything, the effects for Hindus seem to be more positive than the effects for Muslims, which is the opposite of the predictions in Corno et al. (2020). Estimates for Nepal are shown in Table A6. Again, there is no consistent pattern of differences across religious groups.Any differences that do exist across religious groups within countries do not seem to be related to the direction of the marriage transfer. This suggests that the direction of the marriage transfer does not primarily determine the effects of shocks on child marriage outcomes.

7.2 Marriage Transfer Amount

In my model, households reduce child marriage when exposed to a negative transitory income shock in part because transfer amounts cannot fully adjust to clear the market.[56] If marriage transfers are not heavily constrained by historic norms, changes to income and preferences for child marriage could cause changes in the average marriage transfer amount. To test whether marriage transfer amounts respond to shocks, I use data on bride price amounts for individuals in Indonesia, which is available through the Indonesian Family Life Survey (IFLS).

Table 6 shows the impact of a rainfall, temperature, or conflict shock on the average bride price amount. The first column shows the unwinsorized bride price values, and there is a large negative effect for all three types of shocks, though only marginally statistically significant due to the small sample. However, when these values are winsorized in columns (2) and (3) at the 99th and 95th percentile, respectively, the effect shrinks towards zero and becomes statistically insignificant for low rainfall and high temperature shocks in Panel A and Panel B. This suggests that changes are only taking place at the far right tail of the bride price distribution and are not affecting the median or lower parts of the distribution. For conflict exposure in Panel C, the effect remains relatively large, even when bride price amounts are winsorized, suggesting that average bride price responds more to conflict shocks than to rainfall and temperature shocks.[57]

These findings are consistent with my model and explain why the estimated effects of shocks on child marriage are the same in locations where bride price and dowry are practiced: if transfer amounts are not fully flexible, households that experience a decrease in income are less able to pay a marriage transfer, regardless of who makes the payment.

7.3 Impacts on Income

I use several strategies to provide evidence that low rainfall and high temperatures impact child marriage through transitory changes in income. First, I present an instrumental variables approach, which uses rainfall and temperature shocks to instrument for agricultural yields, which act as a proxy for income. Second, I use data on irrigation to examine whether the effects of shocks are smaller in areas with high levels of irrigation.[58]

[56]In my model this restriction is implemented by bounding the marriage transfer amount at zero. The model's predictions still hold if the transfer is bounded away from zero.
[57]Since bride price is only observed for matches that take place, these effects are descriptive and not strictly causal. Figure A2 shows a kernel density of bride price for girls ages 12-17 in years with and without shocks. I also estimate quantile regressions for the bride price amounts, shown in Figure A3.
[58]In the Supplemental Appendix, I examine the effects of rainfall and temperature on district-level Gross Regional Domestic Product and estimate the main specification separately for rural and urban areas.

7.3.1 Instrumental Variables Approach

First, to directly test whether rainfall and temperature shocks primarily impact the annual probability of child marriage through changing income, I implement an instrumental variables (IV) approach, where shocks are used to instrument for endogenous crop yields. The exclusion restriction requires that low rainfall and high temperatures impact child marriage only through changes in agricultural yields. This is unlikely to be strictly true.[59] Since the exclusion restriction may not hold, estimates should be interpreted as suggestive evidence of the relationship between shocks, crop yields, and marriage, but are not strictly causal.

I run a district-level analysis using data on crop yields for India from ICRISAT.[60] The first stage used to create the predicted values for crop yields based on exogenous variation in rainfall and temperature shocks is:

$$CropYield_{dt} = \beta_0 + \beta_1 LowRain_{dt} + \beta_1 HighTemp_{dt} + \gamma_d + \delta_t + \epsilon_{dt} \tag{14}$$

In this equation, $CropYield_{dt}$ represents the crop yield index, $LowRain_{dt}$ is an indicator for a low rainfall year, $HighTemp_{dt}$ is an indicator for a high temperature year, γ_d is a fixed effect for the district, and δ_t is a fixed effect for the year. This equation estimates the impacts of the shocks (low rainfall and high temperature) on crop yields, capturing exogenous variation in crop yields from year to year. The second stage is estimated as:

$$Y_{dt} = \beta_0 + \beta_1 \widehat{CropYield}_{dt} + \gamma_d + \delta_t + \epsilon_{dt} \tag{15}$$

$\widehat{CropYield}_{dt}$ represents the predicted values of crop yields from the first stage. As a placebo test, I run a similar IV for conflict shocks. Since conflict likely does not directly impact agricultural yields, I do not expect to see an effect. The first stage for conflict is:

$$CropYield_{dt} = \beta_0 + \beta_1 AnyConflict_{dt} + \gamma_d + \delta_t + \epsilon_{dt} \tag{16}$$

Table 7 presents the results of the IV regressions, with the main estimates in Panel A and the first-stage estimates in Panel B. The F-statistics for the first stage are also shown in Panel A, demonstrating a strong link between rainfall and temperature shocks and the index of agricultural yields in columns (1) and (2). For conflict in columns (3) and (4), the first stage is quite weak, as expected. The second stage outcome is the

[59] For example, rainfall could cause flooding and destruction of property, influencing child marriage by reducing long-term wealth independent of transitory changes to income.

[60] Crop yields (by hectare) are provided for a number of different crops separately. To capture overall changes in crop yields, I create an index across yields for all types of crops, following Kling et al. (2007). I aggregate the annual probability of child marriage to the district level by taking the number of child marriages that occur in a given year in my sample and dividing by the number of individuals in the district in that year who are on the marriage market (defined as individuals between the ages of 12 and 17 who are not yet married at the start of the year). The data cover 1967-2010 for 302 districts.

district-level probability of child marriage in a given year. The results in columns (1) and (2) show that higher agricultural output leads to higher probabilities of child marriage for girls, though not for boys.[61] This provides evidence that transitory reductions in income decrease female child marriage, while increases in income increase female child marriage. Thus, income is likely a major channel for the impacts of low rainfall and high temperatures on child marriage. The fact that the first stage estimates for conflict are much weaker suggests that exposure to conflict does not impact marriage outcomes through transitory changes in income.

7.3.2 Irrigation as a Mitigator

If low rainfall and high temperatures impact child marriage through reducing agricultural income, factors that mitigate the negative impacts of low rainfall (or high temperatures) on agricultural income should also mitigate the negative impacts of low rainfall (or high temperatures) on child marriage. For example, when areas with high levels of irrigation experience rainfall shocks, the effects of low rainfall on child marriage should not be as strong, since the effects of rainfall on agricultural income will be smaller. On the other hand, irrigation should not mitigate the effects of temperature or conflict shocks, since those shocks do not directly reduce the water supply for crops.[62]

I use ICRISAT data in India to see whether irrigation mitigates the impacts of shocks. If transitory income changes are the true mechanism, irrigation should mitigate the effects of low rainfall but not of temperature or conflict shocks. I estimate the following:

$$Y_{idt} = \beta_0 + \beta_1 Shock_{dt} + \beta_2 Shock_{dt} \times Irr_{dt} + \beta_2 Irr_{dt} + \gamma_d + \delta_t + X_{idt} + \epsilon_{idt} \tag{17}$$

where Irr_{dt} is a continuous measure of irrigation (in 1000s of hectares).

Table A7 shows the estimates of equation (5) for boys and girls in columns (1) and (2), respectively. Aligning with my main results, the coefficients in Panel A are negative for rainfall shocks; however, the interaction between low rainfall and irrigated area has a positive coefficient, indicating that increased irrigation mitigates the impacts of low rainfall on child marriage. The coefficients are statistically significant for girls, but not for boys. This suggests that, at least in the case of India, rainfall shocks impact child marriage by decreasing agricultural income.[63]

[61] This is likely because the main estimates for the impacts of shocks of male child marriage in India are not statistically significant.

[62] High temperatures might be correlated with drought, so irrigation might still have a slight mitigating effect. However, high temperatures themselves can be harmful for crop output, even with plenty of rain. If this is the case, irrigation will not mitigate the effects of temperature shocks.

[63] The estimates of equation (17) for high temperatures and conflict exposure in Panel B and Panel C, respectively, act as a placebo test, since irrigation should not mitigate the impacts of temperature or conflict shocks. The point estimates on the interaction term confirm that irrigation does not mitigate temperature and conflict shocks.

7.4 Risk of Violence

It is likely that an attempt to protect their children, especially daughters, from sexual violence is one factor that drives households to marry their children at a young age (Sharma et al., 2015). If children in a household are at higher risk of experiencing violence, this could increase that household's preferences for child marriage. My model predicts that child marriage will increase as preferences for child marriage increase. Changes in preferences for child marriage cannot be directly observed in my sample, since I do not see the reasons why families choose to have their children marry at a young age. However, given my finding that the annual probability of female (but not male) child marriage increases in the presence of exposure to conflict in India and Indonesia, it is likely that considerations of gender-specific risk of violence may be at play.

For boys, an increased risk of violence may increase participation in armed groups, as suggested by Lawoti (2010). Boys are often recruited, either voluntarily or forcibly, by rebel groups or the government. If this is the case, these boys are no longer on the marriage market, so their preferences for marriage fall. Thus, increased participation in armed groups, which decreases preferences for male child marriage, should cause the annual probability of male child marriage to fall. This is exactly what I find. The fact that female child marriage is increasing in the presence of conflict, while male child marriage is decreasing, also suggests that more young girls are marrying older men.

7.5 Impacts on Migration

Changes in rainfall, temperatures, and conflict could also impact migration in a way that correlates with child marriage. In this subsection, I address potential concerns of bias in my estimates due to differential migration. I test whether changes in migration patterns are causing bias in two ways: First, for Nepal, I assign shocks to each individual based on that individual's birthplace (instead of the district of residence when surveyed) and re-estimate the main specification; second, I use the detailed migration histories in the IFLS to fully account for migration in Indonesia by assigning shocks to individuals based on their actual location in a given year and re-estimating the main specification.[64]

For Nepal, because the district of birth is available in the data, I re-assign shocks to individuals based on their district of birth. Table A9 shows the impacts of shocks on the annual probability of child marriage for individuals in Nepal, with individuals assigned to their district of birth instead of current residence. The coefficient estimates in columns (3) and (4) are similar to those in the main specification (reproduced in columns (1) and (2)), where shocks are assigned based on the current residence. This suggests that migration between an individual's birth and the time that they are surveyed does not bias the estimates in a meaningful way.

[64] In the Supplemental Appendix, I also look at the impacts of shocks on migration as an outcome.

I also use the Indonesian Family Life Survey (IFLS) to correct for migration in Indonesia. A disadvantage of the IFLS is that it is much smaller than the Census. However, it provides much more detailed data and is a panel dataset, meaning that the same individuals are tracked over time. Combined with the detailed data available on each individual's migration history, this makes it possible to see where an individual lived in each year of their lives.[65] Because I observe an individual's location of residence in each year, I can directly match the shocks in each year to where the individual was actually living in that year, as well as to their place of birth or residence at the time of the survey.[66]

Table A10 shows estimates of the main specification using IFLS data in columns (1) and (2), assigning shocks based on individuals' location at the time of the survey. The sample is small, so the effects in panels A and B are not statistically significant. However, for the most part, the coefficients are negative, aligning with my main results. In panel C, the effect of conflict remains statistically significant, increasing the annual probability of child marriage for girls and decreasing the annual probability of child marriage for boys.

In columns (3) and (4) of Table A10, I estimate the same specification, but assign shocks to individuals based on their birthplace. I find that the effects are similar to those in columns (1) and (2). Finally, I fully correct for migration in columns (5) and (6), assigning shocks to individuals based on their actual location in each year. Again, the point estimates are quite similar to those shown in the previous columns. The fact that these results are consistent with my main findings and across specifications suggests that the estimates are not sensitive to migration patterns that individuals experience.[67]

8 Robustness & Additional Outcomes

This section provides robustness checks, showing that the results are not sensitive to the timing of the shocks, the sample used, or the regression specification.[68]

8.1 Impacts of Lagged Shocks

Depending on when the growing season is and when a negative shock occurs, shocks may impact marriage market decisions in the following year instead of the contemporary

[65]I show in Table A8 that most migrants in the IFLS move from a district experiencing a shock to another district experiencing a shock or from a district not experiencing a shock to another district not experiencing a shock. This suggests shock avoidance is not the main driver of migration.

[66]I use adult survey respondents in all 5 waves of the IFLS who answered the migration history questions. I apply similar sample restrictions as for the main sample, keeping only individuals over age 17, below age 50, and who are not married before age 12. This gives me a sample of around 31,000 individuals. Using the marriage and migration histories, I expand the data into an individual-year-level panel and run the main specification in equation (1).

[67]However, the sample is quite small, so I do not have enough power to reject differences between any of the pairs of coefficients, and most of the coefficients are not statistically significant.

[68]I also use alternate shock definitions, district level regressions, corrections for spatial correlation, and survey weights for robustness. Results are shown in the Supplemental Appendix.

year. Alternatively, the impacts of shocks may persist over multiple years. I estimate the impacts of shocks in both the current and previous years on the household's decision to marry their child in the current year. These estimates are shown in Table A1. These regressions show the effect of lagged shocks from the previous three years, conditional on the individual not previously marrying. For the most part, the effects on child marriage are not simply a result of intertemporal substitution from one year to the next, since the impact of shocks in previous years generally has the same sign as the main effect.

8.2 Impacts on Marriage for Young Adults (Ages 18-24)

In addition to impacting child marriage (ages 12-17), rainfall, temperature, and conflict could also impact marriage outcomes for adults, through similar mechanisms as the effects on child marriage. I estimate the impacts on marriage for young adults (ages 18-24), shown in Table A11. The signs of the coefficients are generally consistent with the effects on child marriage, though for exposure to conflict, the coefficients are positive for both men and women. This suggests that the impacts of transitory changes in income on marriage are similar for children and young adults, while the impacts of changes in risk of violence do not always move in the same direction for children and young adults.[69]

8.3 Multiple Shocks

If shocks are correlated, the effects of experiencing multiple different shocks in a single year might not be additive. Panel A of Table A12 shows the estimated effects when including all three shocks in the same regression. These effects are similar to the main results, suggesting that correlations between the shocks are not driving my estimates. Panel B shows the effects of experiencing both a low rainfall and high temperature shock in a given year. When combining the two shocks, effects do not seem to be much stronger than the effects of single shocks. Panel C shows the effects of experiencing low rainfall, high temperatures, and conflict together in a single year. All effects are statistically insignificant and quite small, likely because the magnitudes of the effects are different for each shock and can move in different directions, introducing noise into the estimates when combined.

9 Discussion & Conclusion

In this paper, I find that transitory changes in income and the risk of experiencing conflict-related violence affect the decision-making processes of households regarding marriage for their children. I show that the direction of the effects are similar across different types

[69] I also show the impacts on the distribution of age at marriage and age gap in the Supplemental Appendix.

of matches, while worse types of child marriage matches seem to be less responsive to shocks. This paper is one of the first to examine both male and female child marriage across multiple countries and shocks.

Child marriage for both boys and girls responds to transitory changes in income and the risk of experiencing conflict-related violence. I find that low rainfall and high temperatures decrease child marriage rates for both girls and boys. Exposure to conflict decreases the annual child marriage probability for boys and increases the annual child marriage probability for girls. Low rainfall and high temperatures likely operate primarily through impacts on income, which affect boys and girls similarly. Exposure to conflict likely operates primarily through increasing the risk of violence for girls and participation in conflict for boys, meaning that effects for boys and girls go in opposite directions. This matches the predictions of my theoretical model. Importantly, I find that the direction of the marriage transfer does not predict the direction of the effect. I also find that changes in migration patterns do not drive the changes in marriage outcomes. Rather, the effects on marriage outcomes are robust to accounting for migration, though marriage and migration are likely associated with one another.

Additionally, different types of matches are more or less responsive to shocks. For girls, marriages where the spouses are of similar ages are more likely to be affected by shocks than matches where the spouses have a large spousal age gap. The size of the impact also varies by the age of the child at the time of the shock. Shocks are more likely to defer matches for older boys and girls than for younger boys and girls. This suggests that the worst types of child marriage are also the least marginal. Future work could focus on how these different types of matches impact individuals' welfare later in life.

Understanding how economic shocks impact child marriage is important because of the international consensus towards decreasing child marriage. This is especially true where legal measures against child marriage have not succeeded, such as in India, Indonesia, and Nepal. To reduce child marriage effectively, the incentives behind child marriage must be understood. This paper is one of the first to study male child marriage and contributes to a larger understanding of how the child marriage market operates.

As a word of caution, my findings do not suggest reducing income as a viable strategy for combating child marriage, as this would negatively affect overall welfare. Rather, my findings highlight the importance of including conditionality with income-enhancing policies to ensure that this increased income is not spent on funding child marriage. For example, cash transfers could be made conditional on children in the household remaining unmarried to avoid unintentionally incentivizing child marriage. Future research could investigate how these types of policies could be implemented most effectively.

My findings imply that policymakers should think carefully about how changes to income impact child marriage. While much of the previous literature finds that positive income shocks in the form of cash transfers decrease child marriage or delay marriage (see, for example, the summary in Mathers (2021)), others find that cash transfers increase child marriage (Figueiredo, 2024). My finding that negative transitory income shocks decrease child marriage adds to the evidence that changes to income impact child marriage. Additionally, since negative shocks in this setting and positive shocks in other settings both reduce child marriage, impacts of income shocks may not always be symmetric, and the manner in which the shock occurs (i.e. expected or unexpected, transitory or relatively permanent) may dictate households' responses. Any policy that effectively reduces child marriage will have to take into account the incentives behind child marriage and the overall welfare of individuals and households.

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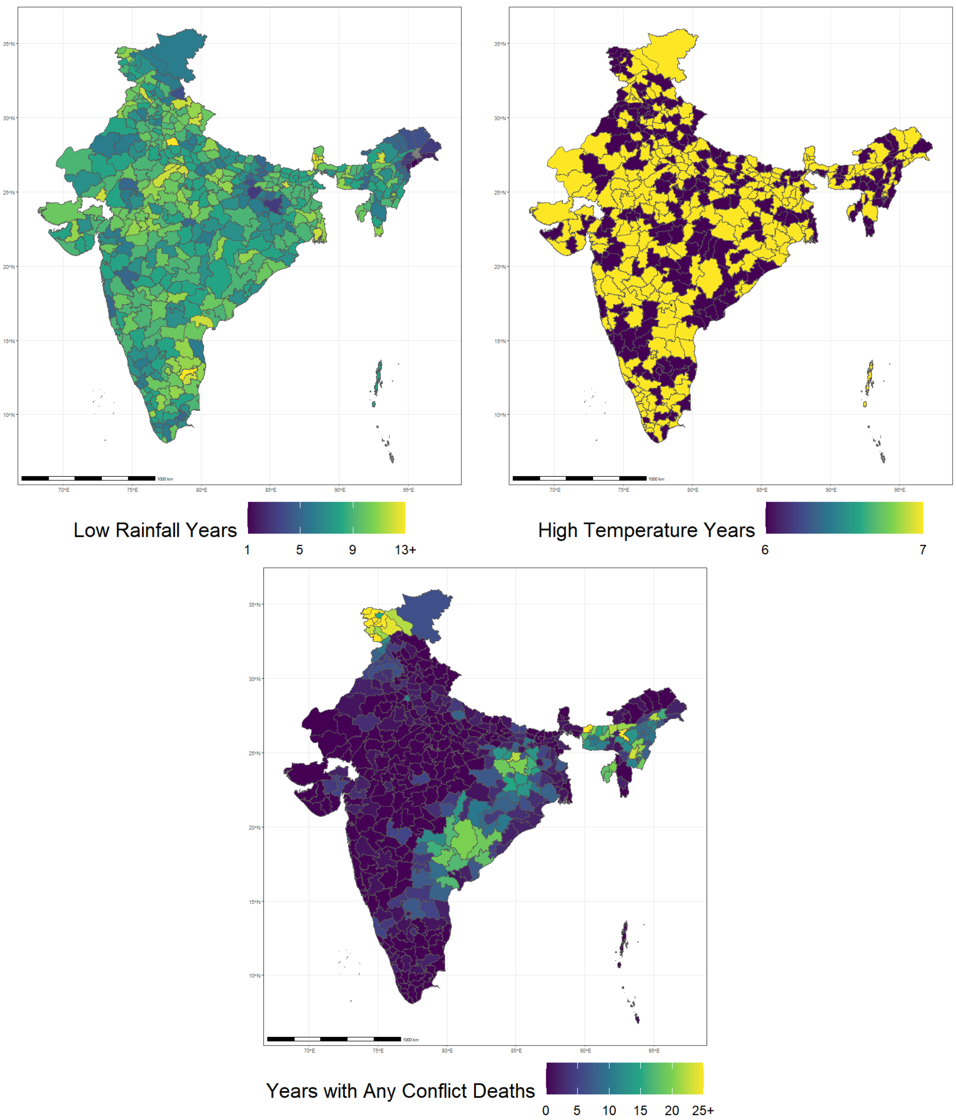
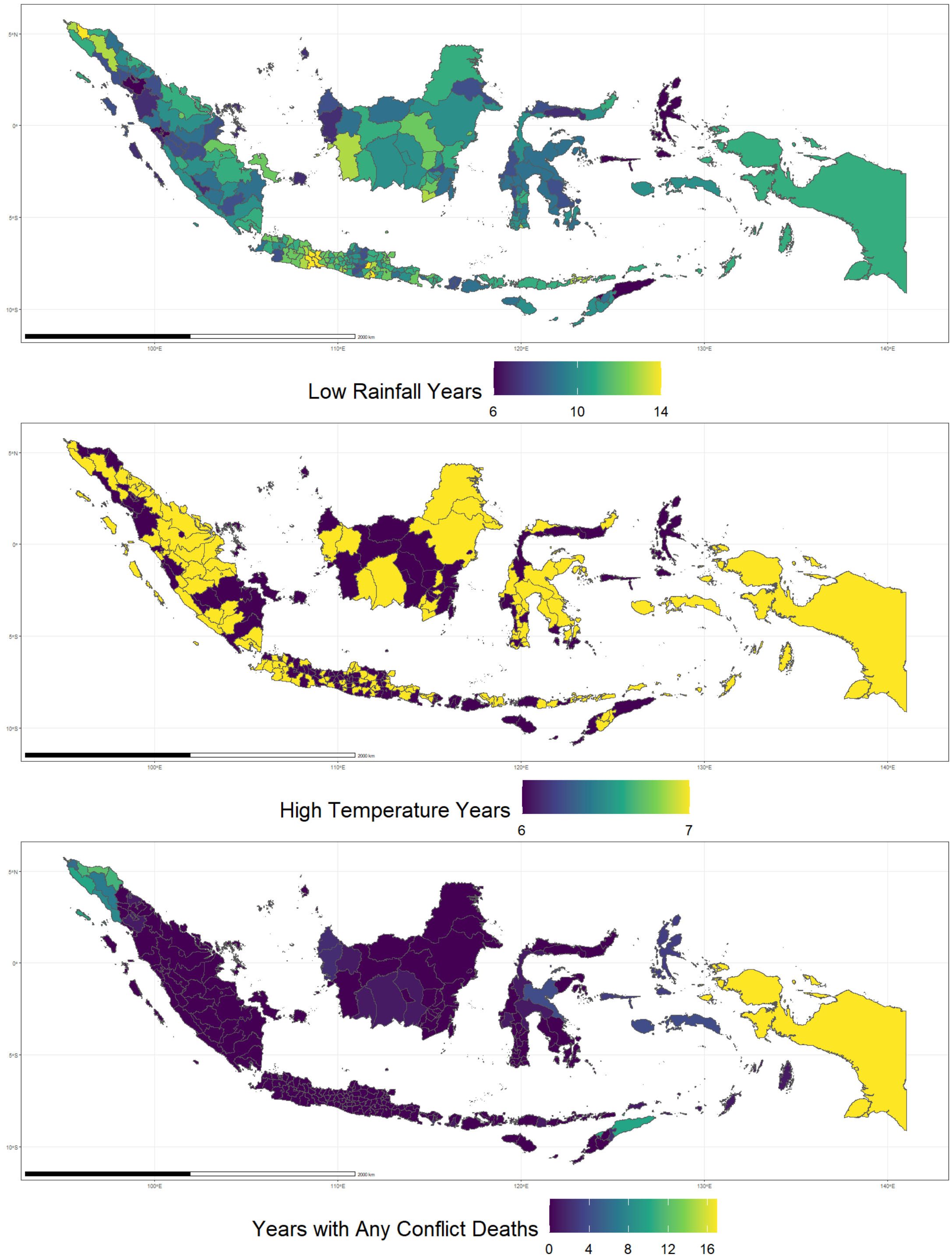
Figures & Tables

Figure 1. Rainfall, Temperature, and Conflict in India by District



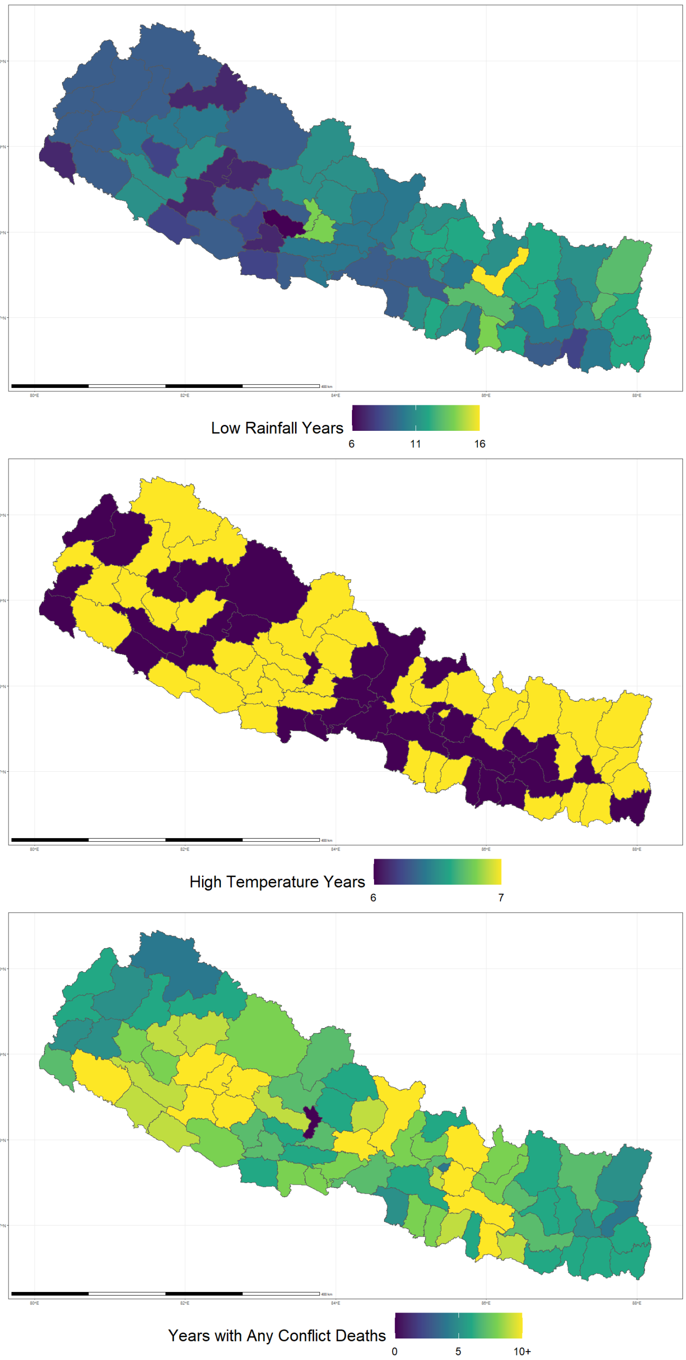
Notes: This figure shows the number of years with a low rainfall shock, high temperature shock, or conflict shock between 1950-2017 by district in India. Rainfall and temperature data come from the University of Delaware and conflict data comes from UCDP. A low rainfall year is defined as one with rainfall below the 15th percentile of the district-specific gamma distribution and a high temperature year is defined as one with temperatures above the 90th percentile of the district-specific distribution.

Figure 2. Rainfall, Temperature, and Conflict in Indonesia by Regency



Notes: This figure shows the number of years with a low rainfall shock, high temperature shock, or conflict shock between 1950-2017 by district in Indonesia. Rainfall and temperature data come from the University of Delaware and conflict data comes from UCDP. A low rainfall year is defined as one with rainfall below the 15th percentile of the district-specific gamma distribution and a high temperature year is defined as one with temperatures above the 90th percentile of the district-specific distribution.

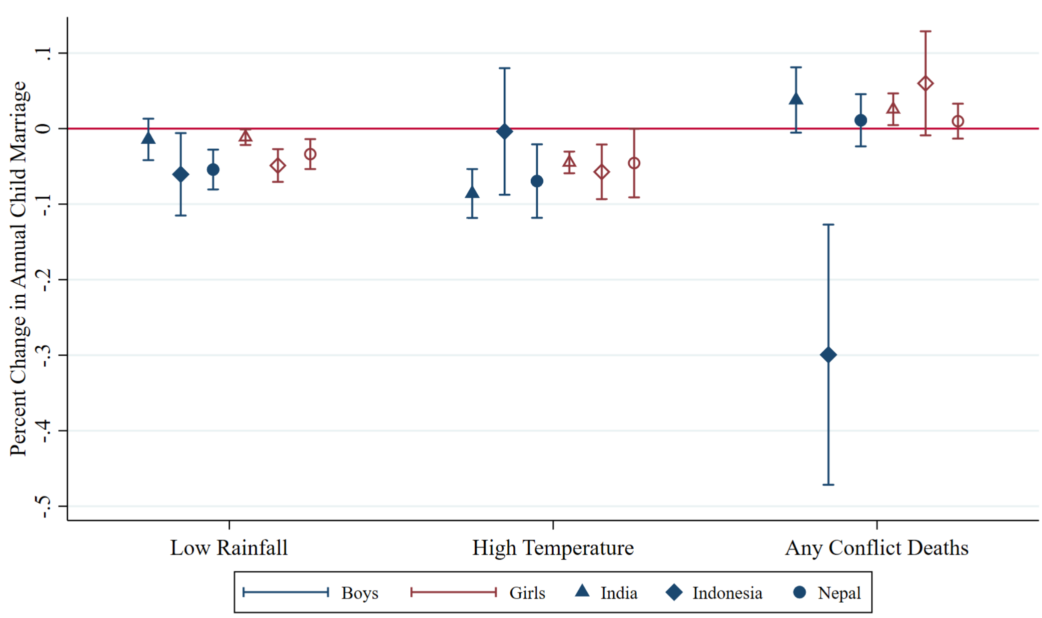
Figure 3. Rainfall, Temperature, and Conflict in Nepal by District



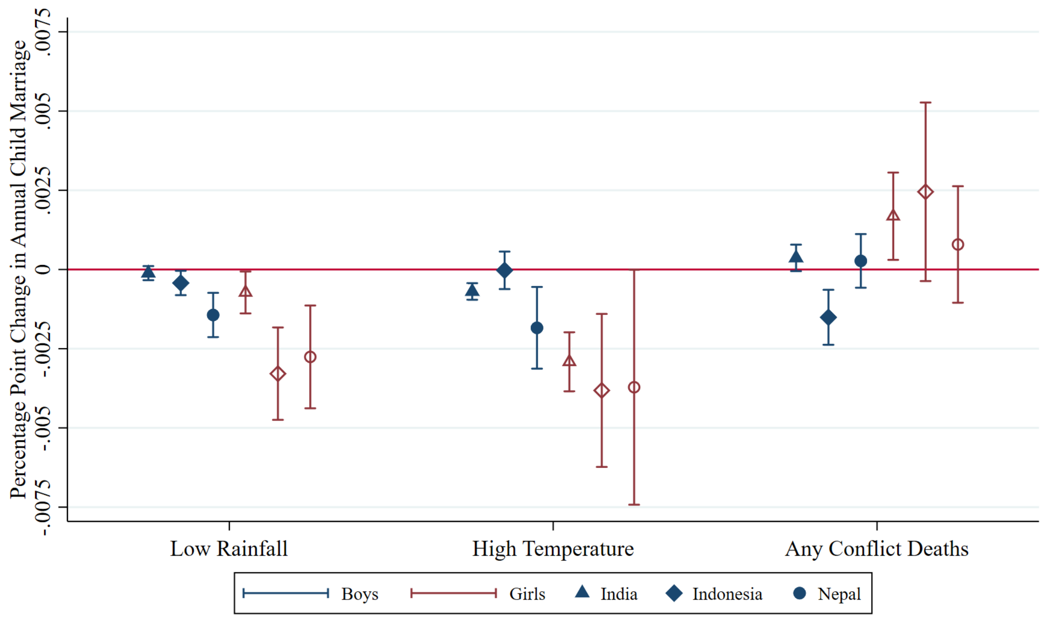
Notes: This figure shows the number of years with a low rainfall shock, high temperature shock, or conflict shock between 1950-2017 by district in Nepal. Rainfall and temperature data come from the University of Delaware and conflict data comes from UCDP. A low rainfall year is defined as one with rainfall below the 15th percentile of the district-specific gamma distribution and a high temperature year is defined as one with temperatures above the 90th percentile of the district-specific distribution.

Figure 4. Main Results: Impacts of Shocks on Child Marriage

Panel A. Effects Relative to the Mean



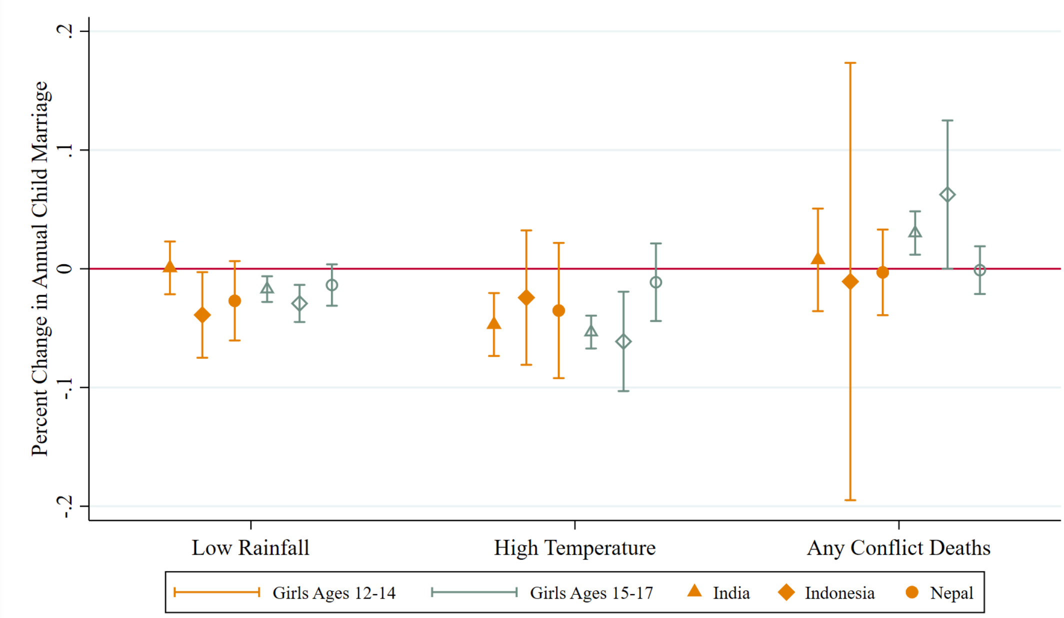
Panel B. Effects as Percentage Points



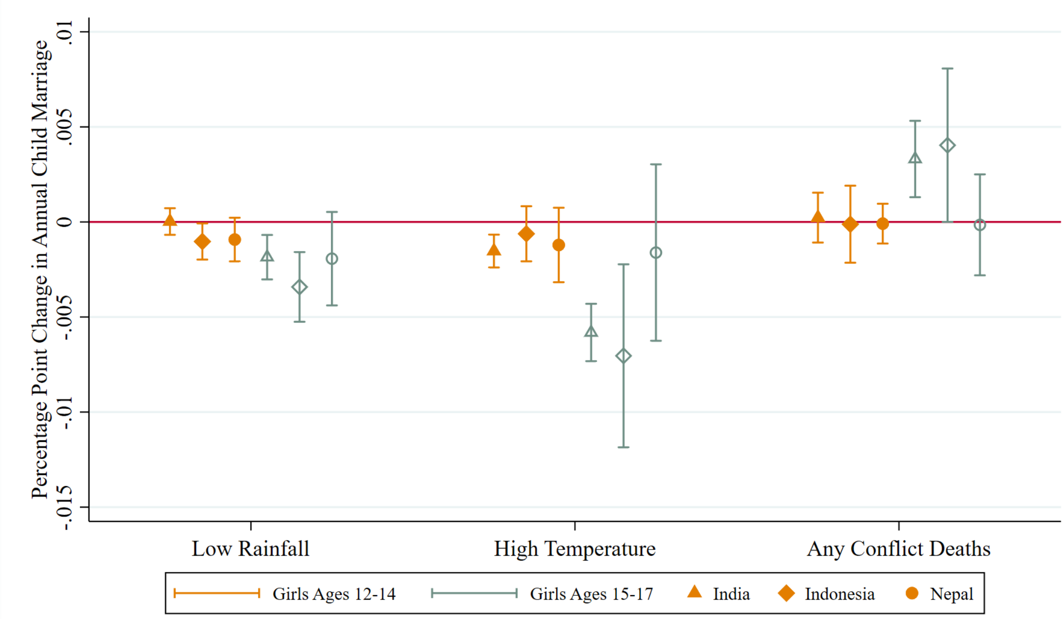
Notes: This figure shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Effects are estimated separately for boys and girls. Panel A shows the effects relative to the mean of the outcome, and Panel B shows the raw estimates as percentage point effects. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level, and 90% confidence intervals are used.

Figure 5. Main Results: Impacts of Shocks on Child Marriage by Age, Girls

Panel A. Effects Relative to the Mean



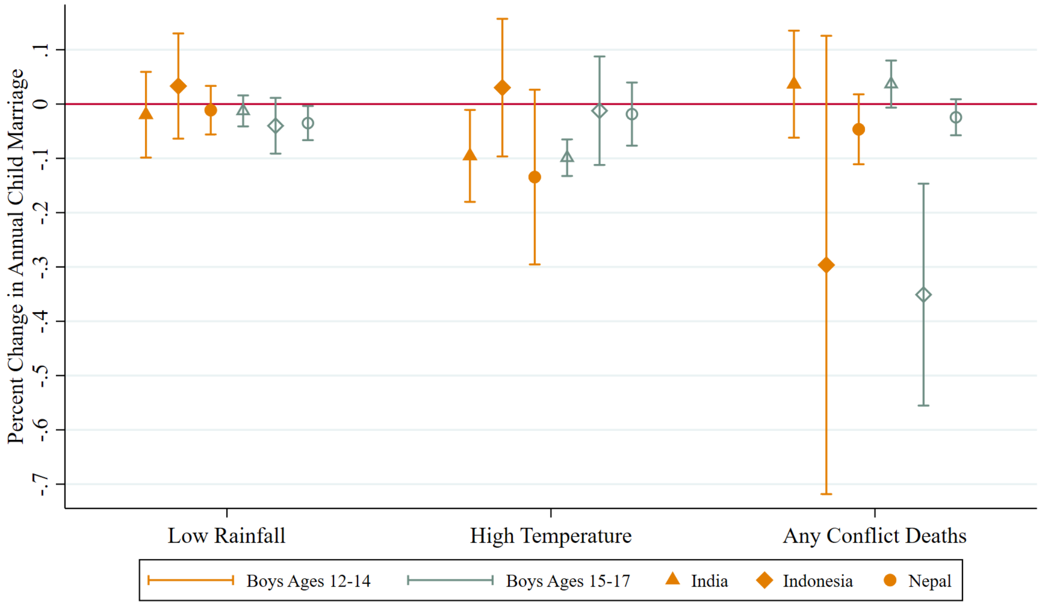
Panel B. Effects as Percentage Points



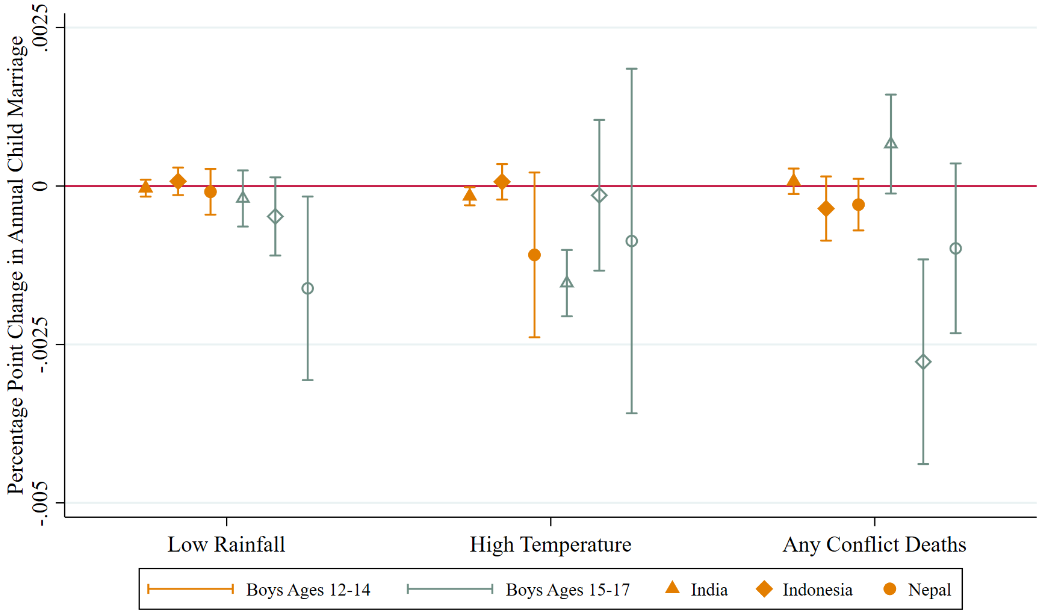
Notes: This figure shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Effects are estimated separately for individuals ages 12-14 and individuals ages 15-17. Panel A shows the effects relative to the mean of the outcome, and Panel B shows the raw estimates as percentage point effects. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level, and 90% confidence intervals are used.

Figure 6. Main Results: Impacts of Shocks on Child Marriage by Age, Boys

Panel A. Effects Relative to the Mean



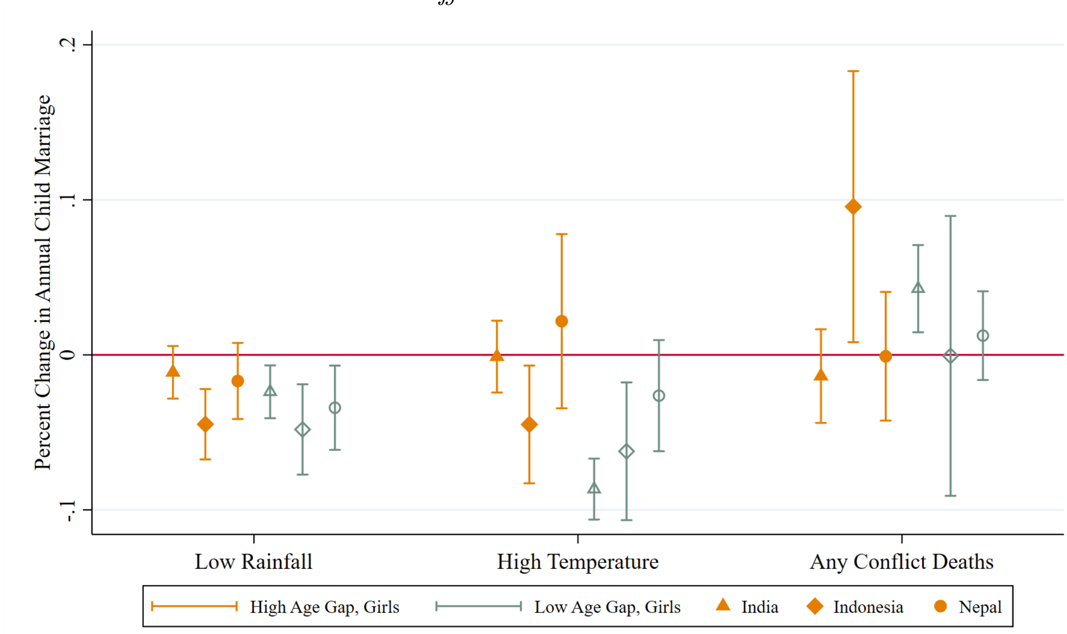
Panel B. Effects as Percentage Points



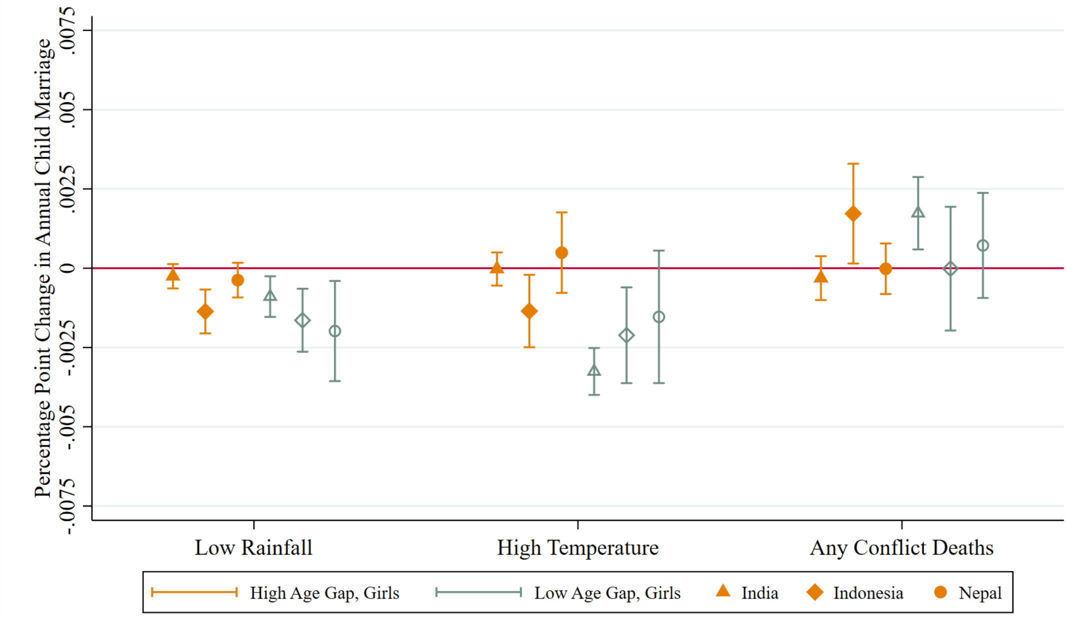
Notes: This figure shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Effects are estimated separately for individuals ages 12-14 and individuals ages 15-17. Panel A shows the effects relative to the mean of the outcome, and Panel B shows the raw estimates as percentage point effects. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data comes from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level, and 90% confidence intervals are used.

Figure 7. Main Results: Impacts of Shocks on Child Marriage by Age Gap, Girls

Panel A. Effects Relative to the Mean



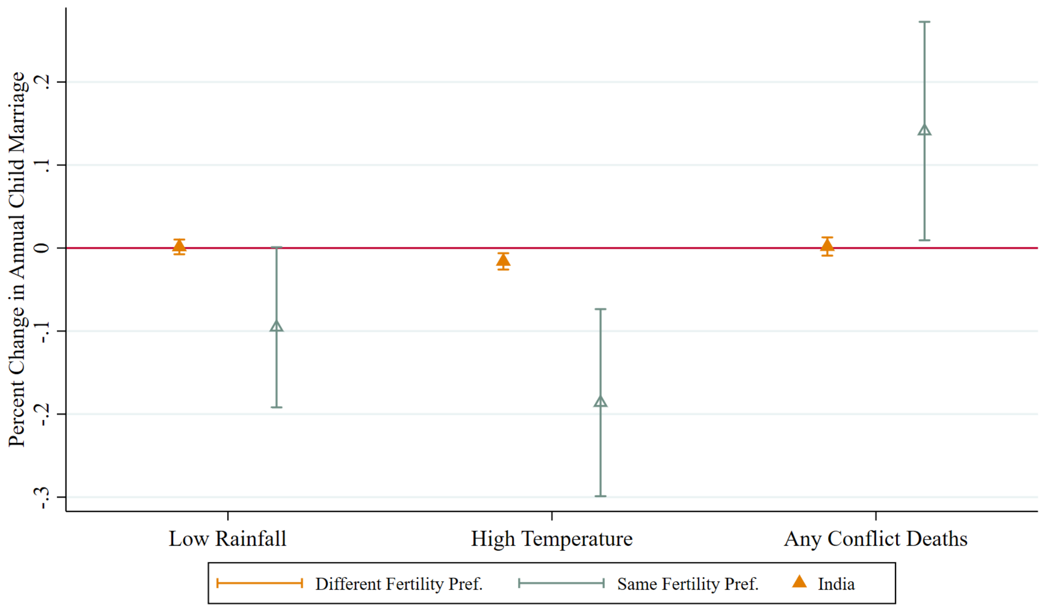
Panel B. Effects as Percentage Points



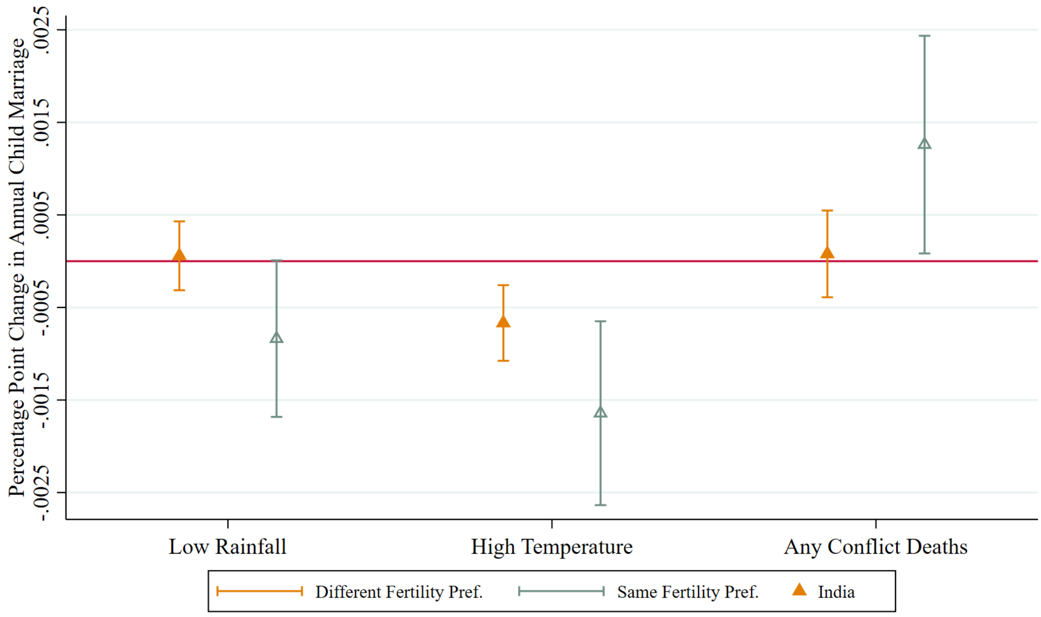
Notes: This figure shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Effects are estimated separately for girls entering matches with a high spousal age gap (more than 5 years) and girls entering matches with a low spousal age gap (5 years or less). Panel A shows the effects relative to the mean of the outcome, and Panel B shows the raw estimates as percentage point effects. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level, and 90% confidence intervals are used.

Figure 8. Main Results: Impacts on Child Marriage by Fertility Preferences, Girls

Panel A. Effects Relative to the Mean



Panel B. Effects as Percentage Points



Notes: This figure shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Effects are estimated separately for girls who have the same fertility preferences as their husbands and girls who have different fertility preferences from their husbands. Panel A shows the effects relative to the mean of the outcome, and Panel B shows the raw estimates as percentage point effects. Data for India come from the DHS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level, and 90% confidence intervals are used.

Table 1. Individual-Level Sample and Marriage Types

	Girls		Boys	
	Mean	*Std. Dev.*	*Mean*	*Std. Dev.*
PANEL A: Full Individual Sample	(1)	(2)	(3)	(4)
Age	30.02	9.3	30.84	10.27
Ever married	0.83	0.38	0.64	0.48
Age at first marriage	18.31	3.56	22.11	4.63
Hindu	0.54	0.5	0.46	0.50
Muslim	0.26	0.44	0.23	0.42
Christian	0.05	0.22	0.04	0.20
Sample from India	0.34	0.47	0.34	0.47
Sample from Indonesia	0.31	0.46	0.31	0.46
Sample from Nepal	0.35	0.48	0.35	0.48
# of low rainfall years ages 12-17	0.75	0.95	0.74	0.94
# of high temperature years ages 12-17	0.43	0.76	0.43	0.77
# of years with any conflict deaths ages 12-17	1.44	1.99	1.39	1.95
Observations	3,633,879		3,146,272	
PANEL B: Matched Individual Sample				
Age gap (husband age - wife age)	4.88	4.44	4.10	3.70
Observations	2,810,383		1,869,344	
PANEL C: Match Types from Individual Sample	**Fraction of Matches**			
Wife < 18, Husband < 18	0.08			
Wife < 18, Husband ≥ 18	0.38			
Wife ≥ 18, Husband < 18	0.01			
Wife ≥ 18, Husband ≥ 18	0.53			

Notes: This table shows descriptives for the full individual-level sample in Panel A, the matched individual-level sample in Panel B, and the types of matches in Panel C. Individual-level data come from the DHS for India and IPUMS International from Indonesia and Nepal. Shock data come from the University of Delaware and UCDP.

Table 2. Demographic Predictors of Child Marriage

	Girls	Boys
PANEL A: Cumulative Marriage Probability by Age	(1)	(2)
Fraction Married By Age 12	0.02	0.00
Fraction Married By Age 13	0.04	0.01
Fraction Married By Age 14	0.09	0.01
Fraction Married By Age 15	0.17	0.03
Fraction Married By Age 16	0.27	0.05
Fraction Married By Age 17	0.38	0.08
Observations	3,633,879	3,146,272
PANEL B: Fraction Married Before 18		
Overall	0.38	0.08
India	0.34	0.05
Indonesia	0.36	0.04
Nepal	0.43	0.15
Hindu	0.41	0.12
Muslim	0.42	0.06
Christian	0.19	0.04
Observations	3,633,879	3,146,272
PANEL C: Marriage Age Gap by Characteristic		
Overall	5.46	1.78
Married Age 12	6.37	0.11
Married Age 13	6.08	0.17
Married Age 14	5.80	1.08
Married Age 15	5.71	1.76
Married Age 16	5.22	1.93
Married Age 17	5.05	2.02
India	5.31	1.54
Indonesia	6.39	0.43
Nepal	4.70	2.31
Hindu	4.92	2.23
Muslim	6.31	1.24
Christian	5.85	0.17
Observations	1,025,717	216,406

Notes: This table shows child marriage rates for girls and boys separately by country, religion, and age. Data for India come from the DHS, and data for Nepal and Indonesia come from IPUMS International. Panel A shows the cumulative probability of marriage by the age of the child, Panel B shows the fraction of individuals married before age 18 by characteristics of the individual, and Panel C shows the age gap between the husband and wife (calculated as husband's age minus wife's age) conditional on the individual being married by age 18.

Table 3. Impact of Shocks on Annual Hazard into Child Marriage

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.0001	-0.0007*	-0.0004*	-0.003***	-0.001***	-0.003***
	(0.0001)	(0.0004)	(0.0002)	(0.0009)	(0.0004)	(0.001)
%Δ (Coefficient / Mean)	-0.014	-0.011	-0.061	-0.049	-0.054	-0.034
Mean of outcome	0.008	0.064	0.007	0.067	0.027	0.082
Observations	5,730,723	6,094,527	5,727,148	6,177,864	6,414,831	6,686,231
PANEL B: Temperature						
High Temperature Year	-0.0007***	-0.003***	-0.00003	-0.004***	-0.002**	-0.004*
	(0.0002)	(0.0006)	(0.0004)	(0.001)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	-0.086	-0.045	-0.004	-0.057	-0.069	-0.046
Mean of outcome	0.008	0.065	0.007	0.067	0.027	0.082
Observations	5,730,723	6,135,400	5,761,652	6,213,561	6,414,831	6,686,231
PANEL C: Conflict						
Any Conflict Deaths	0.0004	0.002**	-0.002***	0.002	0.0003	0.0008
	(0.0003)	(0.0008)	(0.0005)	(0.002)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	0.038	0.026	-0.300	0.060	0.011	0.010
Mean of outcome	0.010	0.066	0.005	0.041	0.025	0.080
Observations	4,137,446	4,708,762	853,513	878,327	3,440,473	3,897,802

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table 4. Annual Hazard into Child Marriage, By Age

	India		Indonesia		Nepal	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall, age 12-14	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.00003	0.00002	0.000007	-0.001*	-0.00009	-0.0009
	(0.00008)	(0.0004)	(0.0001)	(0.0006)	(0.0002)	(0.0007)
%Δ (Coefficient / Mean)	-0.020	0.001	0.003	-0.039	-0.011	-0.027
Mean of outcome	0.00169	0.032	0.00224	0.0264	0.00807	0.0343
Observations	3,079,265	3,505,113	2,881,877	3,336,492	3,297,126	3,692,924
PANEL B: Temperature, age 12-14						
High Temperature Year	-0.0002*	-0.002***	0.00007	-0.0006	-0.001	-0.001
	(0.00009)	(0.0005)	(0.0002)	(0.0009)	(0.0008)	(0.001)
%Δ (Coefficient / Mean)	-0.095	-0.047	0.030	-0.024	-0.135	-0.035
Mean of outcome	0.00169	0.0326	0.00221	0.0256	0.00808	0.0344
Observations	3,079,265	3,527,305	2,899,257	3,355,090	3,297,126	3,692,924
PANEL C: Conflict, age 12-14						
Any Conflict Deaths	0.00007	0.0002	-0.0004	-0.0001	-0.0003	-0.00009
	(0.0001)	(0.0008)	(0.0003)	(0.001)	(0.0002)	(0.0006)
%Δ (Coefficient / Mean)	0.037	0.007	-0.297	-0.011	-0.047	-0.003
Mean of outcome	0.00204	0.0304	0.0012	0.011	0.00632	0.029
Observations	2,164,089	2,611,646	362,385	385,555	1,648,794	2,004,026
PANEL D: Rainfall, age 15-17						
Low Rainfall Year	-0.0002	-0.002***	-0.0005	-0.003***	-0.002*	-0.002
	(0.0003)	(0.0007)	(0.0004)	(0.001)	(0.0009)	(0.001)
%Δ (Coefficient / Mean)	-0.013	-0.017	-0.040	-0.029	-0.035	-0.014
Mean of outcome	0.0155	0.108	0.012	0.117	0.0461	0.141
Observations	2,651,458	2,589,414	2,845,271	2,841,372	3,117,705	2,993,307
PANEL E: Temperature, age 15-17						
High Temperature Year	-0.002***	-0.006***	-0.0001	-0.007**	-0.0009	-0.002
	(0.0003)	(0.0009)	(0.0007)	(0.003)	(0.002)	(0.003)
%Δ (Coefficient / Mean)	-0.099	-0.053	-0.012	-0.061	-0.019	-0.011
Mean of outcome	0.0155	0.109	0.0119	0.115	0.0468	0.142
Observations	2,651,458	2,608,095	2,862,395	2,858,471	3,117,705	2,993,307
PANEL F: Conflict, age 15-17						
Any Conflict Deaths	0.0007	0.003***	-0.003***	0.004*	-0.002	-0.0002
	(0.0005)	(0.001)	(0.001)	(0.002)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	0.037	0.030	-0.351	0.063	-0.024	-0.001
Mean of outcome	0.018	0.11	0.0079	0.0646	0.0404	0.132
Observations	1,973,357	2,097,116	491,128	492,772	1,791,679	1,893,776

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level for individuals ages 12-14 in Panels A-C and ages 15-17 in Panels D-F. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table 5. Impact of Shocks on Annual Hazard into Child Marriage by Age Gap, Girls

	India		Indonesia		Nepal	
	Low Age Gap	***High Age Gap***	***Low Age Gap***	***High Age Gap***	***Low Age Gap***	***High Age Gap***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.0009**	-0.0003	-0.002***	-0.001***	-0.002**	-0.0004
	(0.0004)	(0.0002)	(0.0006)	(0.0004)	(0.0009)	(0.0003)
%Δ (Coefficient / Mean)	-0.024	-0.013	-0.048	-0.045	-0.034	-0.018
Mean of outcome	0.038	0.023	0.034	0.031	0.058	0.022
Observations	4,053,999	4,053,999	5,661,395	5,661,395	4,993,530	4,993,530
PANEL B: Temperature						
High Temperature Year	-0.003***	-0.00003	-0.002**	-0.001*	-0.002	0.0005
	(0.0005)	(0.0003)	(0.0009)	(0.0007)	(0.001)	(0.0008)
%Δ (Coefficient / Mean)	-0.087	-0.001	-0.062	-0.045	-0.034	0.022
Mean of outcome	0.038	0.023	0.034	0.030	0.058	0.023
Observations	4,053,999	4,073,805	5,694,681	5,694,681	4,993,530	4,993,530
PANEL C: Conflict						
Any Conflict Deaths	0.002**	-0.0003	-0.00001	0.002*	0.0007	-0.00002
	(0.0007)	(0.0004)	(0.001)	(0.001)	(0.001)	(0.0005)
%Δ (Coefficient / Mean)	0.043	-0.013	0.000	0.096	0.012	-0.001
Mean of outcome	0.041	0.023	0.022	0.018	0.058	0.019
Observations	3,299,603	3,299,603	846,896	846,896	2,905,358	2,905,358

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage with a particular age gap within that same year at the individual-year level. The age gap is defined as husband's age minus wife's age, and a high age gap is defined as an age gap of more than five years, while a low age gap is defined as an age gap of at most five years. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table 6. Impact of Shocks on Bride Price Amounts, Indonesia

	Bride Price Amount	**Bride Price Amount (Winsorized 1%)**	**Bride Price Amount (Winsorized 5%)**
PANEL A: Rainfall	(1)	(2)	(3)
Low Rainfall Year	-237,773*	-64,836	-19,377
	(137,810)	(65,339)	(31,450)
Mean of outcome	1,190,000	234,829	167,455
Observations	3,635	3,635	3,635
PANEL B: Temperature			
High Temperature Year	-510,563	135,247	85,326
	(502,090)	(165,884)	(70,898)
Mean of outcome	1,152,000	257,026	186,401
Observations	3,635	3,635	3,635
PANEL C: Conflict			
Any Conflict Deaths	-365,931*	-289,043*	-150,192
	(188,222)	(163,497)	(152,334)
Mean of outcome	1,708,000	552,069	377,557
Observations	1,321	1,321	1,321

Notes: This table shows the impact of low rainfall, high temperatures, or conflict violence on bride price amounts for girls ages 12-17 in Indonesia. Individual-level data come from the IFLS, and the amounts in column (1) include the raw amount in the data, the amounts in column (2) are topcoded at the 99th percentile, and the amounts in column (3) are topcoded at the 95th percentile. Shock data come from the University of Delaware and UCDP. Regressions include fixed effects for the district and year. Standard errors are clustered at the district level. $p<0.1^*$, $p<0.05^{**}$, $p<0.01^{***}$.

Table 7. IV Impacts of Crop Yields on District-Level Child Marriage, India

Instrument:	**Low rainfall + High temperature**		**Any conflict deaths**	
	Boys	***Girls***	***Boys***	***Girls***
	(1)	(2)	(3)	(4)
PANEL A: IV Estimates, Outcome = child marriage at the district level (% married)				
Crop Yield Index	-0.002	0.032*	0.013	0.004
	(0.002)	(0.019)	(0.012)	(0.035)
First-Stage F	99.70	97.31	8.366	8.366
Observations	13,146	13,085	6,328	6,328
PANEL B: First Stage, outcome = crop yield index, independent variable = shock				
Low rainfall	-0.077***	-0.077***		
	(0.007)	(0.007)		
High temperature	-0.017**	-0.017*		
	(0.009)	(0.009)		
Any conflict deaths			0.030**	0.030**
			(0.014)	(0.014)
Observations	13,146	13,085	6,328	6,328

Notes: This table shows the IV estimates of the impacts of change in crop yields of child marriage within that same year at the district-year level. The IV estimates are shown in Panel A, and the first stage is shown in Panel B. Low rainfall and high temperature shocks are used as the instruments in columns (1) and (2), and conflict shocks are used as the instrument in columns (3) and (4). Crop yields are calculated using a Kling et al. (2007) index across all available crop types. Data come from the DHS, ICRISAT, the University of Delaware, and UCDP, and the regressions are for India only. All variables are aggregated at the district level. Regressions include state and year fixed effects. Robust standard errors are used. p<0.1*, p<0.05**, p<0.01***.

Appendix A Model Derivations

A.1 Derivation of the Upper Bound of the Marriage Transfer

The payoff to a household of marrying a daughter is:

$$log(y_{g,i}(r) - \tau_i(r)) + \gamma_{g,i}$$

The payoff of deferring marriage is:

$$log(y_{g,i}(r))$$

The threshold for the child marriage decision is:

$$u_{g,i}(c_{g,i}(r, m_{g,i}), m_{g,i} = 1) > u_{g,i}(c_{g,i}(r), m_{g,i} = 0)$$

This provides the upper bound of $\tau_i(r)$ that households with girls are willing to pay:

$$log(y_{g,i}(r) - \tau_i(r)) + \gamma_{g,i} \geq log(y_{g,i}(r))$$

$$log(y_{g,i}(r) - \tau_i(r)) \geq log(y_{g,i}(r)) - \gamma_{g,i}$$
$$e^{log(y_{g,i}(r) - \tau_i(r))} \geq e^{log(y_{g,i}(r)) - \gamma_{g,i}}$$

$$y_{g,i}(r) - \tau_i(r) \geq y_{g,i}(r)e^{-\gamma_{g,i}}$$

$$\tau_i(r) \leq y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}) \equiv \tau_i^{ub}(r)$$

In locations where bride price is practiced, the transfer amount must be weakly negative. Thus, the upper bound of the transfer is given by:

$$\tau_i^{ub}(r) = min\{y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), 0\}$$

A.2 Derivation of the Lower Bound of the Marriage Transfer

The payoff to a household of marrying a son is:

$$log(y_{b,i}(r) + \tau_i(r)) + \gamma_{b,i}(r)$$

The payoff of deferring marriage is:

$$log(y_{b,i}(r))$$

The threshold can be written as follows:

$$u_{b,i}(c_{b,i}(r, m_{b,i}), m_{b,i} = 1) > u_{b,i}(c_{b,i}(r), m_{b,i} = 0)$$

$$log(y_{b,i}(r) + \tau_i(r)) + \gamma_{b,i} \geq log(y_{b,i}(r))$$

$$log(y_{b,i}(r) + \tau_i(r)) \geq log(y_{b,i}(r)) - \gamma_{b,i}(r)$$

$$e^{log(y_{b,i}(r)+\tau_i(r))} \geq e^{log(y_{b,i}(r))-\gamma_{b,i}(r)}$$

$$y_{b,i}(r) + \tau_i(r) \geq y_{b,i}(r)e^{-\gamma_{b,i}}$$

$$\tau_i(r) \geq y_{b,i}(r)(e^{-\gamma_{b,i}} - 1) \equiv \tau_i^{lb}(r)$$

In cases where dowry is practiced and the transfer amount must be weakly positive:

$$\tau_i^{lb}(r) = max\{y_{b,i}(r)(e^{-\gamma_{b,i}} - 1), 0\}$$

A.3 Comparative Statics: Changes in Preferences

This section explains the impacts on the number of matches at the marriage market level and the transfer amounts paid when preferences of households with daughters change, when preferences of households with sons change, and when income changes due to a negative shock to the state of nature.

A.3.1 Changes in Preferences: Households with Daughters

If preferences for female child marriage increase:

1. **For households where previously,** $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$: The upper bound of the range will increase, and the lower bound of the range will not change. Thus, the amount of the transfer will increase, but the number of matches taking place for these pairs of households will not change. If bride price is the prevalent practice and $\tau_i^{ub}(r)$ is close to zero, then the adjustment will not be as dramatic.

2. **For households where previously,** $\tau_i^{ub}(r) < \tau_i^{lb}(r)$: If $\tau_i^{ub}(r)$ increases by enough, the households may be willing to agree on a transfer amount, and the marriage will happen. Mathematically, this will happen if $\Delta\tau_i^{ub}(r) - \Delta\tau_i^{lb}(r) > \tau_i^{lb}(r) - \tau_i^{ub}(r)$, which will be true if the increase in preferences for households with daughters is large enough. Thus, this will increase the quantity of matches among these types of households, as long as preferences shift by enough. If bride price is the prevalent practice and $\tau_i^{ub}(r)$ is close to zero, then the adjustment will not be as dramatic.

For decreases in preferences for child marriage, the opposite will be true, and the number of matches in the market will decrease, as long as the decrease in preferences is large enough.

1. **For households where previously,** $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$: The upper bound of the range will decrease, and the lower bound of the range will not change. Thus, the amount of the transfer will decrease and/or the number of matches taking place will decrease (if the decrease in preferences is such that $\Delta\tau_i^{lb}(r) - \Delta\tau_i^{ub}(r) > \tau_i^{ub}(r) - \tau_i^{lb}(r)$ for at least some households). If bride price is the prevalent practice and $\tau_i^{ub}(r)$ is close to zero, then the adjustment will not be as dramatic.

2. **For households where previously,** $\tau_i^{ub}(r) < \tau_i^{lb}(r)$. The upper bound of the range will decrease, the lower bound will stay the same, and there will be no changes.

A.3.2 Changes in Preferences: Households with Sons

If preferences for male child marriage increase:

1. **For households where previously,** $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$: The lower bound of the range will decrease, and the upper bound of the range will not change. Thus, the amount of the transfer will decrease, but the number of matches taking place for these pairs of households will not change. If $\tau_i^{lb}(r)$ is close to zero in the case of dowry, then the adjustment will not be as dramatic.

2. **For households where previously,** $\tau_i^{ub}(r) < \tau_i^{lb}(r)$: If τ_{lb} decreases by enough, the households may be willing to agree on a transfer amount, and the marriage will happen. Mathematically, this will happen if $\Delta\tau_i^{ub}(r) - \Delta\tau_i^{lb}(r) > \tau_i^{lb}(r) - \tau_i^{ub}(r)$, which will be true if the increase in preferences for households with sons is large enough. Thus, this will increase the quantity of matches among these types of households, as long as preferences shift by enough. If $\tau_i^{lb}(r)$ is close to zero, then the adjustment will not be as dramatic.

If preferences for male child marriage decrease:

1. **For households where previously,** $\tau_i^{ub}(r) \geq \tau_i^{lb}(r)$: The lower bound of the range will increase, and the upper bound of the range will not change. Thus, the amount of the transfer will increase and/or the number of matches taking place will decrease (if the decrease in preferences is such that $\Delta\tau_i^{lb}(r) - \Delta\tau_i^{ub}(r) > \tau_i^{ub}(r) - \tau_i^{lb}(r)$ for at least some households).

2. **For households where previously,** $\tau_i^{ub}(r) < \tau_i^{lb}(r)$. The lower bound of the range will increase, the upper bound will stay the same, and there will be no changes.

A.4 Comparative Statics: Changes in Income

Households in locations where dowry or bride price are practiced will respond to changes in income differently, as shown above. This section shows how a reduction in income impacts households in each situation (where an increase in income would have the opposite effects).

A.4.1 Dowry

Starting with a decrease in income in the case of dowry, where, as established above:

$$\tau_i^{ub}(r) = y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}})$$

$$\tau_i^{lb}(r) = max\{y_{b,i}(r)(e^{-\gamma_{b,i}} - 1), 0\}$$

Suppose that income falls. $\frac{\partial \tau_i^{lb}(r)}{\partial y_{b,i}} \geq 0$ and $\frac{\partial \tau_i^{ub}(r)}{\partial y_{b,i}} > 0$. So, if income falls, both $\tau_i^{lb}(r)$ and $\tau_i^{ub}(r)$ will fall. Imagine a match that would have happened, where $\tau_i^{ub}(r) > \tau_i^{lb}(r)$. This match will no longer happen if $\tau_i^{ub}(r)$ falls enough relative to $\tau_i^{lb}(r)$ that it is no longer true that $\tau_i^{ub}(r) > \tau_i^{lb}(r)$. In other words, the number of marriage matches will decrease if the upper threshold falls enough relative to the lower threshold and closes the gap between the upper and lower thresholds. Marriages will be deferred in response to lower income if, on average,

$$\Delta\tau_i^{lb}(r) - \Delta\tau_i^{ub}(r) > \tau_i^{ub}(r) - \tau_i^{lb}(r)$$

Since $\tau_i^{lb}(r)$ cannot fall below zero and $\Delta\tau_i^{lb}(r) < 0$, it must be true that $\Delta\tau_i^{lb}(r) = max\{\Delta y_{b,i}(r)(e^{-\gamma_{b,i}} - 1), -y_{b,i}(r)(e^{-\gamma_{b,i}} - 1)\}$. Thus, we can re-write the expression as follows:

$$max\{\Delta y_{b,i}(r)(e^{-\gamma_{b,i}} - 1), -y_{b,i}(r)(e^{-\gamma_{b,i}} - 1)\} - \Delta y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}) >$$

$$y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}) - max\{y_{b,i}(r)(e^{-\gamma_{b,i}} - 1), 0\}$$

Assume that households are not near the threshold of zero, then this will imply

$$\Delta y_{b,i}(r) \cdot (e^{-\gamma_{b,i}} - 1) - \Delta y_{g,i}(r) \cdot (1 - e^{-\gamma_{b,i}}) > y_{g,i}(r) \cdot (1 - e^{-\gamma_{b,i}}) - y_{b,i}(r)(e^{-\gamma_{b,i}} - 1)$$

$$(y_{b,i}(r) + \Delta y_{b,i}(r)) \cdot (e^{-\gamma_{b,i}} - 1) > (y_{g,i}(r) + \Delta y_{g,i}(r)) \cdot (1 - e^{-\gamma_{b,i}})$$

As specified above, assume that the distribution of income and the shocks to income are the same for girls and boys. Additionally, assume that income and preferences are i.i.d. Thus, in expectation,

$$E\{(y_{b,i}(r)+\Delta y_{b,i}(r))\cdot(e^{-\gamma_{b,i}}-1)\} > E\{(y_{g,i}(r)+\Delta y_{g,i}(r))\cdot(1-e^{-\gamma_{g,i,}})\}$$

$$E\{(y_{b,i}(r)+\Delta y_{b,i}(r))\}E\{(e^{-\gamma_{b,i}}-1)\} > E\{(y_{g,i}(r)+\Delta y_{g,i}(r))\}E\{(1-e^{-\gamma_{g,i}})\}$$

$$E\{(e^{-\gamma_{b,i}}-1)\} > E\{(1-e^{-\gamma_{g,i}})\}$$

$$E\{(e^{-\gamma_{b,i}}+e^{-\gamma_{g,i}})\} > 2$$

This will be true if preferences for child marriage, for either girls or boys, are small enough (the distribution is far enough to the left). If this is the case, the number of matches should decrease in expectation.[70]

On the other hand, assume that the transfer amounts are near zero. Then

$$max\{\Delta y_{b,i}(e^{-\gamma_{b,i}}-1), -y_{b,i}(r)(e^{-\gamma_{b,i}}-1)\} - \Delta y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}}) >$$

$$y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}}) - max\{y_{b,i}(r)(e^{-\gamma_{b,i}}-1), 0\}$$

$$max\{\Delta y_{b,i}(r)(e^{-\gamma_{b,i}}-1)-\Delta y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}}), -y_{b,i}(r)(e^{-\gamma_{b,i}}-1)-\Delta y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}})\} >$$

$$min\{y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}}) - y_{b,i}(r)(e^{-\gamma_{b,i}}-1), y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}})\}$$

As shown above, given that preferences for child marriage are small enough:

$$E\{\Delta y_{b,i}(r)\cdot(e^{-\gamma_{b,i}}-1)-\Delta y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}})\} > E\{y_{g,i}(r)\cdot(1-e^{-\gamma_{b,i}})-y_{b,i}(r)(e^{-\gamma_{b,i}}-1)\}$$

Thus, with the same preferences, it must be true that

$$E\{max\{\Delta y_{b,i}(r)(e^{-\gamma_{b,i}}-1)-\Delta y_{g,i}(r)\cdot(1-e^{-\gamma_{g,i}}), -y_{b,i}(r)(e^{-\gamma_{b,i}}-1)-\Delta y_{g,i}(r)\cdot(1-e^{-\gamma_{g,i}})\}\} >$$

$$E\{min\{y_{g,i}(r)\cdot(1-e^{-\gamma_{g,i}}) - y_{b,i}(r)(e^{-\gamma_{b,i}}-1), y_{g,i}(r)\cdot(1-e^{-\gamma_{g,i}})\}\}$$

since we are subtracting a weakly larger number from the RHS and adding a weakly larger number to the LHS relative to the previous inequality.

A.4.2 Bride Price

For the case of bride price, the effects are similar:

$$\tau_i^{ub}(r) = min\{y_{g,i}(r)\cdot(1-e^{-\gamma_{g,i}}), 0\}$$

$$\tau_i^{lb}(r) = y_{b,i}(r)\cdot(e^{-\gamma_{b,i}}-1)$$

[70] $\gamma_{b,i} < 0$ and $\gamma_{g,i} > 0$, since dowry is practiced and the transfer amounts are far from zero.

Suppose that income falls. As shown above, $\frac{\partial \tau_i^{lb}(r)}{\partial y_{b,i}} < 0$ and $\frac{\partial \tau_i^{ub}(r)}{\partial y_{b,i}} \leq 0$. So, if income falls, both $\tau_i^{lb}(r)$ and $\tau_i^{ub}(r)$ will increase. Imagine a match that would have happened, where $\tau_i^{ub}(r) > \tau_i^{lb}(r)$. This match will no longer happen if $\tau_i^{lb}(r)$ increases enough relative to $\tau_i^{ub}(r)$ that it is no longer true that $\tau_i^{ub}(r) > \tau_i^{lb}(r)$. In other words, the number of marriage matches will decrease if the lower threshold rises enough relative to the upper threshold and closes the gap between the upper and lower thresholds. Marriages will be deferred in response to lower income if, on average,

$$\Delta\tau_i^{lb}(r) - \Delta\tau_i^{ub}(r) > \tau_i^{ub}(r) - \tau_i^{lb}(r)$$

Since $\tau_i^{lb}(r)$ cannot fall below zero and $\Delta\tau_i^{lb}(r) < 0$, it must be true that $\Delta\tau_i^{ub}(r) = min\{\Delta y_{g,i}(r)(1 - e^{-\gamma_{g,i}}), y_{g,i}(r)(1 - e^{-\gamma_{g,i}})\}$. Thus, we can re-write the expression as follows:

$$\Delta y_{b,i}(r)(e^{-\gamma_{b,i}} - 1) - min\{\Delta y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), y_{g,i}(r) \cdot (1 - e^{-\gamma_{b,i}})\} >$$

$$min\{y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), 0\} - y_{b,i}(r)(e^{-\gamma_{b,i}} - 1)$$

Assume that the households are not near the threshold of zero, then this will imply

$$\Delta y_{b,i}(r) \cdot (e^{-\gamma_{b,i}} - 1) - \Delta y_{g,i}(r) \cdot (1 - e^{-\gamma_{b,i}}) > y_{g,i}(r) \cdot (1 - e^{-\gamma_{b,i}}) - y_{b,i}(r)(e^{-\gamma_{b,i}} - 1)$$

$$(y_{b,i}(r) + \Delta y_{b,i}(r)) \cdot (e^{-\gamma_{b,i}} - 1) > (y_{g,i}(r) + \Delta y_{g,i}(r)) \cdot (1 - e^{-\gamma_{b,i}})$$

$$E\{(y_{b,i}(r) + \Delta y_{b,i}(r)) \cdot (e^{-\gamma_{b,i}} - 1)\} > E\{(y_{g,i}(r) + \Delta y_{g,i}(r)) \cdot (1 - e^{-\gamma_{b,i}})\}$$

$$E\{(y_{b,i}(r) + \Delta y_{b,i}(r))\}E\{e^{-\gamma_{b,i}} - 1)\} > E\{(y_{g,i}(r) + \Delta y_{g,i}(r))\}E1 - e^{-\gamma_{b,i}})$$

$$E\{(e^{-\gamma_{b,i}} - 1)\} > E\{(1 - e^{-\gamma_{b,i}})\}$$

As above, this expression will hold if preferences for child marriage for girls and/or boys are small enough (i.e. the distributions are far enough to the left). In this case, $\gamma_{b,i} > 0$ and $\gamma_{g,i} < 0$, since bride price is practiced and households are far from a transfer value of zero. Thus, as long as preferences for child marriage are small enough, it will be the case that a decrease in income leads to a decrease in the quantity of marriage matches taking place in the market.

On the other hand, assume that the transfer amounts are near zero. Then we get that

$$\Delta y_{b,i}(r)(e^{-\gamma_{b,i}} - 1) - min\{\Delta y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}})\} >$$

$$min\{y_{g,i}(r) \cdot (1 - e^{-\gamma_{g,i}}), 0\} - y_{b,i}(r)(e^{-\gamma_{b,i}} - 1)$$

Since we subtract a smaller number from the LHS and add a smaller number to the RHS relative to the first case given above, this will hold under the same conditions as above.

Thus, the model predicts that the number of matches should decrease in the presence of a negative shock to income, regardless of the direction of the marriage transfer, as long as preferences for child marriage are not too strong. This is true for both the cases of bride price and dowry. The impact on prices is ambiguous and will depend on which matches are deferred, since the prediction is that the bounds of the prices will approach zero (which implies a decrease in prices) but a deferral of some matches (particularly those closest to zero, with the lowest prices).

Appendix B Data Appendix

This section describes in detail the sources of the data and variables that are used in the analysis.

B.1 Shock Data

Rainfall data comes from the University of Delaware, and I use v5.01, which is the most recent version of the rainfall data available at the time of writing. This provides the total monthly precipitation in gridded 0.5x0.5 cells, and precipitation is given in units of cm. The data can be found here: https://downloads.psl.noaa.gov/Datasets/udel.airt.precip/, and the name of the file is precip.mon.total.v501.nc.

Temperature data also comes from the University of Delaware. Again, the monthly temperature data is given in gridded 0.5x0.5 cells. The data can be found here: https://downloads.psl.noaa.gov/Datasets/udel.airt.precip/, and the name of the file is air.mon.mean.v501.nc.

Conflict data comes from the Uppsala Conflict Data program. The specific dataset used is the "UCDP Georeferenced Event Dataset (GED) Global Version 23.1," which has data available for many countries around the world and can be found at https://ucdp.uu.se/downloads/ged/ged231.pdf. I keep only the events that were listed as occurring in India, Indonesia, or Nepal. Incidents must meet several conditions to be included in this dataset, which can be found at https://ucdp.uu.se/downloads/. Data is restricted to incidents where it was possible to calculate the number of fatalities. Conflict incidents are included if they are carried out by an actor that has caused at least 25 events in a single calendar year (for any calendar year; if this requirement is met, the actor is included for all years).

For Nepal, I also use conflict data from the Informal Sector Service Center (INSEC) Conflict Victim's Profile, which provides detailed data on all conflict events between 1996 and 2006, with a few additional events outside of that time period. I scraped this data from https://www.insec.org.np/victim/.

B.2 Geographical Data

One challenge of using geographical data in India, Indonesia, and Nepal is that the geographical boundaries of the districts are changing over time. As a result, I aggregate districts to the earliest time period in the analysis.

- India shapefile for 1993 districts: https://international.ipums.org/international/gis_yrspecific_2nd.shtml

- India shapefile for 1966 districts: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/CHEPOL

- Indonesia shapefile for consistent 1971-2010 districts https://international.ipums.org/international/gis_harmonized_2nd.shtml. I use the consistent 1971-2010 regencies, meaning that I have a total of 257 regencies. These consistent boundaries account for the fact that regency boundaries in Indonesia have changed over time. For the rest of the paper, I will refer to these regencies as districts, since the two units are approximately equivalent.

- Nepal shapefile: https://data.humdata.org/dataset/cod-ab-npl?). There are 77 districts in Nepal, but harmonizing the districts across time results in 72 districts.

B.3 Individual-Level Data

For India, I use DHS data from three rounds from https://dhsprogram.com/Countries/Country-Main.cfm?ctry_id=57&c=India: 1998-1999, 2015-2016, and 2019-2021. While the DHS is representative for women between ages 15 and 49, it is not representative for men. The men's sample is based on men living in the households chosen for the women's sample. However, if these men are not representative of all men in India, then this sample is not representative for men. An additional round of the DHS was collected in 2005-2006, but district identifiers are not available for this dataset. Since geographic information is not available at a fine enough level to match shocks to individuals, these data are not included in the analysis.

For Indonesia and Nepal, I use data from IPUMS International https://international.ipums.org/international-action/variables/group. For Nepal, IPUMS provides an 8.93% sample and a 12% sample of data from the National Population Census 2001 and 2011, respectively. For the 2011 sample in Nepal, the age at marriage is topcoded at 55 years of age, which does not impact my analysis, since I do not consider marriage after age 18 in my main results. Data for Indonesia is available for 1976, 1980, 1990, 1995, and 2005. The 1976 data is not fully representative of three provinces: East Nusa Tenggara, Maluku and Irian Jaya. 1980 and 1990 data come from the full Population Census, and the 1995 and 2000 data come from the Intercensal Population Survey. Information for age at first marriage is only available for women, and it is not available for the other years of the Census besides 1980 and 1990.

B.4 Supplemental Data

- IFLS data comes from https://www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS/download.html. I use adult survey respondents in all 5 waves of the IFLS who answered the migration history questions.

- ICRISAT irrigation data http://data.icrisat.org/dld/src/crops.html. The amount of irrigated land and the crop yields are provided in 1000s of hectares at the 1966 district level. To match these data with my shock data and individual-level data, I use the 1966-1971 shapefile from Smith et al. (2019); https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/CHEPOL, accessed 16 March 2024. to create a crosswalk between 1966 and 1993 districts in India. Some of the districts are not covered by ICRISAT (only 312 districts have data available), so the data does not cover the entire country.

- GRDP data comes from https://www.bps.go.id/en.

Appendix C Matching Couples

This section explains the assumptions and restrictions that are made to create the sets of matched couples and the extrapolation of age at marriage for men.

C.1 India

For India I use DHS data, where the sample is nationally representative of women but is not necessarily representative of men, since only the men who lived in the households of sampled women are surveyed. The sample of individuals who responded to the men's survey, which contains the question on age at first cohabitation for men, is much smaller than the women's sample.[71] This means that, if I try to match the individuals who responded to the men's survey and the women's survey, many of the women will not have matches. The full sample of men are available in the household roster, but do not have information on age at first marriage.

To overcome this issue, I use the men in the household roster and the women in the women's survey. From the women's survey, I drop all women who are under age 18; who have their marital status listed as divorced, widowed, or separated; who are listed as married but do not have a link to a spouse within the household; who are listed as married but missing age at marriage or married before age 12; or who were born before 1955. This results in 873,743 women who are married.

For the men, I drop men in the household survey younger than 18 or who have missing age; men whose marital status is widowed, divorced, separated, or missing/unknown; men whose birth year was before 1955; and those who have duplicate IDs within a household (this is relevant only for men in the 1998 household survey and requires dropping 18 observations). This results in 1,245,256 men who are married. I match the men and women and get 705,880 matched couples (a match rate of 80.8% for women and 56.7% for men).

After this, I also drop all of the women with duplicate husband IDs (indicating polygamous households), which is an additional 2,242 women. From the matched couples, I assume that the woman and man in each couple were each other's first marriage, and I use the age at first marriage of the woman to extrapolate the year at first marriage of her husband. This is, of course, subject to error, because some of these individuals in the sample have remarried, either due to divorce or the death of their spouse. I drop the 887 men whose age at first marriage is calculated to be below 12, then I add back the single individuals for a total of 1,164,039 men and 914,004 women, married and single, where all of the married individuals are matched with partners.

Once this is done, I have a sample with all of the matched married men from the

[71] In fact, for the 1998 wave of the DHS, there is no men's recode.

household roster, all of the single men from the household roster, all of the matched married women from the women's survey, and all of the single women from the women's survey. I also have the information necessary to calculate the age gap for all of the married individuals.[72]

C.2 Indonesia

For Indonesia, age at first marriage is only reported for women. However, since this is Census data, I can match men and women and extrapolate age at first marriage for men.

To do this, I drop individuals who are below 18, women who are missing information on age at first marriage, and individuals have a marital status of separated, divorced, widowed, or unknown/missing. I also drop women who have a spouse in the household but age at first marriage equal to zero (indicating no marriage) or who list an age at first marriage but do not have a spouse in the household. Finally, I drop individuals whose marital status is listed as single but who have a link to a spouse in the household.

Individuals who are not married obviously do not need to be matched, so they are included in the sample as is. From the women's side, I drop observations where multiple women in the same household and year are matching to the same man, effectively removing cases of polygamy (only 50 observations in my data). This results in 1,884,999 women and 2,011,756 men who should theoretically match (though not all of these will match, because, for example, some of them have spouses that are under 18). Women who are married are matched to men who are married by the year, household number, and spouse location in the household. This results in a match of 1,878,145 couples, a 99.6% match rate for women and 93.4% match rate for men. Finally, I drop cases where the wife's age or husband's age are missing or where the wife was married before age 12 (25, 392, and 32,383 observations, respectively). This leaves a total of 1,845,459 matched couples.

In my analysis, I only use the subset of couples where the individual of interest was born in 1955 or later.[73] Thus, I drop women who were born before 1955 (1,120,382 observations) and add back in single women (298,535 observations). This leaves me with a total of 1,064,719 women in the matched sample, 766,184 of whom are matched with spouses (the remainder are unmarried). I drop men who were married before age 12 (18,944 observations) and who were born before 1955 (1,430,032 observations) and add back in single men (529,244 observations). This leaves me with a total of 992,027 men, 462,783 of whom are matched with spouses. For the men, I use the spousal matches

[72] For regressions that require information on both the husband and wife, I use this matched sample of men and women. For regressions that do not require information on both spouses, I use the matched sample only for men (and use the information directly from the women's recode to run the regressions for women, since they have age at first marriage directly recorded).

[73] While this does significantly reduce my sample, I do this so that the samples across the three countries are more comparable in terms of the time period that is being analyzed. However, I allow the spouse to be born before 1955, so that I do not arbitrarily limit my sample to individuals with smaller age gaps.

generate their age at first marriage, under the assumption that the marriage that they are in is their first and their wife's first. This becomes the main sample for men in Indonesia.[74]

C.3 Nepal

For the IPUMS data for Nepal, I have age at marriage for both men and women. Following the normal sample restrictions as in the main sample, I drop all individuals who are below 18, who are missing information on age at marriage or were married before age 12, or who are married and missing a line number for their spouse. I also drop all who have their marital status listed as polygamous, separated, divorced, widowed, or unknown/missing from the dataset. I also drop all who have a spouse location but no age at marriage or an age at marriage but no spouse location. This leaves 967,576 men and 981,843 women who should be able to match with spouses. I match all these married men and married women, dropping the married individuals who do not match. 920,767 couples match, for a match rate of 95.2% for men and 93.8% for women. I then include the single individuals (255,326 men and 152,576 women) and drop all individuals born before 1955 (204,976 men and 133,694 women). This leaves me with 971,117 men and 939,649 women.

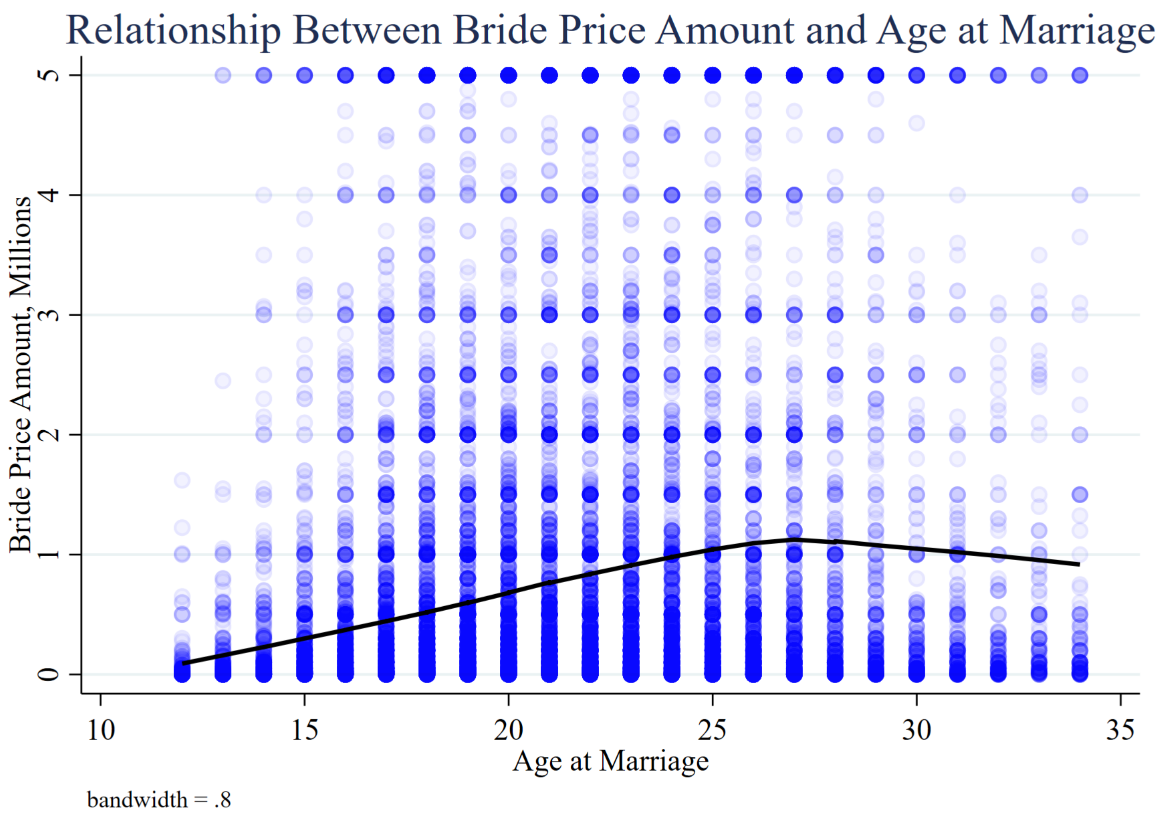
This results in a sample of all married men with matches, all married women with matches, all single men, and all single women. I then use the information on the matched couples to create a variable for the age gap between the husband and wife (husband's age - wife's age).[75]

[74]As is the case in India, I use the matched sample whenever outcomes for both the husband and wife are needed in the regression, such as calculations of the spousal age gap. For the regressions that just include outcomes for boys/men, I use the matched sample. For regressions that just include outcomes for girls/women, I use the raw, unmatched data, since age of marriage is directly reported for the girls/women but not for the men/boys.

[75]When running the regressions that require outcomes for both the husband and wife, I use the matched sample. When running regressions that require outcomes for only boys or girls separately, I use the unmatched data, since age at first marriage is reported for all adults in the sample.

Appendix D Additional Tables and Figures

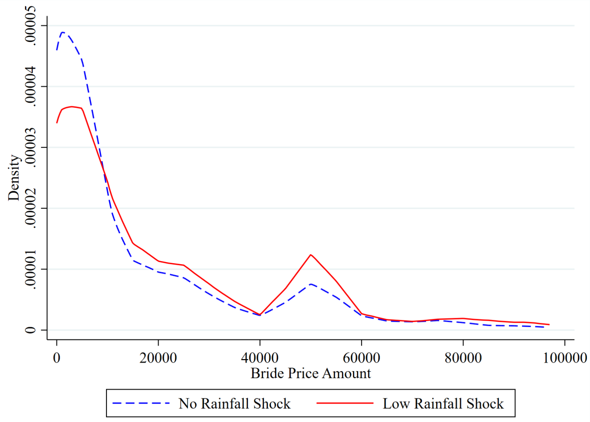
Figure A1. Relationship between Bride Price and Age in Indonesia



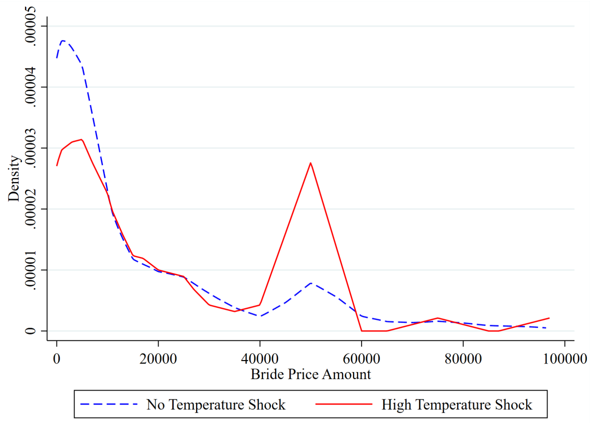
Notes: This figure shows the relationship between bride price amount and age at marriage for women in Indonesia who were married between the ages of 12-34 from 1966-2005. Data come from the IFLS, and units are in Indonesia rupiah (IDR), with bride price amount capped at 5 million IDR to avoid outliers.

Figure A2. Lower Bride Price Distribution by Shocks

Panel A. Bride Price by Rainfall Shock



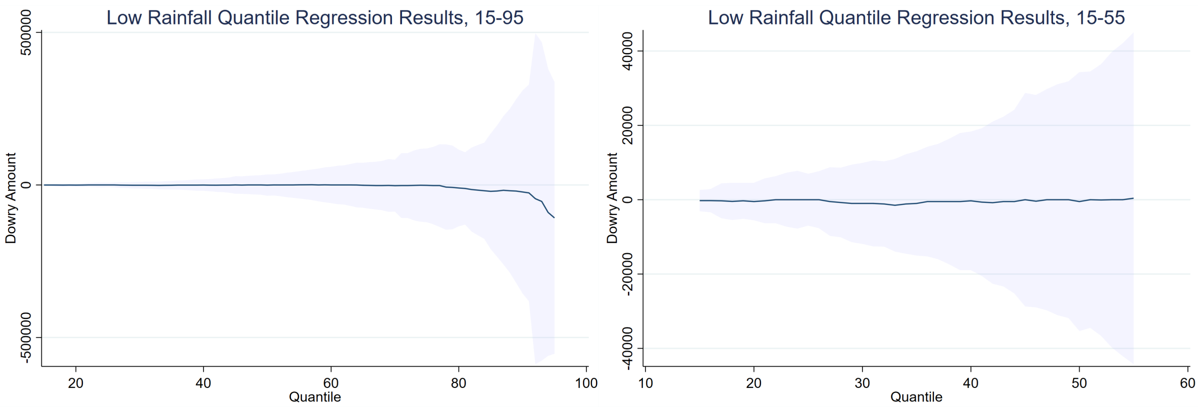
Panel A. Bride Price by Temperature Shock



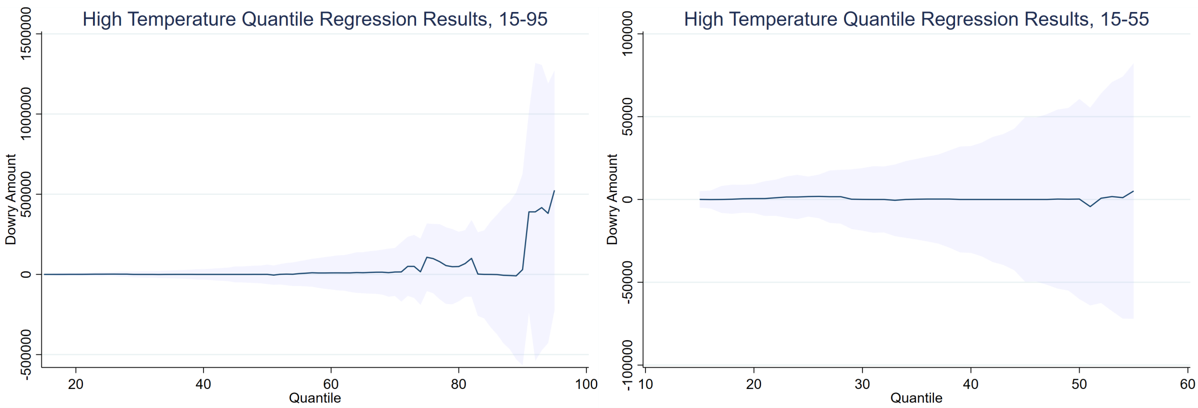
Notes: This figure shows a kernel density of bride prices in the IFLS sample, separately for indivduals who experienced a shock and individuals who did not. Only marriages taking place between ages 12-17 are included. Individual-level data come from the IFLS, and units are in Indonesia rupiah (IDR). Shock data come from the University of Delaware. A triangular kernel is used, and the distribution of prices is capped at 100,000 to avoid outliers.

Figure A3. Bride Price Quantile Regression Results

Panel A. Low Rainfall on Bride Price



Panel B: High Temperature on Bride Price



Notes: This figure shows the impacts of a low rainfall shock (in Panel A) or a high temperature shock (in Panel B) on bride price amounts in Indonesia for girls married ages 12-17 between 1966-2005. Data come from the IFLS, and units are in Indonesia rupiah (IDR). The first graph in each panel shows results from the 15th to 95th quantiles, and the second graph in each panel shows a zoomed-in view of the 15th-55th quantiles.

Table A1. Annual Probability of Child Marriage, with Lagged Shocks

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year (t)	-0.0001	-0.0007*	-0.0004*	-0.003***	-0.001***	-0.003***
	(0.0001)	(0.0004)	(0.0002)	(0.0009)	(0.0004)	(0.0009)
Low Rainfall Year (t-1)	-0.0001	-0.0003	-0.0004**	-0.002*	-0.0003	-0.001
	(0.0001)	(0.0004)	(0.0002)	(0.001)	(0.0003)	(0.0009)
Low Rainfall Year (t-2)	0.000005	0.0002	0.00002	-0.001	-0.0003	-0.0005
	(0.0001)	(0.0004)	(0.0002)	(0.0008)	(0.0003)	(0.0007)
Low Rainfall Year (t-3)	0.0002*	0.0008**	-0.0002	-0.002	-0.0006*	-0.002**
	(0.0001)	(0.0004)	(0.0002)	(0.001)	(0.0004)	(0.0007)
Mean of outcome	0.008	0.064	0.007	0.068	0.027	0.083
Observations	5,730,723	6,094,527	5,727,148	6,177,864	6,414,831	6,686,231
PANEL B: Temperature						
High Temp Year (t)	-0.0007***	-0.003***	-0.00002	-0.004**	-0.002**	-0.003
	(0.0002)	(0.0006)	(0.0004)	(0.001)	(0.0007)	(0.002)
High Temp Year (t-1)	-0.0001	-0.002***	-0.00008	0.0001	-0.001*	-0.005**
	(0.0002)	(0.0006)	(0.0002)	(0.002)	(0.0007)	(0.002)
High Temp Year (t-2)	-0.0007***	-0.002***	-0.0002	-0.0003	-0.002*	-0.006**
	(0.0002)	(0.0006)	(0.0003)	(0.001)	(0.001)	(0.003)
High Temp Year (t-3)	-0.0003	-0.0009	0.0003	0.002	-0.002*	-0.007***
	(0.0002)	(0.0007)	(0.0004)	(0.001)	(0.001)	(0.002)
Mean of outcome	0.008	0.066	0.007	0.067	0.027	0.082
Observations	5,766,592	6,135,400	5,761,652	6,213,561	6,414,831	6,686,231
PANEL C: Conflict						
Any Conflict Deaths (t)	0.0003	0.0003	-0.001**	0.004**	0.0003	-0.0004
	(0.0007)	(0.0007)	(0.0005)	(0.002)	(0.0005)	(0.001)
Any Conflict Deaths (t-1)	0.001*	0.001*	-0.0008	0.002	-0.0004	0.0002
	(0.0006)	(0.0006)	(0.0007)	(0.002)	(0.0005)	(0.001)
Any Conflict Deaths (t-2)	0.0006	0.0006	0.000009	0.003	-0.0001	0.0006
	(0.0007)	(0.0007)	(0.0007)	(0.002)	(0.0007)	(0.001)
Any Conflict Deaths (t-3)	-0.00004	-0.00004	-0.0004	-0.0008	0.0001	0.001
	(0.0007)	(0.0007)	(0.0006)	(0.003)	(0.0006)	(0.001)
Mean of outcome	0.010	0.064	0.005	0.039	0.024	0.077
Observations	3,667,151	4,150,251	608,107	620,103	2,875,006	3,282,668

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage at the individual-year level. I include both contemporaneous shocks and lagged shocks in the previous three years. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A2. Annual Hazard into Child Marriage with Alternate Conflict Measure, Nepal

	UCDP Conflict		**INSEC Conflict**	
	Boys	***Girls***	***Boys***	***Girls***
	(1)	(2)	(3)	(4)
Any Conflict Deaths	0.0003	0.0008	0.0006	0.0009
	(0.0005)	(0.001)	(0.0007)	(0.001)
%Δ (Coefficient / Mean)	0.011	0.010	0.028	0.012
Mean of outcome	0.025	0.080	0.022	0.077
Observations	3,440,473	3,897,802	2,260,400	2,614,537

Notes: This table shows the impacts of a year with exposure to conflict on the probability of child marriage within that same year at the individual-year level. Individual-level data come from IPUMS. Shock data come from UCDP (shown in columns (1) and (2)) and from INSEC (shown in columns (3) and (4)). Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. $p<0.1$*, $p<0.05$**, $p<0.01$***.

Table A3. Annual Probability of Child Marriage, By Husband's Relative Fertility Preferences

	India	
	Same Preferences	***Different or Unknown Preferences***
PANEL A: Rainfall	(1)	(2)
Low Rainfall Year	-0.0008	0.00006
	(0.0005)	(0.0002)
%Δ (Coefficient / Mean)	-0.020	0.007
Mean of outcome	0.042	0.009
Observations	1,888,876	1,888,876
PANEL B: Temperature		
High Temperature Year	-0.002***	-0.0007***
	(0.0006)	(0.0002)
%Δ (Coefficient / Mean)	-0.039	-0.076
Mean of outcome	0.042	0.009
Observations	1,898,593	1,898,593
PANEL C: Conflict		
Any Conflict Deaths	0.001*	0.00008
	(0.0007)	(0.0003)
%Δ (Coefficient / Mean)	0.030	0.009
Mean of outcome	0.043	0.009
Observations	1,658,291	1,658,291

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level by the match between the husband and wife's fertility preferences. Husbands and wives are considered to have different fertility preferences if the husband prefers more or fewer children than his wife or if the wife does not know the husband's fertility preferences. Individual-level data come from the DHS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A4. Annual Hazard into Child Marriage by Religion, India

	Hindu		Muslim		Christian	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	0.0001	-0.0006	-0.0007*	-0.002**	-0.0003	-0.001*
	(0.0002)	(0.0005)	(0.0004)	(0.001)	(0.0006)	(0.0009)
%Δ (Coefficient / Mean)	0.010	-0.009	-0.074	-0.035	-0.034	-0.043
Mean of outcome	0.012	0.069	0.010	0.063	0.008	0.034
Observations	3,072,924	4,544,287	492,040	751,655	284,082	475,589
PANEL B: Temperature						
High Temperature Year	-0.0001	-0.003***	-0.001*	-0.003**	-0.00002	0.0017
	(0.0003)	(0.0006)	(0.0007)	(0.001)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	-0.008	-0.047	-0.117	-0.048	-0.003	0.050
Mean of outcome	0.012	0.070	0.010	0.064	0.008	0.034
Observations	3,072,924	4,560,090	492,040	766,773	284,082	483,268
PANEL C: Conflict						
Any Conflict Deaths	0.0006	0.002**	0.0009	0.002	-0.001	-0.002**
	(0.0004)	(0.001)	(0.0006)	(0.002)	(0.0008)	(0.001)
%Δ (Coefficient / Mean)	0.039	0.034	0.079	0.036	-0.102	-0.068
Mean of outcome	0.015	0.070	0.012	0.069	0.010	0.036
Observations	2,047,276	3,479,163	340,990	607,442	186,857	373,346

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level by religion. Individual-level data come from the DHS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A5. Annual Hazard into Child Marriage by Religion, Indonesia

	Hindu		Muslim		Christian	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	0.0006	0.008**	-0.0004	-0.004***	-0.0003	-0.00002
	(0.002)	(0.004)	(0.0003)	(0.001)	(0.0004)	(0.002)
%Δ (Coefficient / Mean)	0.093	0.200	-0.055	-0.049	-0.071	-0.001
Mean of outcome	0.006	0.040	0.007	0.080	0.004	0.026
Observations	102,660	113,325	3,637,158	4,036,916	405,965	447,818
PANEL B: Temperature						
High Temperature Year	-0.002	0.005	0.0003	-0.002	-0.001***	0.0001
	(0.002)	(0.006)	(0.0004)	(0.002)	(0.0005)	(0.002)
%Δ (Coefficient / Mean)	-0.268	0.118	0.036	-0.021	-0.321	0.004
Mean of outcome	0.007	0.041	0.007	0.080	0.004	0.027
Observations	102,660	113,325	3,637,158	4,036,916	405,965	447,818
PANEL C: Conflict						
Any Conflict Deaths	-	-	-0.001	0.002	-0.006	-0.001
	-	-	(0.001)	(0.007)	(0.006)	(0.014)
%Δ (Coefficient / Mean)	-	-	-0.425	0.041	-2.665	-0.064
Mean of outcome	-	-	0.003	0.055	0.002	0.020
Observations	-	-	113,848	118,486	14,150	15,409

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level by religion. Individual-level data come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A6. Annual Hazard into Child Marriage by Religion, Nepal

	Hindu		Muslim		Christian	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.001***	-0.003***	-0.003**	-0.001	-0.0009	-0.003
	(0.0004)	(0.001)	(0.001)	(0.003)	(0.002)	(0.004)
%Δ (Coefficient / Mean)	-0.054	-0.036	-0.069	-0.011	-0.037	-0.035
Mean of outcome	0.027	0.085	0.041	0.116	0.023	0.074
Observations	5,255,337	5,418,774	247,584	216,294	68,096	78,831
PANEL B: Temperature						
High Temperature Year	-0.002**	-0.004	-0.0002	0.011**	-0.003	-0.012**
	(0.0009)	(0.002)	(0.004)	(0.005)	(0.003)	(0.006)
%Δ (Coefficient / Mean)	-0.072	-0.044	-0.006	0.091	-0.135	-0.167
Mean of outcome	0.027	0.085	0.041	0.116	0.024	0.074
Observations	5,255,337	5,418,774	247,584	216,294	68,096	78,831
PANEL C: Conflict						
Any Conflict Deaths	0.0002	0.0008	-0.003	-0.008	0.003	0.003
	(0.0005)	(0.001)	(0.002)	(0.005)	(0.003)	(0.005)
%Δ (Coefficient / Mean)	0.010	0.010	-0.075	-0.066	0.143	0.042
Mean of outcome	0.024	0.082	0.041	0.123	0.023	0.076
Observations	2,824,021	3,179,157	138,531	129,546	40,156	50,290

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level by religion. Individual-level data come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A7. Annual Hazard into Child Marriage Interacted with Irrigation, India

	India	
	Boys	*Girls*
PANEL A: Rainfall	(1)	(2)
Low Rainfall Year	-0.00005	-0.002***
	(0.0002)	(0.0006)
Irrigation Area (%)	0.001	-0.013**
	(0.001)	(0.006)
Low Rainfall x Irrigation Area	0.0003	0.005***
	(0.0007)	(0.002)
Mean of outcome	0.009	0.070
Observations	4,328,607	4,651,939
PANEL B: Temperature		
High Temperature Year	-0.0001	-0.002**
	(0.0003)	(0.001)
Irrigation Area (%)	0.002	-0.012**
	(0.001)	(0.006)
High Temperature x Irrigation Area	-0.001*	-0.0008
	(0.0008)	(0.003)
Mean of outcome	0.009	0.072
Observations	4,343,754	4,669,139
PANEL C: Conflict		
Any Conflict Deaths	0.0004	0.003*
	(0.0006)	(0.002)
Irrigation Area (%)	-0.004**	-0.011*
	(0.002)	(0.006)
Any Conflict Deaths x Irrigation Area	-0.002	-0.013***
	(0.001)	(0.004)
Mean of outcome	0.011	0.070
Observations	3,107,799	3,572,885

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Indicators for the three types of shocks are interacted with a continuous measure of the irrigated area in the district. Districts are aggregated based on 1966 levels. Data for India come from the DHS, shock data come from the University of Delaware and UCDP, and irrigation data come from ICRISAT. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A8. Migration by Experience of Shocks, Indonesia

	Shock to Shock	Shock to No Shock	No Shock to Shock	No Shock to No Shock	Total
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)
# Migrants	2900	285	222	9781	13,188
Percent of Total	21.99	2.16	1.68	74.17	
PANEL B: Temperature					
# Migrants	637	95	85	12371	13,188
Percent of Total	4.83	0.72	0.64	93.80	
PANEL C: Conflict					
# Migrants	48	15	17	8916	8,996
Percent of Total	0.53	0.17	0.19	99.11	

Notes: This table shows a descriptive summary of the types of migrants in the IFLS sample. Column (1) shows the number and percent of migrants who moved from one district that experienced a shock to another district that experienced a shock. Column (2) shows the number and percent of migrants who moved from one district that experienced a shock to another district that did not experience a shock. Column (3) shows the number and percent of migrants who moved from one district that did not experience a shock to another district that experienced a shock. Column (4) shows the number and percent of migrants who moved from one district that did not experience a shock to another district that did not experience a shock. Column (5) shows the total number of migrants for which data on shocks is available. Individual-level data come from the IFLS, and shock data come from the University of Delaware and UCDP.

Table A9. Annual Hazard into Child Marriage & Migration by Birthplace, Nepal

	Marriage by Current Residence		Marriage by Birthplace		Marriage & Migration		Marriage & No Migration	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Low Rainfall Year	-0.001***	-0.003***	-0.001***	-0.003***	-0.00006	-0.0003	-0.001***	-0.002***
	(0.0004)	(0.001)	(0.0004)	(0.0009)	(0.0001)	(0.0003)	(0.0004)	(0.0008)
%Δ (Coefficient / Mean)	-0.054	-0.034	-0.048	-0.032	-0.022	-0.019	-0.050	-0.036
Mean of outcome	0.027	0.082	0.027	0.082	0.003	0.017	0.024	0.064
Observations	6,414,831	6,686,231	6,092,559	6,179,699	6,092,559	6,179,699	6,092,559	6,179,699
PANEL B: Temperature								
High Temperature Year	-0.002**	-0.004*	-0.003***	-0.004**	-0.0001	-0.0007	-0.003***	-0.004*
	(0.0008)	(0.002)	(0.0008)	(0.002)	(0.0002)	(0.0006)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	-0.069	-0.046	-0.101	-0.054	-0.038	-0.039	-0.108	-0.059
Mean of outcome	0.027	0.082	0.027	0.081	0.003	0.018	0.024	0.063
Observations	6,414,831	6,686,231	6,253,390	6,360,762	6,253,390	6,360,762	6,253,390	6,360,762
PANEL C: Conflict								
Any Conflict Deaths	0.0003	0.0008	0.00004	0.001	0.00003	0.0006	0.000007	0.0004
	(0.0005)	(0.001)	(0.0005)	(0.001)	(0.0001)	(0.0004)	(0.0005)	(0.0009)
%Δ (Coefficient / Mean)	0.011	0.010	0.002	0.012	0.016	0.035	0.000	0.007
Mean of outcome	0.025	0.080	0.025	0.079	0.002	0.016	0.023	0.063
Observations	3,440,473	3,897,802	3,353,426	3,725,287	3,353,426	3,725,287	3,353,426	3,725,287

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. In columns (1)-(4), the outcome is any child marriage in a given year. In columns (5) and (6), the outcome is the probability of child marriage and migration in a given year. In columns (7) and (8), the outcome is the probability of child marriage and no migration in a given year. In columns (1)-(2), the shocks are assigned based on the individuals location of residence at the time of the survey. In columns (3)-(8), shocks are assigned based on the individual's birthplace. Individual-level data come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. $p<0.1$*, $p<0.05$**, $p<0.01$***.

Table A10. Impact of Shocks on Annual Hazard into Child Marriage & Migration, Indonesia

	Annual Probability of Marriage, Survey Residence		Annual Probability of Marriage, Birthplace		Annual Probability of Marriage, Migration by Year		Annual Probability of Migration	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Low Rainfall Year	-0.0006	-0.001	-0.0003	-0.001	-0.0005	0.00008	0.002	-0.006**
	(0.0009)	(0.003)	(0.0009)	(0.003)	(0.0009)	(0.003)	(0.003)	(0.003)
%Δ (Coefficient / Mean)	-0.090	-0.032	-0.040	-0.027	-0.077	0.002	0.060	-0.209
Mean of outcome	0.007	0.038	0.007	0.038	0.007	0.039	0.026	0.027
Observations	89,231	85,357	89,245	85,246	89,619	85,783	55,650	58,944
PANEL B: Temperature								
High Temperature Year	-0.00001	-0.005	-0.0001	-0.004	-0.0001	-0.002	-0.001	0.005
	(0.002)	(0.003)	(0.002)	(0.004)	(0.001)	(0.004)	(0.004)	(0.005)
%Δ (Coefficient / Mean)	-0.002	-0.133	-0.015	-0.101	-0.020	-0.043	-0.044	0.204
Mean of outcome	0.006	0.037	0.006	0.037	0.006	0.037	0.026	0.027
Observations	89,231	85,357	89,245	85,246	89,619	85,783	55,650	58,944
PANEL C: Conflict								
Any Conflict Deaths	0.001	0.013	-0.004***	0.012	-0.004***	0.016**	0.031	0.004
	(0.003)	(0.012)	(0.0009)	(0.010)	(0.0008)	(0.008)	(0.032)	(0.016)
%Δ (Coefficient / Mean)	0.458	0.506	-1.408	0.486	-1.483	0.622	0.978	0.120
Mean of outcome	0.003	0.025	0.003	0.025	0.003	0.025	0.031	0.036
Observations	45,358	48,270	45,352	48,146	45,539	48,484	24,361	26,261

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage or migration within that same year at the individual-year level for Indonesia for individuals between the ages of 12 and 17. Individual-level data come from the IFLS. Shock data come from the University of Delaware and UCDP. In columns 1-6, the outcome is an indicator for whether an individual experienced child marriage in a given year. In columns 7 and 8, the outcome is an indicator for whether an individual migrated in a given year. Columns 1 and 2 show assign shocks to individuals based on their location of residence at the time of the survey, columns 3 and 4 assign shocks to individuals based on their location of birth, and columns 5 - 8 assign shocks to individuals based on their actual location in each year. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. $p<0.1$*, $p<0.05$**, $p<0.01$***.

Table A11. Impact of Shocks on Annual Hazard into Marriage, Ages 18-24

	India		Indonesia		Nepal	
	Men	***Women***	***Men***	***Women***	***Men***	***Women***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.002***	-0.004***	-0.002	0.001	-0.003	-0.0007
	(0.0005)	(0.001)	(0.001)	(0.002)	(0.003)	(0.003)
%Δ (Coefficient / Mean)	-0.021	-0.020	-0.016	0.007	-0.019	-0.003
Mean of outcome	0.0798	0.178	0.0971	0.179	0.153	0.238
Observations	3,696,981	2,115,320	2,233,748	1,239,027	2,998,112	1,072,588
PANEL B: Temperature						
High Temperature Year	-0.001*	-0.0001	-0.006***	-0.005*	-0.0009	-0.002
	(0.0007)	(0.001)	(0.002)	(0.003)	(0.002)	(0.004)
%Δ (Coefficient / Mean)	-0.017	-0.001	-0.057	-0.030	-0.006	-0.010
Mean of outcome	0.0787	0.176	0.0967	0.177	0.152	0.239
Observations	3,720,674	2,135,633	2,249,964	1,249,648	2,998,112	1,667,744
PANEL C: Conflict						
Any Conflict Deaths	0.004***	0.010***	0.006***	0.005	-0.0008	0.001
	(0.0008)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
%Δ (Coefficient / Mean)	0.049	0.053	0.071	0.033	-0.006	0.006
Mean of outcome	0.0883	0.18	0.0868	0.15	0.149	0.238
Observations	2,975,320	1,904,183	672,894	426,592	1,817,830	1,072,588

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of marriage within that same year at the individual-year level for individuals ages 18-24 and on the marriage market. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table A12. Annual Probability of Child Marriage, with Multiple Shocks

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A:	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.00005	-0.001***	0.00006	-0.001	-0.0004	-0.0008
	-0.0002	(0.0004)	(0.0003)	(0.0008)	(0.0004)	(0.0009)
High Temperature Year	-0.0005**	-0.002***	-0.0007	-0.004***	-0.002**	-0.003*
	(0.0002)	(0.0006)	(0.0004)	(0.001)	(0.0006)	(0.002)
Any Conflict Deaths	0.0004	0.002**	-0.001**	0.004*	0.0002	0.0007
	(0.0003)	(0.0008)	(0.0006)	(0.002)	(0.0005)	(0.001)
Mean of outcome	0.010	0.067	0.005	0.043	0.023	0.077
Observations	4,112,149	4,681,827	842,748	866,982	2,260,400	2,614,537
PANEL B:						
Low Rain & High Temps	-0.0003	-0.001	-0.001***	-0.007***	-0.001	-0.002
	(0.0003)	(0.0008)	(0.0004)	(0.002)	(0.0009)	(0.003)
Mean of outcome	0.008	0.064	0.007	0.067	0.026	0.081
Observations	5,766,592	6,135,400	5,761,652	6,213,561	6,414,831	6,686,231
PANEL C:						
Low Rain, High Temps, & Conflict	-0.00005	-0.001	-0.0009	0.0008	-0.0002	-0.001
	(0.0006)	(0.002)	(0.0007)	(0.003)	(0.001)	(0.002)
Mean of outcome	0.009	0.064	0.005	0.041	0.023	0.077
Observations	4,137,446	4,708,762	853,513	878,327	3,440,473	3,897,802

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Panel A shows the impacts for each type of shock separately, Panel B shows the impacts of an indicator for either low rainfall or high temperatures (or both), and Panel C shows the impacts of an indicator for any type of shock. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. This table uses p<0.1*, p<0.05**, p<0.01***.

Supplemental Appendix

A Details on Institutional Background

This section describes the institutional background of child marriage, the prevalence of child marriage, and agricultural outcomes in India, Indonesia, and Nepal.

A.1 Institutional Background: India

Child marriage is quite common in India, and marriage transfers are also common. In 1993, 49% of girls were married before age 18, and in 2006, 7% of boys were married before age 18 (Gausman et al., 2024).[76] There is a mixture of dowry and bride price practiced depending on religious traditions and geographical region of the country, but the prevailing custom is dowry transfers from the bride's family to the groom's family. It used to be that bride price was practiced more in the south, while dowry was practiced more in the north, but gradually during the 1900s dowry began to spread to all areas of the country (Bhat and Halli, 1999, Dalmia and Lawrence, 2005, Anderson, 2007).

In India, the legal age of marriage is 18 for women and 21 for men, with no parental consent required. However, policies have recently been proposed to increase legal age of marriage for women to 21, but debates over these policies have not progressed over recent months Chakrabarty (2023), Pew Research Center (2016). In any case, marriage under age 18 is illegal for all individuals in India. The legal minimum marriage age was 15 for girls and 18 for boys in 1949, increasing to 18 for girls and 21 for boys in 1978.

Much of the population of India, especially in rural areas, is reliant on agriculture for their income. Agricultural output is subject to rainfall amounts and temperatures. Since much of the farmland in India is not irrigated, crops like rice, sugar, pulses, cotton, maize, and soya are negatively impacted by low rainfall (Jazeera, 2023, Rajshekhar, 2011). Similarly, increasing temperatures result in potential reductions in crop yields for wheat, soybeans, potatoes, wheat, and groundnuts (Birewar, 2023, Beillard and Singh, 2022). This implies that the effects of low rainfall shocks and high temperature shocks on annual probability of child marriage should move in the same direction, since both types of shocks will have similar impacts on income.

Conflict has occurred in India for several reasons. For example, conflict incidents in India during the relevant time period include incidents related to the conflict between India and Pakistan (Schumann, 2023). Additionally, ethnic-related conflict has historically been an issue (Roychoudhury, 2015).

[76] This has since decreased; prevalence in 2021 was 22% for girls and 2% for boys (Gausman et al., 2024).

A.2 Institutional Background: Indonesia

In Indonesia, 16% of girls are married before age 18 (2% before age 15), and 5% of boys are married before age 18 (Girls Not Brides, 2024). Net marriage transfers in Indonesia move from the groom's family to the bride's family, indicating that bride price is the most common type of transfer. However, there is some variation in the degree to which bride price is practiced, and while many families do practice bride price, others do not make net transfers upon the occasion of marriage. Previous work has found that dowry is not practiced in any region in Indonesia, and all regions practice either net marriage transfers from the groom's family to the bride's family or zero net transfers between the two households (Ashraf et al., 2020).

Until 2019, the legal age of marriage was 16 for women and 19 for men with parental consent. However, recently the legal age of marriage for women has also been raised to 19 with parental consent. Both men and women can marry at age 21 without parental consent required (UNICEF, 2019). As in India, this means that marriage before age 18 is currently not legal for any individuals.

Agriculture is also a major factor in income for many households in Indonesia, and this means that income is also highly dependent on rainfall and temperatures. Some of Indonesia's top crops include rice, coffee, and palm oil. In Indonesia, rising temperatures could reduce rice yields, thus reducing agricultural income (Yuliawan and Handoko, 2016). Rainfall is also important to rice production, so disruptions in rainfall would reduce rice yield (Matsuura and Sakagami, 2022). Oil palm production decreases with droughts or with increases in temperatures (Paterson et al., 2015). The same is true for coffee, where droughts and rising temperatures cause decreases in production (Stockholm Environment Institute, 2023, Sujatmiko and Ihsaniyati, 2018). This would suggest that lower rain and higher temperatures would both have the effect of reducing household income, meaning that the impacts of both types of shocks should be in the same direction.

Indonesia also experienced conflict violence during this time period. For example, there has been a series of civil conflict incidents and ethno-communal violence, which took place between 1998 and 2003 (Barron et al., 2016).[77]

A.3 Institutional Background: Nepal

In Nepal, net marriage transfers move from the bride's family to the groom's family, also known as dowry. This is likely to be stronger in the Terai region, which borders India. As in most contexts, dowry is increasing in age and education (Human Rights Watch, 2016, Bhandari, 2019). Marriage in Nepal is legal at age 20, with marriage at 18 legal with parental consent. However, 37% of girls in Nepal get married before they turn 18

[77]Since 2003, levels of conflict violence have declined drastically. This matches higher reported numbers of conflict incidents between 1998 and 2004.

(Human Rights Watch, 2016, Food and Agriculture Organization of the United Nations (FAO), 2021). The number is smaller but still significant for boys, at around 10% (United Nations Children's Fund (UNICEF), 2019).

Nepal is the only country in South Asia with a high prevalence of child marriage among both genders (UNICEF, 2017). Empirically, young boys in Nepal are likely to marry girls of similar ages, while young girls tend to marry either boys their age or older men (Choe et al., 2001). According to Choe et al. (2001), most women in Nepal do not view early marriage as desirable, with the majority of individuals who married at a young age later reporting that they did so because it was their parents' desire.[78]

In Nepal, higher rainfall and higher temperatures have been found to increase rice and potato production (Chandio et al., 2021, Dawadi et al., 2022). This suggests that low rainfall would lower agricultural income, while higher temperatures would increase agricultural income in areas where rice is mainly grown. However, other crops, such as maize, millet, and wheat, fare worse in higher temperatures, meaning that in areas where these crops are grown, higher temperatures could reduce agricultural output (Dawadi et al., 2022, Maharjan et al., 2013). There is also evidence that elevation could matter in terms of the response of crop output to temperature increases. As a result, low rainfall and higher temperatures could move in opposite directions or the same direction in terms of the impact on child marriage in Nepal, depending on which crops are grown most.

Between 1996 and 2006, Nepal underwent a civil conflict, which was initiated against the monarchy by the Communist Party of Nepal, a Maoist insurgency group (Do and Iyer, 2010). Conflict incidents occurred across all of Nepal, with most districts experiencing deaths, injuries, and kidnappings by both the Communist Party of Nepal and the government of Nepal. Roughly 30% of the Maoist army in Nepal was comprised of boys ages 14-18, which could have had a large impact on child marriage rates, lending credence to this mechanism (Valente, 2014, Lawoti, 2010). Additionally, it is likely that the civil conflict in Nepal resulted in a decrease in income, especially for the individuals who made a living through agricultural production. Areas under Maoist control were taxed more heavily for their production, and damage to road infrastructure meant that it was more difficult to transport food to markets (Upreti, 2010). Thus, the civil conflict could be considered a shock to income.

B Comparison to Corno et al. (2020)

Corno et al. (2020) study a similar question, where they estimate the impacts for female child marriage of experiencing a rainfall shock. They use version v3.01 of the Univer-

[78]While child marriage is illegal in Nepal, the prohibition against child marriage is not well-enforced. Officials often do not refuse to register the marriage, especially when parental consent is given (Human Rights Watch, 2016, Collin and Talbot, 2023).

sity of Delaware rainfall data and estimate the effects of rainfall shocks on female chile marriage for both India and the countries in Africa that have GPS data available in the DHS, and they find that in countries where bride price is practiced (Africa in their paper), experiencing a low rainfall year increases probability of child marriage for girls. In locations where dowry is practiced (India in their paper), experiencing a low rainfall year decreases the probability of child marriage for girls.

I replicate their results using the University of Delaware v5.01 monthly precipitation data. The v5.01 data is the updated version of the v3.01 data used by Corno et al. (2020), and changes include more input data, corrected historical imputation of values, and slightly different analysis methods.

For India, I use the 1993 shapefile available through IPUMS International. This is important because the sample frame for the 1998-1999 DHS came from the list of villages at in the 1991 Census and was stratified at the village level.[79] Thus, districts in the 1998-1999 DHS match the districts in the 1991 Census. Corno et al. (2020) mention that they match 440 out of 675 districts in India, indicating that they are using a newer shapefile with more districts. This shapefile is from at least 2011, since in 2011 there were 640 districts in India, with more being added over time.[80] However, because some of the districts in India change and are split over time, assigning individuals who are answering for the 1993 districts to 20xx districts will result in some individuals who are incorrectly located. Additionally, the district areas will be different, meaning that rainfall measures might be inaccurate as well. I re-estimate the shocks using the correct shapefile and compare these to the main results.

Panel A of Table S1 shows the results for India, for individuals under age 18. The first column shows the estimates directly from their paper, calculated using their replication data. The second column shows estimates using the same DHS survey and rainfall data from the University of Delaware rainfall data, calculated using the corrected shapefiles. Due to the fact that the total rainfall amounts are slightly different, the results that I find change a lot. The third column shows the same estimates as the second column, but with rainfall data from v5.01. Columns (4) - (6) match the data used in columns (1) - (3), but use observed year fixed effects instead of birth cohort fixed effects. Panel B shows the estimates when I include individuals ages 18+ instead of just 25+, but still restrict to the 1998 wave of the DHS. Panel C shows the estimates when the additional waves of the DHS (2015-2016 and 2019-2022) are added to the analysis.

I also re-estimate the shocks for Africa, though in this case the locations are based on GPS coordinates and not subject to the same concerns regarding shapefile. These recalculated estimates are found in Table S2. I find a much more consistent story comparing the estimates in Corno et al. (2020) and the estimates with my version of the v3.01 rainfall

[79]India had 467 districts in 1991 and 466 in 1993.

[80]When I use what I believe to be the same shapefile, my results are nearly identical to theirs.

and v5.01 rainfall data. Note that for this replication, I use the DHS data provided in the replication package on *Econometrica*'s website. However, I cannot match all points to the rainfally data.

C Additional Results

C.1 Methodology for Individual-Level Results

I can also estimate the effects of shocks at the individual level. To determine exposure to shocks, for each type of shock I sum up the number of years in which the individual experienced that particular shock between the ages of 12 and 17. The individual-level outcomes include an indicator for whether the individual was married before age 18, the continuous age at marriage, and an indicator for migration from location of birth.

For the individual-level analysis, I run the following regression:

$$Y_{idt} = \beta_0 + \beta_1 TotalShocks_{dt} + \gamma_d + \delta_t + X_{idt} + \epsilon_{idt} \tag{18}$$

Most of the terms in equation (18) are defined in the same way as in equation (1) above. The major difference between equation (1) and equation (18) is that the unit of analysis in the second regression is at the individual level. In equation (18), outcomes include overall probability of child marriage, age at first marriage, and probability of migration.

$TotalShocks_{dt}$ is the sum of the shocks that the individual experienced between the ages of 12 and 17, inclusive. This is true regardless of whether or not the individual married before turning 18. For example, if the individual experienced a low rainfall year at age 12, got married at age 14, and experienced a second low rainfall year at age 17, the value of $TotalShocks_{dt}$ would be 2, even though the individual only experienced one shock before marriage. This ensures that the number of potential shocks is comparable across individuals who married as children and those who did not, whereas comparing only the years before marriage would lead to a mechanical relationship between marriage outcomes and the number of shocks experienced. Estimates from this regression show how much shocks change the lifetime probability of experiencing child marriage.[81]

Estimates from equation (18) should align with those from equation (1) as follows: If a rainfall shock causes a decrease in the annual probability of child marriage as estimated in equation (1), an increased number of rainfall shocks should also reduce the overall probability of child marriage in equation (18). Similarly, the average age at marriage should increase, since individuals are delaying marriage.[82] One caveat is that the analysis

[81] Overall child marriage will change if individuals who are nearly 18 push their marriage past age 18 or individuals who are much younger than age 18 delay their marriage significantly.

[82] Note that this would not be the case if shocks increased hazard into marriage at ages 18 or older.

for age at marriage is only run for the subsample of individuals who are already married at the time of the survey. While the sample is restricted to individuals who are at least age 18, there are many individuals who remain single. This could cause bias, since individuals in the sample who are not yet married are not included. This would be a concern if individuals are differentially more or less likely to be single at the time of the survey based on their experience of shocks during adolescence.

I also investigate impacts on migration. While the main sample does not include information on whether the timing of migration is concurrent with marriage, I do have information on location of birth for individuals in Nepal and Indonesia.[83] In Nepal, this information is available at the district level, while for Indonesia information on location of birth is available only at the region level. I construct two outcome variables: an indicator for whether an individual's location at the time of birth is the same as the location at the time of the survey (for both Nepal and Indonesia) and a measure of the distance between the individual's location of birth and location of residence at the time of the survey (for Nepal only). This allows me to see changes in migration patterns at the individual level for individuals who experience additional shocks between the ages of 12-17.

C.2 Individual-level Outcomes

This subsection discusses how shocks change marriage patterns at the individual level, aggregating both shocks and outcomes to the individual level across all years of adolescence. I estimate equation (18) with both probability of child marriage and age at first marriage as outcomes. Table S3 shows the impacts of each additional shock year experienced between the ages of 12-17 on the overall probability of child marriage. The effects of an additional low rainfall year are shown in Panel A. For the most part, there is no impact of low rainfall on overall child marriage. An additional high temperature year decreases the probability of experiencing child marriage in India and Nepal (Panel B). An additional year with any conflict deaths decreases the probability of child marriage for boys in Indonesia and increases the probability of child marriage for girls in Indonesia and Nepal (Panel C). For the most part, these results are consistent with the findings at the individual-year level, though weaker.

The change in probability of child marriage will mainly reflect individuals who are pushed from marrying at age 17 to marrying at age 18 or later (or marrying at age 16 to marrying at age 18 or later, etc.). However, it is unlikely that individuals who would marry at a very young age in the counterfactual (e.g. age 12) will delay their marriage past age 18 in the presence of a shock. Thus, the individual-level analysis only captures part of the change of marriage patterns.

Age at marriage helps to capture another part of the picture. However, since not

[83]In the IFLS, I also have the individual's full migration history, starting at age 12.

all individuals in the sample are married and marriage in this setting is endogenous, the results of the estimation are not strictly causal. Table S4 shows the impact of an additional shock year on the average age at marriage. The results are slightly more mixed, with rainfall in Panel A decreasing age at first marriage in Indonesia and increasing age at first marriage in Nepal. In Panel B, an additional high temperature year increases age at first marriage in India and Nepal and decreases age at first marriage in Indonesia. Conflict seems to have a much weaker effect, only lowering age at first marriage for boys in India. Again, since these results are conditional on marriage, selection into marriage could be driving the differences in effects across the countries. Additionally, all of the impacts are quite small, with magnitudes of 0.004-0.1 years, or the equivalent of around a month at the most. Compared to an average age of marriage of around 18 for girls and 20-24 for boys, these effects are not large.

Overall, these findings suggest that there are changes in the probability of child marriage in a given year when a household experiences a shock as well as changes in the overall probability of child marriage over an individual's lifetime when they experience a shock between the ages of 12 and 17. The following section explores the mechanisms behind the effects of low rainfall, high temperatures, and exposure to conflict on child marriage outcomes.

C.3 Income as a Mechanism

C.3.1 Rural vs. Urban Areas

As an additional way to test whether agricultural income is a main mechanism for the impacts of shocks on child marriage, I divide the sample between individuals living in rural vs. urban areas and run the regression separately for each group. In theory, individuals in urban areas should be less dependent on agricultural income, and, consequently, less responsive to rainfall and temperature shocks. Individuals in rural areas, on the other hand, should be more dependent on agriculture and more responsive to rainfall and temperature shocks and their effects on agricultural yields.

Panels A-C of Table S6 show the main results for rural areas, and Panels D-F show the results for urban areas. As expected, the effects of rainfall and temperature shocks are stronger in more rural areas, with the coefficients for rural areas generally being larger and more statistically significant than the coefficients for urban areas. This aligns with the hypothesis that rainfall and temperature shocks operate mainly through the channel of changing agricultural income, which is more important in rural areas.[84]

[84]Note that this pattern is not as strong when looking at the effects as a percentage of the underlying child marriage rates, since child marriage is also much more common in rural areas.

C.3.2 Gross Regional Domestic Product

To further assess whether shocks have a negative impact on aggregate income, I estimate the impact of low rainfall, high temperatures, and conflict on Gross Regional Domestic Product (GRDP) at the district level in Indonesia using data from 2010-2017.[85] I include district fixed effects in my regressions, as well as province-specific linear time trends, and standard errors are clustered at the district level. The results are shown in Table S5. Low rainfall decreases GRDP, while high temperatures and exposure to conflict do not have a statistically significant effect (though the coefficients are positive). One caveat regarding this estimation is that GRDP is measured for all industries, not just for the agricultural sector. Additionally, the standard errors are large in columns (2) and (3), meaning that large negative effects of high temperature shocks on GRDP, for example, cannot be ruled out.

C.4 Conflict by Rural vs. Urban Areas

Risk of violence associated with conflict may vary across rural and urban areas. The response to conflict shocks should be stronger in areas where the risk of violence is higher. Panel C of Table S6 shows the impacts of conflict exposure in rural areas, and Panel F shows the impacts of conflict exposure in urban areas. Notably, for girls in India, exposure to conflict increases the annual probability of child marriage in rural areas and decreases the annual probability of child marriage in urban areas. This could occur if the risk of experiencing conflict-related violence varies based on population density.

C.5 Migration

C.5.1 Indonesia and Nepal: Migration Probability

For individuals in Nepal, the data include their birthplace at the district level, and for individuals in Indonesia, the data include their birthplace at the province level. I estimate whether individuals are more likely to move away from their location of birth if they experienced a shock.[86] Table S7 shows the impact of experiencing an additional shock between the ages of 12-17 on the probability that the individual lives in a difference province or district than the one they were born in.[87] Since shocks are assigned based on the district of residence at the time of the survey in columns (1) and (2), these estimates should be interpreted as changes in the likelihood that an individual moved to a district

[85]These years are outside the sample period. However, GRDP for Indonesia at the district level is not available for earlier years.

[86]For India, I only have information on the individual's district of current residence, so I do not include India in the analysis.

[87]Note that it is not possible to see the timing of when the individual migrated. Thus, this migration may not be due to marriage. Additionally, individuals who were born outside the country or who moved outside the country will not be included in this analysis.

given that this district experienced negative shocks during the years when that individual was ages 12-17. In Panel A, an additional low rainfall year increases the probability of migrating to that district for both boys and girls in Nepal. In Panel B, an additional high temperature year decreases the probability of migrating to that district in Indonesia and increases it for boys in Nepal. In Panel C, an additional year with any conflict deaths decreases the probability of migrating to that district for girls in Nepal. Results are similar when shocks are assigned based on location of birth, shown for Nepal in column (3).[88] Thus, migration is clearly correlated with experiencing shocks, though the impacts are not the same across all subgroups.[89]

Using data from the Indonesian Family Life Survey I also directly test whether shocks influence the probability that an individual migrates in a given year. These estimates are shown in columns (7) and (8) of Table A10.[90] I find some evidence of minor changes in migration patterns for girls as a result of experiencing a low rainfall shock, though caution should be taken with many of the other point estimates, since the sample is quite small and the confidence intervals are quite large. All of these approaches provide evidence that migration is not causing bias in my estimates.

C.6 Marriage & Migration

I estimate the effect on the annual probability of both marriage and migration (or marriage and no migration) for Nepal in Table A9. The outcome variable in columns (5) and (6) is an indicator for an individual marrying in the observed year and migrating.[91] The outcome variable in columns (7) and (8) is an indicator for the individual getting married in the observed year and not migrating. The results in Panel A and Panel B show that marriages where an individual does not eventually migrate are the ones being delayed, suggesting a relative shift towards marriages that require migration when low rainfall or high temperatures are experienced. This aligns with the findings that experiencing shocks increases the probability of migration for individuals in Nepal.

[88] The fact that the coefficients in columns (2) and (3) generally move in the same direction is likely because individuals tend to move from areas that received a shock to other areas that also received a shock.

[89] Columns (7) and (8) of Table S7 show impacts on the distance that individuals migrate from their place of birth, where distance is measured in meters between district centroids. The fact that all coefficients are positive, though not all are statistically significant, suggests that individuals are more likely to migrate and to migrate further when they experience a shock. This is especially true for boys, with only small impacts on migration distance for girls when experiencing a high temperature year. The magnitudes of all of these point estimates are relatively small.

[90] The outcome variable is an indicator for the individual's district of current residence being different than the district of residence in the previous year. Intra-district moves will not be captured using this approach.

[91] Note that it is not possible to know the exact timing of the migration, only whether or not they currently live in the same district that they were born in.

C.7 Additional Robustness

C.7.1 Alternate Shock Definitions

Since there is no consensus in the literature regarding exactly how low rainfall and high temperature shocks should be defined, I show that my estimates are robust to various shock definitions. Figure S7 shows estimates of my main specification with various cutoffs for rainfall along the gamma distribution, from the first to the 49th percentile. These results are shown separately for boys and girls in India, Indonesia, and Nepal. Figure S7 shows that the choice of the 15th percentile as a cutoff does not greatly impact the estimates, since the results hold across cutoffs. The effects of shocks are generally stronger when only more severe rainfall shocks are considered and weaker when less severe rainfall shocks are also considered.

I do the same for temperature in Figure S8, where cutoffs range from the 50th to the 95th percentile. Again, the results are consistent for both boys and girls in all three countries, with more severe shocks resulting in larger effects. Once again, this suggests that my findings are not dependent on a particular definition of temperature shock.

Additionally, I calculate the cutoffs for low rainfall and high temperature years using the entire time span of data available from the University of Delaware (1900-2017). I find that the estimated effects do not change when using the entire 118 years of data to create the cutoffs for low rainfall and high temperatures, as seen in Table S8. This suggests that the results are not sensitive to the rainfall and temperature distributions.

Since temperature and rainfall change over time, there may be concerns that the effects of rainfall and temperature shocks are simply picking up time trends in the data. Including fixed effects for the year corrects for changes in the response to shocks over time. However, I also re-define the shocks using a de-trended version of the rainfall and temperature data. I define a low rainfall year as a year with rainfall below the 15th percentile of the district-specific rainfall values from 1950-2017. I define a high temperature year as a year with temperatures above the 90th percentile of the district-specific temperature values from 1950-2017. The results are shown in Table S9 and are consistent with the main estimates.

Next, I calculate exposure to conflict using an indicator for any conflict incidents, the continuous number of conflict deaths, and the continuous number of conflict incidents in Table S8. The results generally align with the main specification, though, especially for the continuous shocks, the point estimates are quite small and almost never statistically significant. The biggest difference compared to the main results is that using conflict incidents instead of conflict deaths results in a statistically significant (though small) decrease in child marriage for both girls and boys in Nepal.

Finally, to compare effects across the three different types of shocks, I measure the shock intensity based on standard deviations from the district-specific mean. The interpretations of the coefficients are different from the main effects, since here each coefficient represents the impact of a standard deviation increase in rainfall, temperature, or conflict on the annual probability of child marriage. To align with the main results, it is expected that the coefficients on rainfall should be positive, the coefficients on temperature should be negative, and the coefficients on conflict events should be positive for girls and negative for boys. These results can be found in Table S10. In this table, the effects of a one standard deviation increase in rainfall are positive, and the effects of a one standard deviation increase in temperatures are negative, aligning with the main results. However, the coefficients for conflict are generally statistically insignificant and close to zero, except for the effect on girls in Nepal, which is negative. This suggests that there may be non-linearities in the impacts of conflict deaths on child marriage.

C.7.2 District Level Regressions

In addition to investigating the impacts of shocks on the annual probability of child marriage at the individual level, it is important to understand how the district-level marriage market is affected. To estimate this, I aggregate the number of marriages taking place among individuals ages 12-24 in the district in any given year.[92] I then run regressions at the district level to estimate the impact of rainfall, temperature, and conflict shocks on the number of matches taking place in that district in a given year. The results of this analysis are shown in Panels A-C of Table S11. Aligning with the main results, low rainfall and high temperatures decrease the number of marriages that occur at the district level. Conflict, on the other hand, increases the number of marriages overall in India and decreases the number of marriages overall (for both men and women) in Indonesia and Nepal. Since this does not completely align with the effects on the individual probability of child marriage in a given year, these results also suggest that there are changes in both the quantity and composition of matches that take place.[93]

When looking only at child marriage at the district level, the signs of the effects are very similar to those of the individual-level regressions, though they are generally less precisely measured. These estimates are shown in Panels D-F of Table S11. This further suggests that, while rainfall and temperature shocks seem to impact marriage similarly for all ages, conflict shocks have different impacts on individuals of different ages, leading to differences in the estimates shown in Table 3 and Panels A-C of Table S11.

[92] I choose this age range to include as many individuals as possible while still retaining a large sample. To examine marriages up until age 25, I must drop individuals below age 25 at the time of the survey. Thus, to include marriages at later ages in the regression, I would need to drop larger fractions of the sample. Additionally, the majority of individuals in the sample are married by age 24.

[93] This is also consistent with the fact that conflict seems to have different effects for individuals between ages 12-17 versus 18-24, as seen in Table 3 and Table A11.

C.7.3 Impacts on Marriage Distribution

Since the main results show impacts on the probability of marriage and of certain types of matches occurring, it is reasonable to ask whether all types of matches are being affected equally. I test whether the composition of marriages within the district is changing, and I find that, in most cases, low rainfall, high temperatures, and exposure to conflict impact both the distribution of age at marriage and the distribution of spousal age gaps.

Figure S1 shows the distributions for men in India, Figure S2 for women in India, Figure S3 for men in Indonesia, Figure S4 for women in Indonesia, Figure S5 for men in Nepal, and Figure S6 for women in Nepal. Each figure shows, for each of the three types of shocks, the distribution of age at marriage and of age gaps between husbands and wives in the data for each given country, plotted separately for years when a shock was and was not experienced in the district. I conduct a Kolomogorov-Smirnov test for equality between the two distributions, and the p-values are reported in each panel. In most cases, it is possible to reject that the two distributions are the same.

The fact that the distribution of age at first marriage and of age gaps between spouses in the marriage market changes in the presence of a shock suggests that the composition of marriages is affected, not just the quantity of marriages taking place.

C.7.4 Accounting for Spatial Correlation

Due to the nature of the shocks, there is likely spatial correlation within the data. I use two strategies to account for this. First, I assign the shocks and cluster the standard errors at the district level. By clustering the standard errors at the district level, I am accounting for the fact that outcomes for individuals in the same district are likely correlated, including any spatial correlation.

However, there may also be concerns regarding spatial correlation across districts. To account for this, I run the district-level regressions described in the subsection above and cluster the standard errors across different distances, as in Colella et al. (2023).[94] Table S12 shows the standard errors allowing for spatial correlation up to 100km and 200km. While the standard errors do increase in size for some of the specifications, the increases are not extremely large. In some specifications, the standard errors even shrink. This suggests that spatial correlation is likely not a major concern for my main estimates.

C.7.5 Including Survey Weights

In the main specification, each individual is given equal weight. However, to ensure that the individuals in the regression are representative of the population (which would be

[94] This is achieved by assigning each individual to the location of the district centroid, since I do not have more precise information on individuals' locations. Running these regressions correcting for spatial correlation across districts is computationally infeasible due to the large number of observations and the fact that a distance matrix is required to be generated for all pairs of observations.

a concern if there is heterogeneity in the treatment effects), I include as an analytical weight the individual sampling probability. These regression results can be found in Table S13. The signs and magnitudes of the coefficients are generally consistent with the effects found in the main results in Table 3, suggesting that the representativeness of the sample is not a concern in this setting.

C.7.6 Indonesia without 1976

As mentioned above, the individual-level data from Indonesia in 1976 are not representative of the entire country, since the 1976 Intercensal Survey does not fully cover three of Indonesia's provinces: East Nusa Tenggara, Maluku and Irian Jaya. To ensure that my findings are not driven by the fact that my sample is not fully representative, I run the regressions for Indonesia without individuals surveyed in 1976. The results are found in Table S14, and the estimates are quite similar to my main results. This suggests that the effects are not driven by the lack of representativeness in the 1976 data.

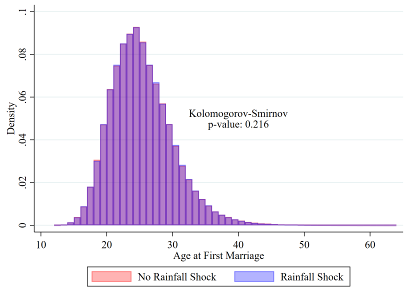
C.7.7 Samples with Various Age Cutoffs

To ensure that my results are not being driven by the choice of using only individuals who are age 18 and older, I also run the main specification with individuals who are age 21 and older and for individuals who are age 25 and older. These results are found in Table S15. For the most part, the effects are similar, especially for the sample of individuals age 21 and older. The main difference is the impact of high temperatures in Panel E of Table S15 on probability of child marriage for boys, where the coefficient is positive and marginally significant instead of negative. However, further restricting the sample generally does not change the results, showing that my estimates are robust to different sample definitions.

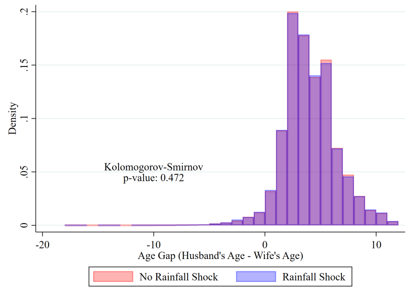
Supplemental Appendix Tables & Figures

Figure S1. Age at Marriage & Age Gap Distribution, Men in India

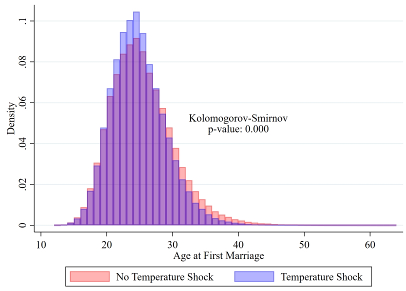
(a) Age Married, Low Rainfall



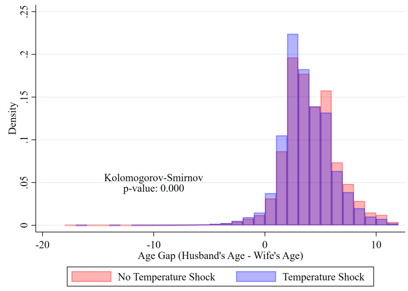
(b) Age Gap, Low Rainfall



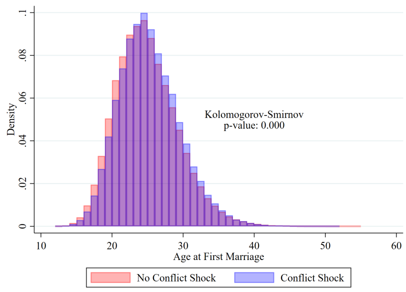
(c) Age Married, High Temperature



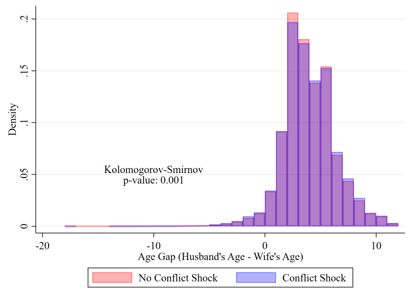
(d) Age Gap, High Temperature



(e) Age Married, Conflict

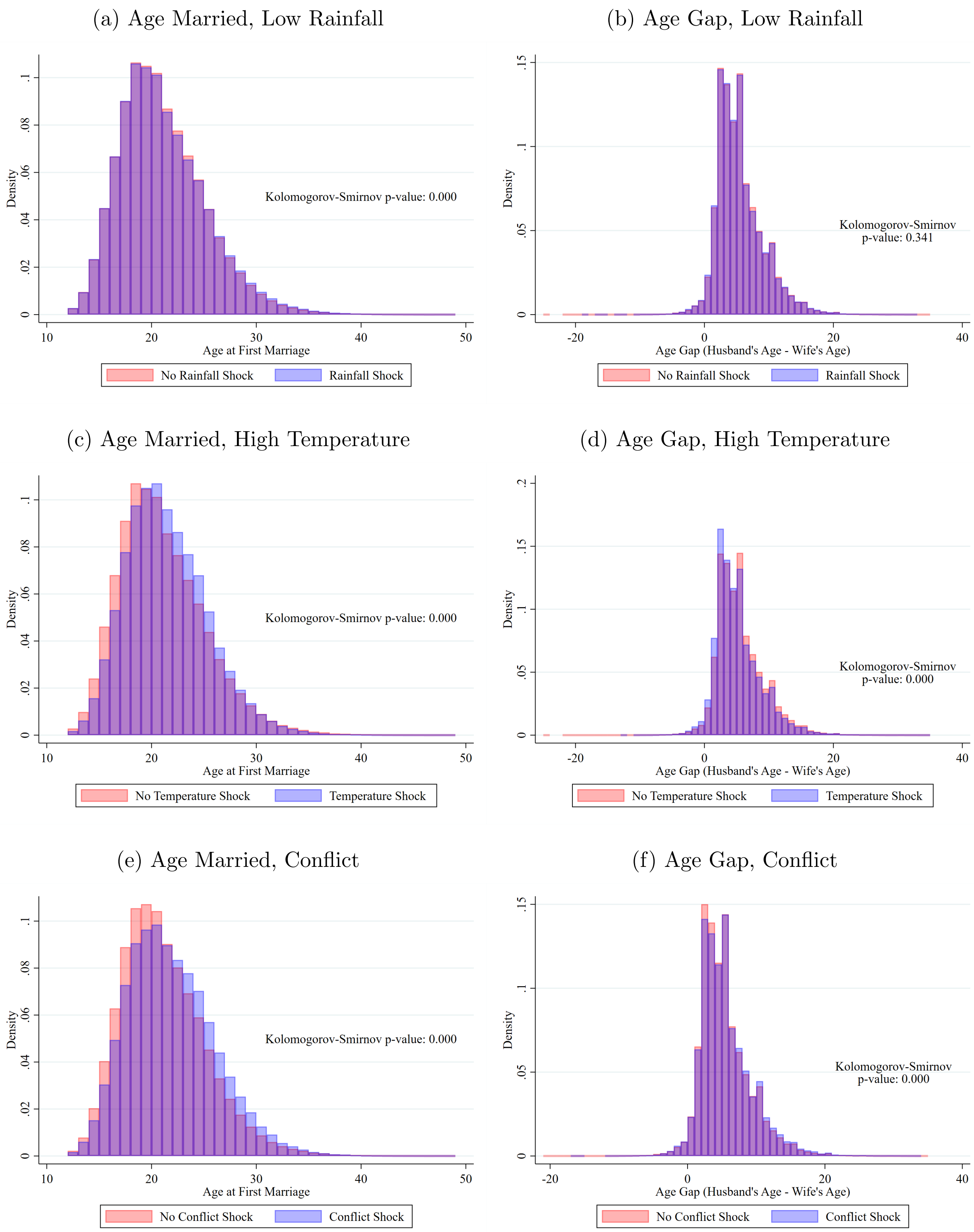


(f) Age Gap, Conflict



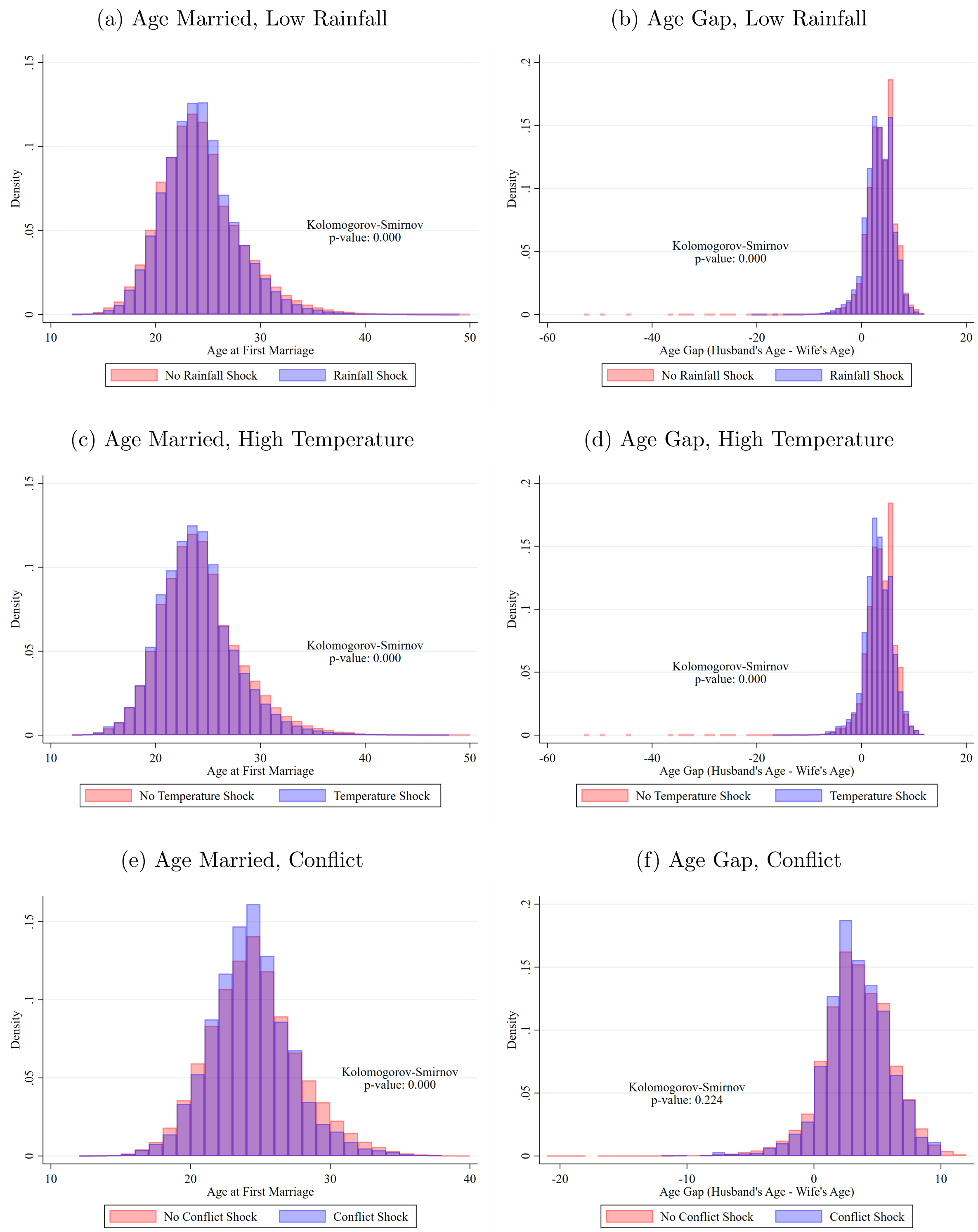
Notes: These figures show distribution of age at first marriage (in Panels A, C, and E) and age gap between spouses (in Panels B, D, and F) for men in India in years with and without low rainfall, high temperatures, and exposure to conflict. Data come from the University of Delaware, UCDP, and the DHS. P-values for a Kolomogorov-Smirnov test of equality between the pairs of distributions are shown on each Panel.

Figure S2. Age at Marriage & Age Gap Distribution, Women in India



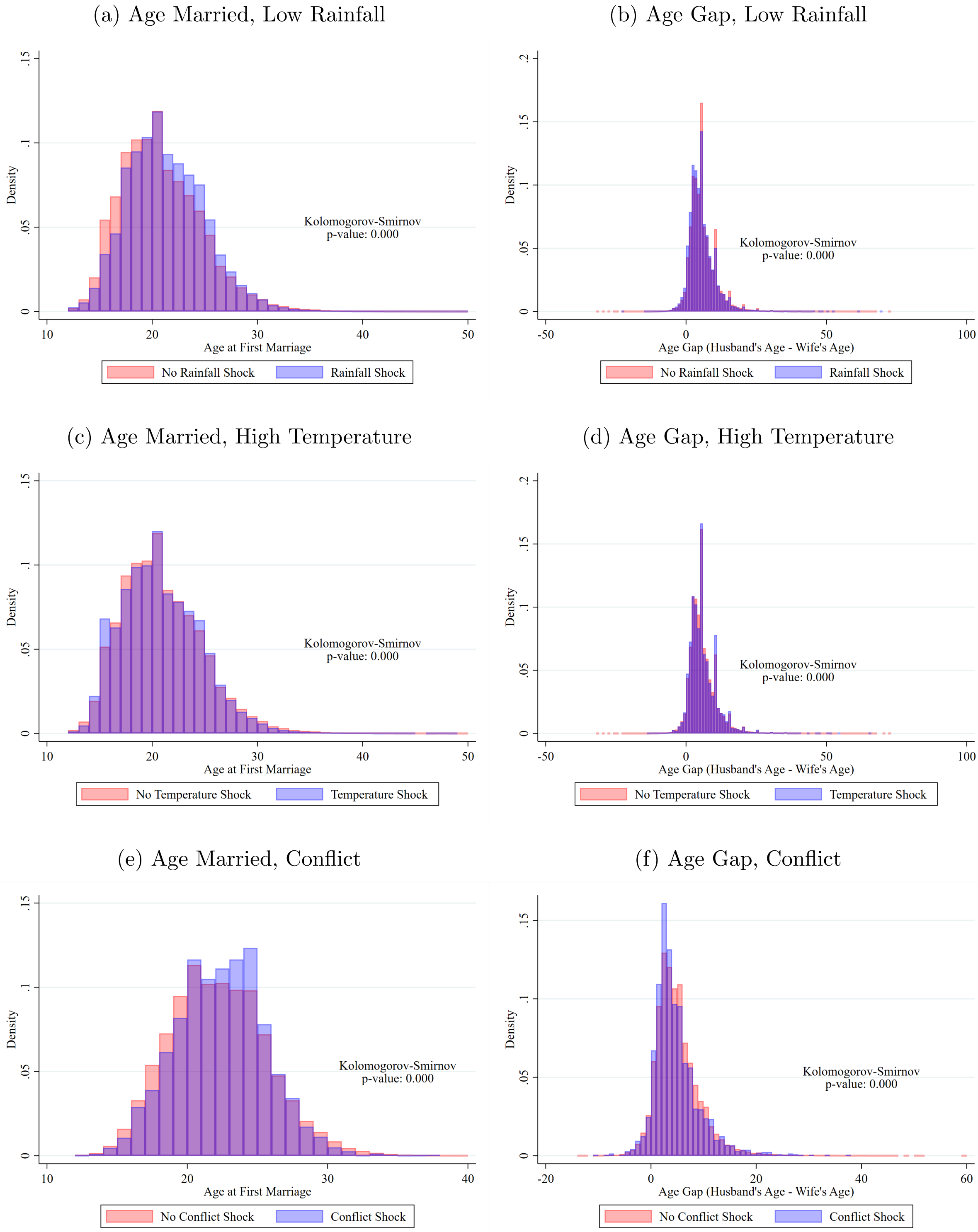
Notes: These figures show distribution of age at first marriage (in Panels A, C, and E) and age gap between spouses (in Panels B, D, and F) for women in India in years with and without low rainfall, high temperatures, and exposure to conflict. Data come from the University of Delaware, UCDP, and the DHS. P-values for a Kolomogorov-Smirnov test of equality between the pairs of distributions are shown on each Panel.

Figure S3. Age at Marriage & Age Gap Distribution, Men in Indonesia



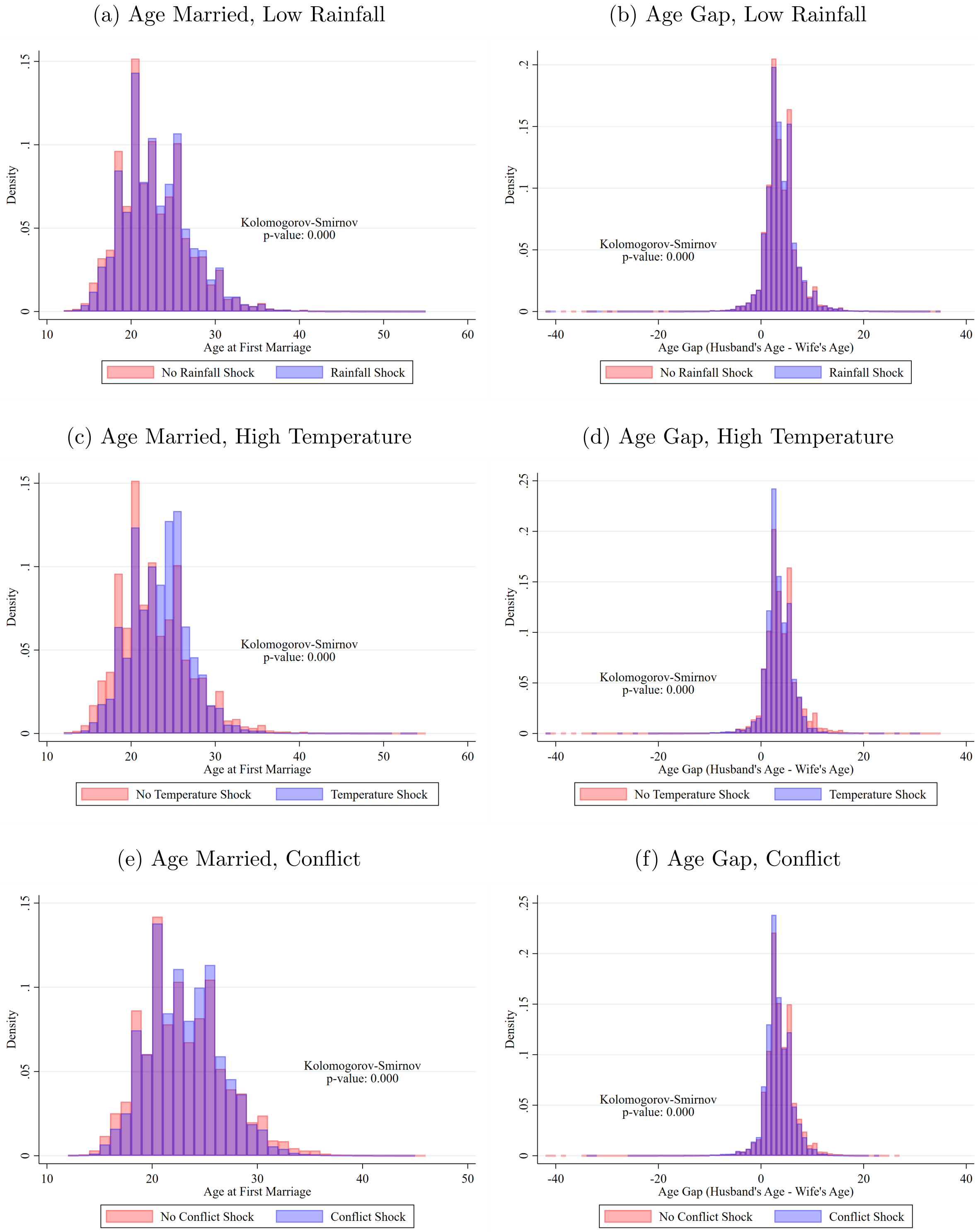
Notes: These figures show distribution of age at first marriage (in Panels A, C, and E) and age gap between spouses (in Panels B, D, and F) for men in Indonesia in years with and without low rainfall, high temperatures, and exposure to conflict. Data come from the University of Delaware, UCDP, and IPUMS International. P-values for a Kolomogorov-Smirnov test of equality between the pairs of distributions are shown on each Panel.

Figure S4. Age at Marriage & Age Gap Distribution, Women in Indonesia



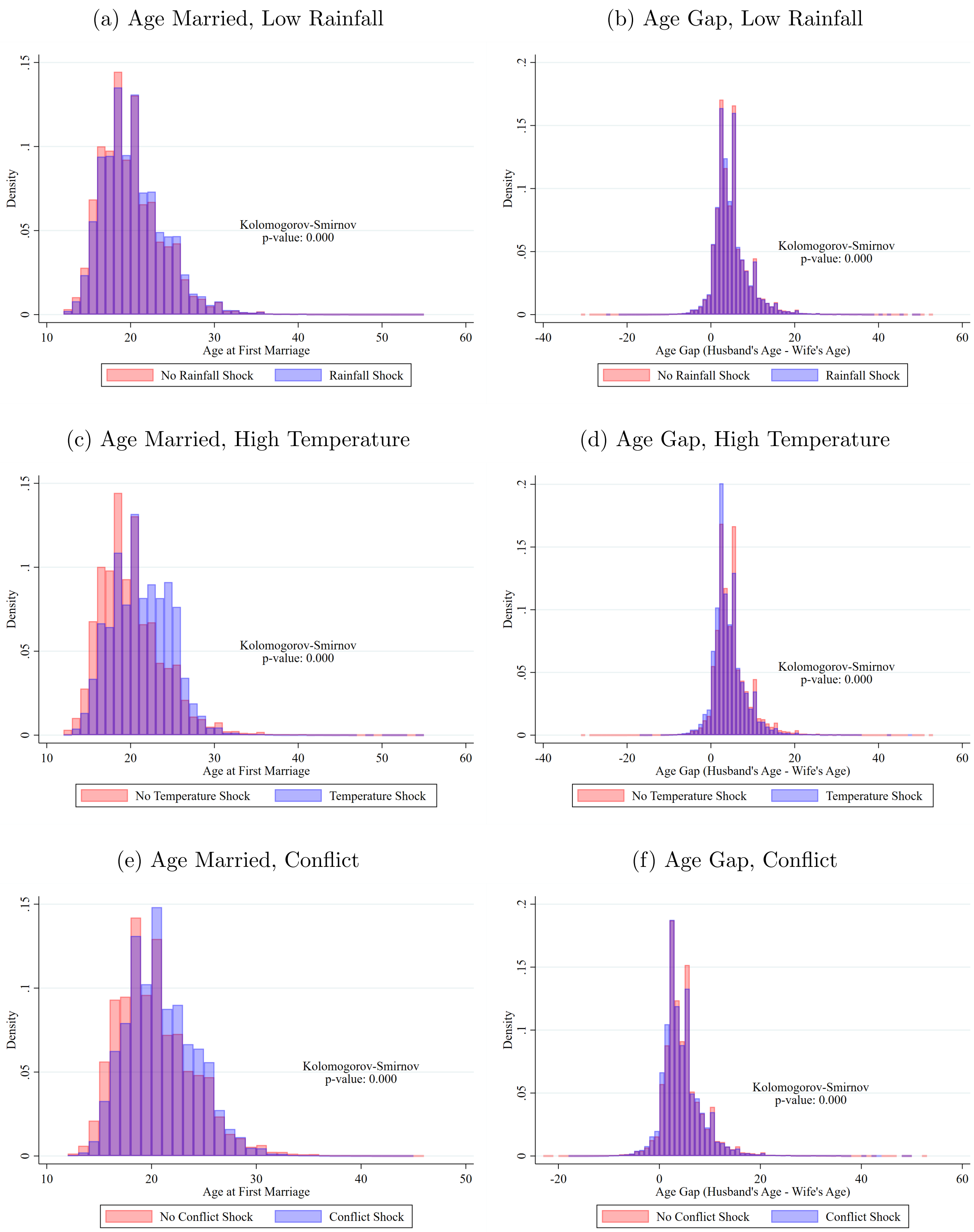
Notes: These figures show distribution of age at first marriage (in Panels A, C, and E) and age gap between spouses (in Panels B, D, and F) for women in Indonesia in years with and without low rainfall, high temperatures, and exposure to conflict. Data come from the University of Delaware, UCDP, and IPUMS International. P-values for a Kolomogorov-Smirnov test of equality between the pairs of distributions are shown on each Panel.

Figure S5. Age at Marriage & Age Gap Distribution, Men in Nepal



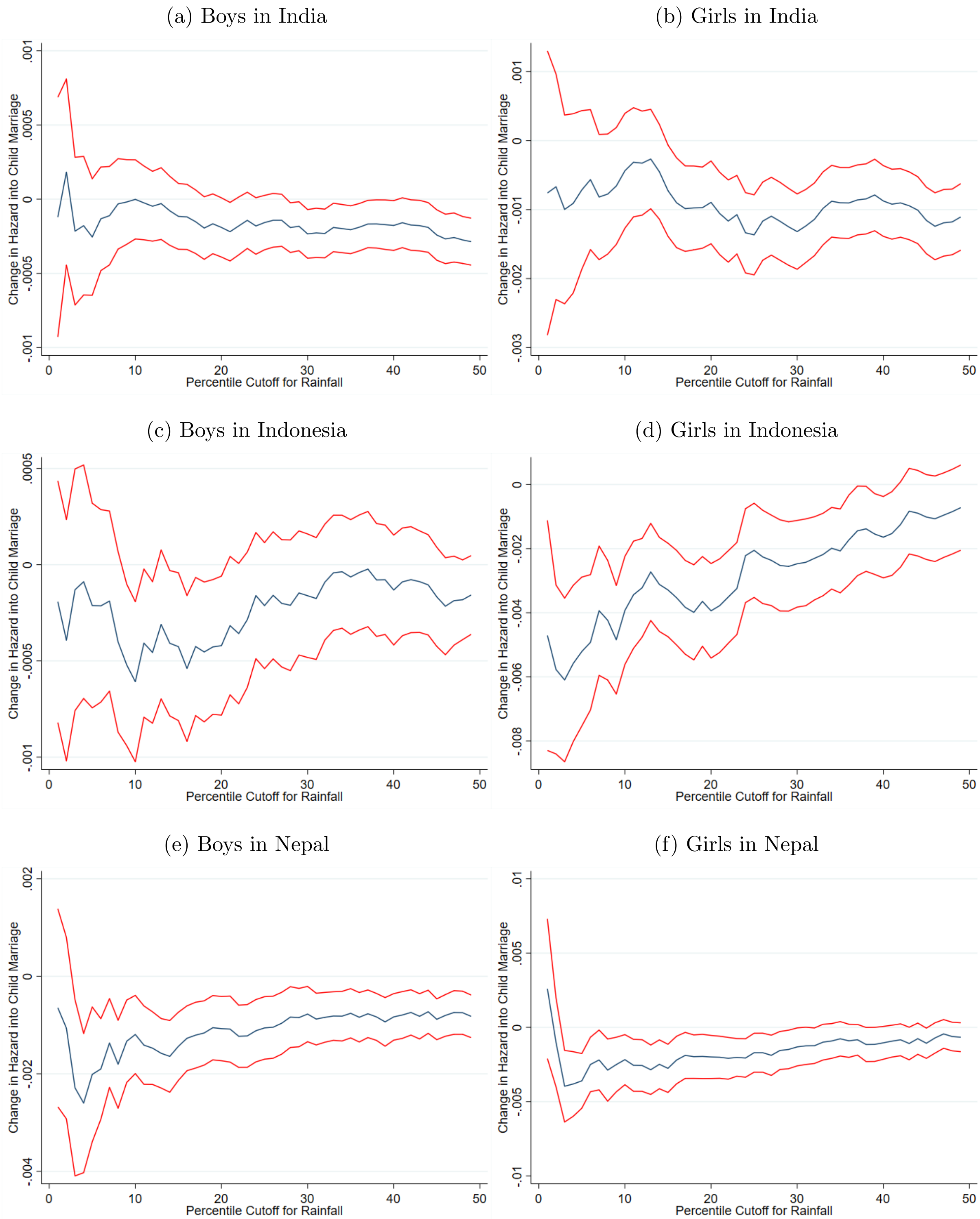
Notes: These figures show distribution of age at first marriage (in Panels A, C, and E) and age gap between spouses (in Panels B, D, and F) for men in Nepal in years with and without low rainfall, high temperatures, and exposure to conflict. Data come from the University of Delaware, UCDP, and IPUMS International. P-values for a Kolomogorov-Smirnov test of equality between the pairs of distributions are shown on each Panel.

Figure S6. Age at Marriage & Age Gap Distribution, Women in Nepal



Notes: These figures show distribution of age at first marriage (in Panels A, C, and E) and age gap between spouses (in Panels B, Dm abd F) for women in Nepal in years with and without low rainfall, high temperatures, and exposure to conflict. Data come from the University of Delaware, UCDP, and IPUMS International. P-values for a Kolomogorov-Smirnov test of equality between the pairs of distributions are shown on each Panel.

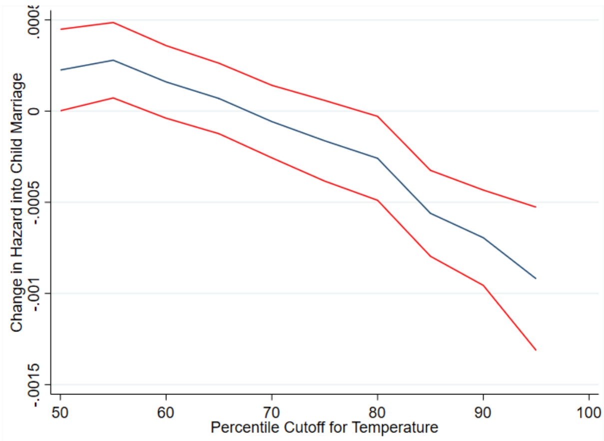
Figure S7. Robustness for Rainfall Cutoffs



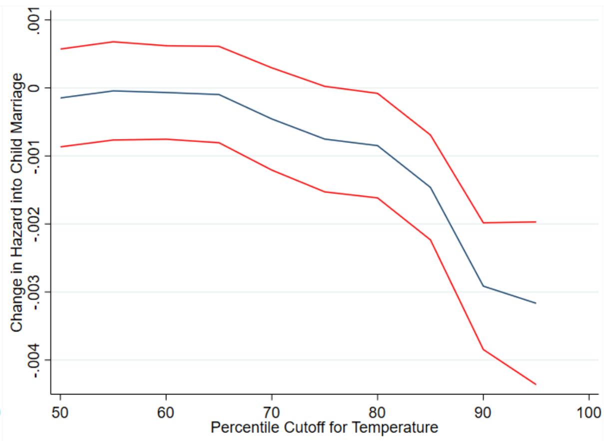
Notes: These figures show the effects of low rainfall for boys and girls in each country using different cutoffs along the gamma distribution. Data come from the University of Delaware, IPUMS International, and the DHS. Confidence intervals are at the 90% level.

Figure S8. Robustness for Temperature Cutoffs

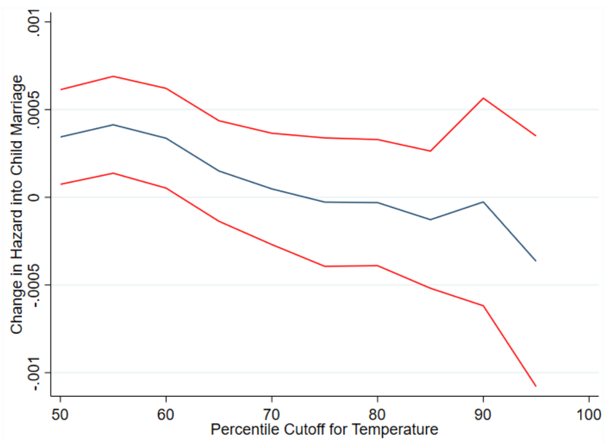
(a) Boys in India



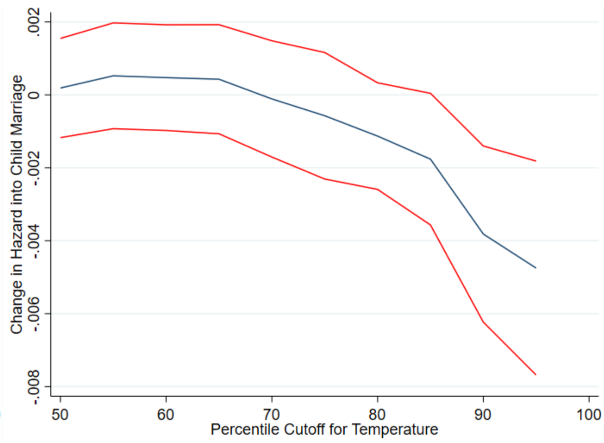
(b) Girls in India



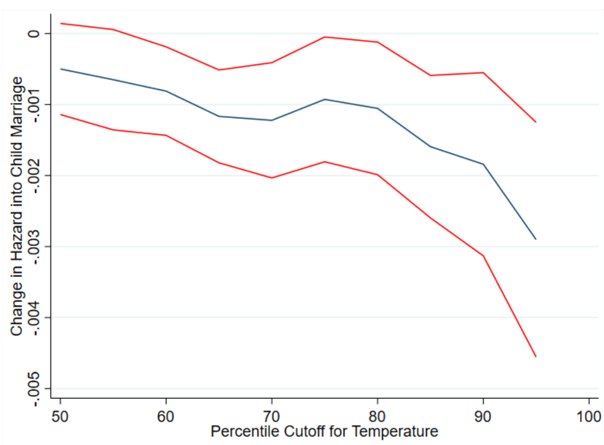
(c) Boys in Indonesia



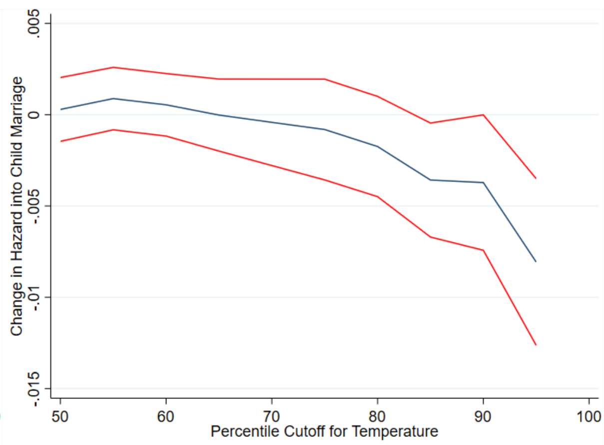
(d) Girls in Indonesia



(e) Boys in Nepal



(f) Girls in Nepal



Notes: These figures show the effects of high temperature for boys and girls in each country using different cutoffs along the temperature distribution. Data come from the University of Delaware, IPUMS International, and the DHS. Confidence intervals are at the 90% level.

Table S1. Shocks on Probability of Child Marriage vs. Corno et al. (2020), India

	Corno Specification			**Preferred Specification**		
	Corno	**v3.01 data**	**v5.01 data**	**Corno**	**v3.01 data**	**v5.01 data**
PANEL A: 25+, '98 Only	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.005***	0.001	0.0008	-0.004**	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Observations	329,586	324,541	324,541	329,586	324,541	324,541
Observed Age FE	X	X	X	X	X	X
District FE	X	X	X	X	X	X
Birth Year FE	X	X	X			
Calendar Year FE				X	X	X
PANEL B: 18+, '98 Only						
Low Rainfall Year	-	0.001	0.002	-	0.003*	0.003**
	-	(0.001)	(0.001)	-	(0.001)	(0.001)
Observations	-	423,084	423,084	-	423,084	423,084
Observed Age FE	X	X	X	X	X	X
District FE	X	X	X	X	X	X
Birth Year FE	X	X	X			
Calendar Year FE				X	X	X
PANEL C: 18+, '98-'22						
Low Rainfall Year	-	-0.002***	-0.001**	-	-0.001***	-0.0002
	-	(0.0004)	(0.0004)	-	(0.0004)	(0.0004)
Observations	-	6,135,470	6,135,470	-	6,135,470	6,135,470
Observed Age FE	X	X	X	X	X	X
District FE	X	X	X	X	X	X
Birth Year FE	X	X	X			
Calendar Year FE				X	X	X

Notes: Columns (1) and (4) use replication data provided by Corno et al. (2020). Columns (2) and (5) use data from the 1998-1999 DHS in India, 0.5X0.5 gridded monthly precipitation data from the University of Delaware (v3.01), and the 1993 version of the shapefile of India from IPUMS International to reconstruct the shocks. In Columns (3) and (6), I use v5.01 of the University of Delaware rainfall data. Columns (1)-(3) show the regression specification used in Corno et al. (2020), while the specification in Columns (4)-(6) shows a version with year fixed effects instead of birth year fixed effects. In Panel B, I include individuals who are ages 18-24 at the time of the survey. In Panel C, I use data from the 1998-1999, 2015-2016, and 2019-2022 DHS in India, 0.5x0.5 gridded monthly precipitation data from the University of Delaware (v5.01), and the 1993 version of the shapefile of India from IPUMS International to construct the shocks. These estimate differ from my main effects because I do not include individuals born from 1950-1954 in my main specification to allow for optimal comparability across samples.

Table S2. Shocks on Probability of Child Marriage vs. Corno et al. (2020), Africa

	Corno Specification			**Preferred Specification**		
	Corno	**v3.01 data**	**v5.01 data**	**Corno**	**v3.01 data**	**v5.01 data**
PANEL A: Ages 12-24	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	0.004***	0.003**	0.002*	0.002**	0.001	0.0003
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	2,461,413	1,888,867	1,941,733	2,461,409	1,888,867	1,941,733
Observed Age FE	X	X	X	X	X	X
District FE	X	X	X	X	X	X
Birth Year FE	X	X	X			
Calendar Year FE				X	X	X
PANEL B: Ages 12-17						
Low Rainfall Year	0.003**	0.002	0.001	0.001	-0.00002	-0.0005
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	1,702,218	1,372,141	1,406,812	1,702,214	1,372,141	1,406,812
Observed Age FE	X	X	X	X	X	X
District FE	X	X	X	X	X	X
Birth Year FE	X	X	X			
Calendar Year FE				X	X	X

Notes: Columns (1) and (4) use replication data provided by Corno et al. (2020). Columns (2) and (5) use individual-level data from Corno et al. (2020) and 0.5X0.5 gridded monthly precipitation data from the University of Delaware (v3.01). In Columns (3) and (6), I use v5.01 of the University of Delaware rainfall data. Columns (1)-(3) show the regression specification used in Corno et al. (2020), while the specification in Columns (4)-(6) shows a version with year fixed effects instead of birth year fixed effects. Panel A shows the effects on marriage between ages 12-24, and Panel B shows the effects on marriage between ages 12-17.

Table S3. Impact of Shocks on Overall Probability of Child Marriage, Individual Level

	India		Indonesia		Nepal	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	0.0003	0.001	0.0007	0.001	-0.0009	-0.0007
	(0.0005)	(0.001)	(0.0005)	(0.002)	(0.001)	(0.002)
%Δ (Coefficient / Mean)	0.007	0.003	0.016	0.003	-0.006	-0.002
Mean of outcome	0.043	0.327	0.044	0.381	0.158	0.441
Observations	699,190	1,240,667	967,916	1,136,051	1,105,448	1,257,161
PANEL B: Temperature						
High Temperature Year	-0.002***	-0.014***	-0.0002	0.002	-0.005	-0.022***
	(0.0008)	(0.002)	(0.001)	(0.003)	(0.003)	(0.004)
%Δ (Coefficient / Mean)	-0.045	-0.036	-0.005	0.005	-0.030	-0.049
Mean of outcome	0.049	0.374	0.042	0.364	0.165	0.458
Observations	699,190	1,240,667	967,916	1,136,051	1,105,448	1,257,161
PANEL C: Conflict						
Any Conflict Deaths	0.0004	0.0002	-0.006**	0.008*	-0.00003	0.003***
	(0.0008)	(0.002)	(0.002)	(0.004)	(0.0008)	(0.001)
%Δ (Coefficient / Mean)	0.008	0.001	-0.239	0.042	0.000	0.008
Mean of outcome	0.051	0.332	0.023	0.190	0.127	0.402
Observations	414,963	887,395	106,272	113,010	503,248	622,528

Notes: This table shows the impacts experiencing years with low rainfall, high temperatures, or exposure to conflict between the ages of 12-17 on the overall probability of child marriage at the individual level. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and survey year fixed effects, as well as controlling for the age of the individual at the time of the survey. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table S4. Impact of Shocks on Age at First Marriage, Individual Level

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.026	0.015	-0.096***	-0.086***	0.040***	0.014
	(0.020)	(0.012)	(0.014)	(0.014)	(0.010)	(0.011)
Mean of outcome	23.79	18.88	22.33	18.04	20.77	17.99
Observations	699,190	1,063,974	450,884	845,745	853,601	1,107,675
PANEL B: Temperature						
High Temperature Year	0.060**	0.095***	-0.120***	-0.133***	0.025	0.071*
	(0.025)	(0.019)	(0.028)	(0.027)	(0.091)	(0.041)
Mean of outcome	24.24	18.85	22.27	18.07	20.87	18.03
Observations	699,190	1,063,974	450,884	845,745	853,601	1,107,675
PANEL C: Conflict						
Any Conflict Deaths	-0.065***	-0.014	0.041	-0.039	0.027	-0.012
	(0.024)	(0.016)	(0.050)	(0.042)	(0.018)	(0.013)
Mean of outcome	22.88	18.86	21.23	18.70	20.63	18.06
Observations	414,963	750,654	25,844	57,774	275,821	486,159

Notes: This table shows the impacts experiencing years with low rainfall, high temperatures, or exposure to conflict between the ages of 12-17 on the age of marriage at the individual level. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and survey year fixed effects, as well as controlling for the age of the individual at the time of the survey. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table S5. Impacts of Shocks on GRDP, Indonesia

	GRDP		
	(1)	(2)	(3)
Low Rainfall	-135,477**		
	(66,285)		
High Temperature		71,015	
		(261,609)	
Any Conflict Deaths			452,137
			(1,111,000)
Mean of Outcome	16,000,000	16,300,000	17,200,000
Observations	3,812	3,828	3,828
Province FE	X	X	X
Linear time trend by province	X	X	X

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on district-level GRDP in Indonesia. I include both district fixed effects and linear time trends by province. Data on GRDP come from BPS - Statistics, Indonesia, the Indonesian Bureau of Statistics, and is available from 2010-2017. Shock data come from the University of Delaware and UCDP. Standard errors are clustered at the district level. $p<0.1^*$, $p<0.05^{**}$, $p<0.01^{***}$.

Table S6. Impact of Shocks on Annual Hazard into Child Marriage, Rural & Urban

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall, Rural	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.0002	-0.0007	-0.0007**	-0.004***	-0.002***	-0.003***
	(0.0002)	(0.0005)	(0.0003)	(0.001)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	-0.023	-0.010	-0.083	-0.045	-0.054	-0.033
Mean of outcome	0.009	0.070	0.008	0.080	0.030	0.086
Observations	4,030,420	4,338,447	3,743,664	4,122,813	4,816,592	3,023,828
PANEL B: Temperature, Rural						
High Temperature Year	-0.0008***	-0.003***	0.0002	-0.004**	-0.003***	-0.004*
	(0.0002)	(0.0007)	(0.0004)	(0.002)	(0.0009)	(0.002)
%Δ (Coefficient / Mean)	-0.081	-0.048	0.019	-0.047	-0.084	-0.048
Mean of outcome	0.010	0.072	0.008	0.080	0.030	0.087
Observations	4,039,099	4,348,769	3,771,280	4,151,261	4,816,592	5,192,167
PANEL C: Conflict, Rural						
Any Conflict Deaths	0.0007**	0.003***	-0.002***	0.002	0.0001	0.0006
	(0.0003)	(0.001)	(0.0007)	(0.002)	(0.0006)	(0.001)
%Δ (Coefficient / Mean)	0.574	0.046	-0.353	0.027	0.004	0.007
Mean of outcome	0.011	0.073	0.007	0.055	0.029	0.086
Observations	2,907,070	3,359,274	495,756	498,232	2,562,661	3,023,828
PANEL D: Rainfall, Urban						
Low Rainfall Year	0.0001	-0.0008	0.00003	-0.003***	-0.0004	-0.001
	(0.0002)	(0.0005)	(0.0002)	(0.0008)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	0.029	-0.016	0.006	-0.062	-0.026	-0.020
Mean of outcome	0.005	0.048	0.004	0.041	0.015	0.058
Observations	1,700,303	1,756,080	1,983,484	2,055,051	1,598,239	873,974
PANEL E: Temperature, Urban						
High Temperature Year	-0.0005***	-0.002***	-0.0005	-0.003**	-0.00002	0.002
	(0.0002)	(0.0007)	(0.0003)	(0.001)	(0.001)	(0.003)
%Δ (Coefficient / Mean)	-0.109	-0.041	-0.105	-0.064	-0.001	0.028
Mean of outcome	0.005	0.049	0.004	0.040	0.016	0.064
Observations	1,727,493	1,786,631	1,990,372	2,062,300	1,598,239	1,494,064
PANEL F: Conflict, Urban						
Any Conflict Deaths	-0.0002	-0.002**	0.0004	0.002	0.0003	0.0008
	(0.0003)	(0.0009)	(0.0005)	(0.002)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	-0.031	-0.043	0.143	0.086	0.020	0.014
Mean of outcome	0.006	0.048	0.003	0.023	0.013	0.058
Observations	1,230,376	1,349,488	357,757	380,095	877,812	873,974

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Panels A-C show the effects for individuals from rural areas, while Panels D-F show the effects for individuals from urban areas. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. $p<0.1^*$, $p<0.05^{**}$, $p<0.01^{***}$.

Table S7. Probability of Migration, Individual Level

	Indonesia, Probability of Migration (Current Residence)		**Nepal, Probability of Migration (Current Residence)**		**Nepal, Probability of Migration (Birthplace)**		**Nepal, Distance of Migration (Birthplace)**	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Low Rainfall Year	-0.001	-0.001	0.006*	0.005**	0.003*	0.003*	118.1	230.8
	(0.002)	(0.002)	(0.003)	(0.002)	(0.001)	(0.002)	(206.0)	(203.1)
%Δ (Coefficient / Mean)	-0.008	-0.011	0.034	0.024	0.014	0.013	0.006	0.011
Mean of outcome	0.151	0.133	0.175	0.228	0.177	0.229	19,486	20,170
Observations	967,255	1,135,353	1,077,841	1,193,353	1,077,841	1,193,353	1,077,841	1,193,353
PANEL B: Temperature								
High Temperature Year	-0.015***	-0.011***	0.019**	0.009	0.010***	0.002	1,270***	495.6*
	(0.003)	(0.003)	(0.007)	(0.006)	(0.002)	(0.003)	(300.0)	(263.4)
%Δ (Coefficient / Mean)	-0.106	-0.088	0.108	0.038	0.056	0.010	0.065	0.024
Mean of outcome	0.14	0.122	0.173	0.231	0.184	0.241	19,476	20,729
Observations	967,255	1,135,353	1,077,841	1,193,353	1,077,841	1,193,353	1,077,841	1,193,353
PANEL C: Conflict								
Any Conflict Deaths	-0.005	-0.002	-0.0008	-0.002**	0.003***	0.001	318.9**	30.33
	(0.006)	(0.008)	(0.001)	(0.0009)	(0.001)	(0.001)	(124.5)	(96.68)
%Δ (Coefficient / Mean)	-0.041	-0.020	-0.004	-0.009	0.017	0.005	0.017	0.001
Mean of outcome	0.112	0.117	0.176	0.237	0.171	0.233	19,123	20,950
Observations	106,201	112,942	490,770	594,904	490,770	594,904	490,770	594,904

Notes: This table shows the impacts experiencing years with low rainfall, high temperatures, or exposure to conflict between the ages of 12-17 on the probability of migration and distance of migration at the individual level. In columns (1)-(6), the outcome is the probability of migration at the individual level. In columns (7)-(8), the outcome is the distance of migration for the subset of individuals in Nepal who report a different district of birth and district of current residence, measured as the Euclidean distance in meters between district centroids. In columns (1)-(4), individuals are matched with shocks based on their district of current residence. In columns (5)-(8), individuals are matched with shocks based on their district of birth. Individual-level data come from the DHS and IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and survey year fixed effects, as well as controlling for the age of the individual at the time of the survey. Standard errors are clustered at the district level. $p<0.1$*, $p<0.05$**, $p<0.01$***.

Table S8. Annual Probability of Child Marriage, Robustness

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.0001	-0.0002	-0.0004	-0.007*	-0.001***	-0.002**
	(0.0001)	(0.0004)	(0.0005)	(0.002)	(0.0004)	(0.0009)
%Δ (Coefficient / Mean)	-0.012	-0.003	-0.058	-0.153	-0.050	-0.025
Mean of outcome	0.008	0.064	0.007	0.043	0.027	0.082
Observations	5,766,592	6,094,527	126,632	127,115	6,414,831	6,686,231
PANEL B: Temperature						
High Temperature Year	-0.0006***	-0.002***	-0.0003	-0.003**	-0.002**	-0.003
	(0.0001)	(0.0005)	(0.0003)	(0.001)	(0.0006)	(0.002)
%Δ (Coefficient / Mean)	-0.073	-0.024	-0.042	-0.042	-0.057	-0.036
Mean of outcome	0.008	0.066	0.007	0.067	0.027	0.082
Observations	5,766,592	6,094,527	5,761,652	6,213,561	6,414,831	6,686,231
PANEL C: Any Conflict						
Any Conflict Incidents	0.0003	0.001*	-0.0002	0.0003	-0.00007*	-0.0001***
	(0.0002)	(0.0008)	(0.0001)	(0.0004)	(0.00004)	(0.00005)
%Δ (Coefficient / Mean)	0.032	0.021	-0.035	0.007	-0.003	-0.002
Mean of outcome	0.010	0.066	0.005	0.041	0.025	0.080
Observations	4,137,446	4,708,762	853,513	878,327	3,440,473	3,897,802
PANEL D: Conflict Deaths						
Number of Conflict Deaths	-0.000002	0.00002	-0.002***	0.002	-0.00002	-0.0001
	(0.000008)	(0.00003)	(0.0006)	(0.002)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	0.000	0.000	-0.299	0.061	-0.001	-0.002
Mean of outcome	0.010	0.066	0.005	0.041	0.025	0.080
Observations	4,137,446	4,708,762	853,513	878,327	3,440,473	3,897,802
PANEL E: Conflict Incidents						
Number of Conflict Incidents	-0.0000003	-0.000006	-0.00003	0.00001	-0.000008	-0.000007
	(0.0000007)	(0.000005)	(0.000005)	(0.00002)	(0.000007)	(0.00002)
%Δ (Coefficient / Mean)	0.000	0.000	-0.006	0.000	0.000	0.000
Mean of outcome	0.010	0.041	0.005	0.041	0.025	0.080
Observations	4,137,446	878,327	853,513	878,327	3,440,473	3,897,802

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Low rainfall and high temperatures are defined using the full series of data from 1900-2017. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. This table uses p<0.1*, p<0.05**, p<0.01***.

Table S9. Impact of Shocks on Annual Hazard into Child Marriage, De-trended Shocks

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.000009	-0.0007*	-0.0004*	-0.003***	-0.0009**	-0.003**
	(0.0001)	(0.0004)	(0.0002)	(0.0009)	(0.0004)	(0.001)
%Δ (Coefficient / Mean)	-0.001	-0.010	-0.057	-0.045	-0.035	-0.037
Mean of outcome	0.007	0.061	0.007	0.067	0.026	0.081
Observations	6,860,673	7,235,976	5,761,652	6,213,561	6,414,831	6,686,231
PANEL B: Temperature						
High Temperature Year	-0.0005***	-0.001**	0.0004	0.001	-0.003***	-0.005**
	(0.0001)	(0.0006)	(0.0003)	(0.001)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	-0.071	-0.016	0.057	0.015	-0.115	-0.061
Mean of outcome	0.007	0.061	0.007	0.067	0.026	0.082
Observations	6,860,673	7,235,976	5,761,652	6,213,561	6,414,831	6,686,231

Notes: This table shows the impacts of a year with low rainfall and high temperatures on the probability of child marriage within that same year at the individual-year level. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table S10. Impact of Standard Deviation Changes on Annual Hazard into Marriage

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall, Standard Deviations	0.00006	0.0005***	0.00008	0.001***	0.0006***	0.0007*
	(0.00005)	(0.0002)	(0.00007)	(0.0005)	(0.0001)	(0.0004)
%Δ (Coefficient / Mean)	0.007	0.008	0.011	0.021	0.076	0.011
Mean of outcome	0.008	0.064	0.007	0.066	0.008	0.064
Observations	5,730,723	6,094,527	5,761,652	6,213,561	5,730,723	6,094,527
PANEL B: Temperature						
Temperature, Standard Deviations	-0.00007	-0.0006*	0.0001	-0.0003	-0.002***	-0.002
	(0.00008)	(0.0003)	(0.00009)	(0.0005)	(0.0004)	(0.001)
%Δ (Coefficient / Mean)	-0.009	-0.009	0.021	-0.005	-0.190	-0.031
Mean of outcome	0.008	0.064	0.007	0.066	0.008	0.064
Observations	5,730,723	6,094,527	5,761,652	6,213,561	5,730,723	6,094,527
PANEL C: Conflict						
Conflict Deaths, Standard Deviations	0.00009	0.0003	-0.0002	0.0008	-0.0004	-0.001***
	(0.00008)	(0.0002)	(0.0002)	(0.0007)	(0.0002)	(0.0003)
%Δ (Coefficient / Mean)	0.011	0.004	-0.024	0.012	-0.047	-0.018
Mean of outcome	0.008	0.064	0.007	0.066	0.008	0.064
Observations	4,331,702	4,911,420	853,513	878,327	4,331,702	4,911,420

Notes: This table shows the impacts of a one standard deviation increase in rainfall, temperatures, or conflict deaths on the probability of marriage within that same year at the individual-year level. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table S11. Impacts on Number of Marriages at the District Level, Ages 12-24

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall, Ages 12-24	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	0.32	0.07	-3.39***	-3.72***	-6.43	-0.21
	(0.26)	(0.41)	(0.73)	(0.93)	(6.58)	(7.30)
%Δ (Coefficient / Mean)	0.019	0.002	-0.126	-0.083	-0.035	-0.001
Mean of outcome	16.67	36.40	26.90	44.67	186.1	308.6
Observations	20,249	19,958	9,628	9,628	3,168	1,512
PANEL B: Temperature, Ages 12-24						
High Temperature Year	0.39	-0.88	-3.08***	-3.16	-17.16*	-28.03*
	(0.38)	(0.65)	(1.11)	(2.46)	(9.13)	(14.81)
%Δ (Coefficient / Mean)	0.025	-0.025	-0.121	-0.075	-0.089	-0.104
Mean of outcome	15.87	35.24	25.48	41.91	192.5	268.3
Observations	20,381	20,067	9,666	9,666	3,168	3,168
PANEL C: Conflict, Ages 12-24						
Any Conflict Deaths	1.69***	3.28***	-2.46*	-7.60***	-21.85***	-24.84***
	(0.52)	(0.79)	(1.26)	(2.07)	(6.81)	(8.250)
%Δ (Coefficient / Mean)	0.059	0.055	-0.163	-0.378	-0.096	-0.080
Mean of outcome	28.70	59.35	15.08	20.13	228.5	308.6
Observations	9,840	9,827	4,012	4,012	1,512	1,512
PANEL D: Rainfall, Ages 12-17						
Low Rainfall Year	0.08	-0.04	-0.35	-1.46	-4.03**	-8.15**
	(0.05)	(0.23)	(0.24)	(1.07)	(1.94)	(3.80)
%Δ (Coefficient / Mean)	0.037	-0.002	-0.112	-0.061	-0.079	-0.045
Mean of outcome	2.19	17.69	3.124	23.9	51.19	180.7
Observations	20,227	19,949	7,836	7,836	2,664	1,080
PANEL E: Temperature, Ages 12-17						
High Temperature Year	-0.06	-1.16***	0.08	0.84	3.07	7.41
	(0.07)	(0.35)	(0.34)	(2.41)	(4.02)	(7.70)
%Δ (Coefficient / Mean)	-0.028	-0.066	0.027	0.038	0.060	0.047
Mean of outcome	2.156	17.60	2.995	22.33	51.37	157.2
Observations	20,359	20,058	7,867	7,867	2,664	2,664
PANEL F: Conflict, Ages 12-17						
Any Conflict Deaths	0.10	0.75*	-1.94*	1.70	1.94	4.57
	(0.12)	(0.43)	(1.07)	(2.36)	(2.15)	(4.24)
%Δ (Coefficient / Mean)	0.027	0.028	-1.603	0.222	0.038	0.025
Mean of outcome	3.727	27.14	1.21	7.66	51.09	180.7
Observations	9,819	9,819	2,223	2,223	1,080	1,080

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the number of marriages within that same year at the district level. Panels A-C include marriages for all individuals ages 12-24. while Panels D-F only include marriages for individuals ages 12-17. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects. Standard errors are clustered at the district level. This table uses $p<0.1$*, $p<0.05$**, $p<0.01$***.

Table S12. Impacts on District-Level Child Marriage, Correcting for Spatial Correlation

	India		Indonesia		Nepal	
	Boys	***Girls***	***Boys***	***Girls***	***Boys***	***Girls***
PANEL A: Rainfall, Ages 12-24	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	0.32	0.07	-3.39	-3.72	-6.43	-0.21
SE, district level	(0.26)	(0.41)	(0.73)***	(0.93)***	(6.48)	(7.20)
SE, spatial correlation <100km	(0.36)	(0.57)	(1.14)***	(1.33)***	(10.29)	(11.08)
SE, spatial correlation <200km	(0.46)	(0.68)	(1.42)**	(1.35)***	(6.05)	(4.67)
Observations	20,249	19,958	9,628	9,628	3,168	1,512
PANEL B: Temp, Ages 12-24						
High Temperature Year	0.39	-0.88	-3.08	-3.16	-17.16	-28.03
SE, district level	(0.37)	(0.65)	(1.11)***	(2.45)	(9.00)*	(14.60)*
SE, spatial correlation <100km	(0.45)	(0.80)	(1.87)*	(4.08)	(13.76)	(20.97)
SE, spatial correlation <200km	(0.54)	(0.93)	(2.19)	(5.14)	(10.36)*	(16.44)*
Observations	20,381	20,067	9,666	9,666	3,168	3,168
PANEL C: Conflict, Ages 12-24						
Any Conflict Deaths	1.69	3.28	-2.46	-7.60	-21.85	-24.84
SE, district level	(0.52)***	(0.79)***	(1.25)**	(2.06)***	(6.72)***	(8.14)***
SE, spatial correlation <100km	(0.55)***	(0.84)***	(1.31)*	(2.14)***	(3.98)***	(4.37)***
SE, spatial correlation <200km	(0.69)**	(1.07)***	(1.37)*	(2.50)***	(4.64)***	(5.85)***
Observations	9,840	9,827	4,012	4,012	1,512	1,512
PANEL D: Rainfall, Ages 12-17						
Low Rainfall Year	0.08	-0.04	-0.35	-1.46	-4.03	-8.15
SE, district level	(0.05)*	(0.23)	(0.24)	(1.06)	(1.91)**	(3.75)**
SE, spatial correlation <100km	(0.07)	(0.30)	(0.31)	(1.63)	(2.00)**	(4.37)*
SE, spatial correlation <200km	(0.09)	(0.36)	(0.19)*	(1.29)	(2.32)*	(2.60)***
Observations	20,227	19,949	7,836	7,836	2,664	1,080
PANEL E: Temp, Ages 12-17						
High Temperature Year	-0.06	1.16	0.08	0.84	3.07	7.41
SE, district level	(0.07)	(0.35)***	(0.34)	(2.40)	(3.97)	(7.59)
SE, spatial correlation <100km	(0.09)	(0.49)**	(0.43)	(3.29)	(6.40)	(8.29)
SE, spatial correlation <200km	(0.13)	(0.68)*	(0.27)	(3.99)	(4.26)	(9.65)
Observations	20,359	20,058	7,867	7,867	2,664	2,664
PANEL F: Conflict, Ages 12-17						
Any Conflict Deaths	0.10	0.75	-1.94	1.70	1.94	4.57
SE, district level	(0.11)	(0.43)*	(1.06)*	(2.35)	(2.13)	(4.18)
SE, spatial correlation <100km	(0.12)	(0.49)	(1.06)*	(2.36)	(2.51)	(3.97)
SE, spatial correlation <200km	(0.14)	(0.56)	(1,17)*	(2.62)	(0.92)**	(2.95)
Observations	9,819	9,819	2,223	2,223	1,080	1,080

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the number of marriages within that same year at the district level. Panels A-C include marriages for all individuals ages 12-24. while Panels D-F only include marriages for individuals ages 12-17. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects. Standard errors are shown clustered at the district level, allowing for spatial correlation within 100km, and allowing for spatial correlation within 200km, following Colella et al. (2023). This table uses p<0.1*, p<0.05**, p<0.01***.

Table S13. Impact of Shocks on Annual Hazard into Child Marriage, with Sample Weights

	India		Indonesia		Nepal	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.0002	-0.0008	-0.0003	-0.003***	-0.001**	-0.002**
	(0.0002)	(0.0005)	(0.0003)	(0.0009)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	-0.030	-0.013	-0.039	-0.037	-0.048	-0.030
Mean of outcome	0.008	0.064	0.007	0.067	0.027	0.080
Observations	5,730,723	6,094,527	5,727,148	6,177,864	6,414,831	3,897,802
PANEL B: Temperature						
High Temperature Year	-0.0006***	-0.003***	-0.0007*	-0.005***	-0.002**	-0.003
	(0.0002)	(0.0007)	(0.0004)	(0.001)	(0.0008)	(0.002)
%Δ (Coefficient / Mean)	-0.074	-0.039	-0.104	-0.078	-0.065	-0.042
Mean of outcome	0.008	0.065	0.007	0.067	0.027	0.082
Observations	5,766,592	6,135,400	5,761,652	6,213,561	6,414,831	6,686,231
PANEL C: Conflict						
Any Conflict Deaths	0.0001	0.0003	-0.0009*	0.005***	0.0004	0.0008
	(0.0002)	(0.0009)	(0.0005)	(0.001)	(0.0006)	(0.001)
%Δ (Coefficient / Mean)	0.012	0.004	-0.182	0.123	0.018	0.010
Mean of outcome	0.010	0.066	0.005	0.041	0.025	0.080
Observations	4,137,446	4,708,762	853,513	878,327	3,440,473	3,897,802

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Sample weights are included as analytical weights. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table S14. Annual Hazard into Child Marriage, Indonesia without 1976

	Indonesia	
	Boys	***Girls***
PANEL A: Rainfall	(1)	(2)
Low Rainfall Year	-0.0004*	-0.003***
	(0.0002)	(0.0009)
%Δ (Coefficient / Mean)	-0.063	-0.048
Mean of outcome	0.007	0.067
Observations	5,669,907	6,116,941
PANEL B: Temperature		
High Temperature Year	-0.00005	-0.004***
	(0.0004)	(0.001)
%Δ (Coefficient / Mean)	-0.007	-0.057
Mean of outcome	0.007	0.067
Observations	5,704,411	6,152,638
PANEL C: Conflict		
Any Conflict Deaths	-0.002***	0.002
	(0.0005)	(0.002)
%Δ (Coefficient / Mean)	-0.300	0.060
Mean of outcome	0.005	0.041
Observations	853,513	878,327

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Individual-level data come from IPUMS, and individuals surveyed in 1976 are dropped from the regression. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. p<0.1*, p<0.05**, p<0.01***.

Table S15. Annual Probability of Child Marriage, Individuals Ages 21+ & 25+

	India		Indonesia		Nepal	
	Boys	*Girls*	*Boys*	*Girls*	*Boys*	*Girls*
PANEL A: Rainfall, 21+	(1)	(2)	(3)	(4)	(5)	(6)
Low Rainfall Year	-0.00002	-0.0004	-0.0006*	-0.003***	-0.002***	-0.003***
	(0.0001)	(0.0004)	(0.0003)	(0.001)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	-0.002	-0.006	-0.069	-0.040	-0.064	-0.035
Mean of outcome	0.008	0.065	0.009	0.068	0.028	0.085
Observations	5,444,067	5,866,925	4,045,177	4,252,583	5,467,704	5,693,856
PANEL B: Temperature, 21+						
High Temperature Year	-0.0005***	-0.002***	0.00002	-0.001	-0.002	-0.004
	(0.0002)	(0.0006)	(0.0004)	(0.002)	(0.001)	(0.003)
%Δ (Coefficient / Mean)	-0.060	-0.031	0.002	-0.017	-0.056	-0.045
Mean of outcome	0.008	0.066	0.009	0.067	0.028	0.085
Observations	5,444,067	5,906,411	4,071,730	4,279,951	5,467,704	5,693,856
PANEL C: Conflict, 21+						
Any Conflict Deaths	0.0004	0.002**	-0.0004	0.003	-0.0003	-0.0009
	(0.0003)	(0.0008)	(0.001)	(0.002)	(0.0006)	(0.001)
%Δ (Coefficient / Mean)	0.035	0.029	-0.064	0.063	-0.011	-0.011
Mean of outcome	0.010	0.066	0.007	0.046	0.027	0.084
Observations	3,910,738	4,538,956	567,335	593,207	2,532,893	2,939,799
PANEL D: Rainfall, 25+						
Low Rainfall Year	0.00006	-0.0006	-0.0007	-0.002	-0.002***	-0.003***
	(0.0001)	(0.0004)	(0.0004)	(0.001)	(0.0005)	(0.001)
%Δ (Coefficient / Mean)	0.006	-0.009	-0.071	-0.028	-0.067	-0.039
Mean of outcome	0.009	0.069	0.009	0.069	0.030	0.090
Observations	4,790,035	5,150,290	2,504,969	2,542,477	4,592,401	4,655,163
PANEL E: Temperature, 25+						
High Temperature Year	-0.0001	-0.001	0.0006	0.0003	0.002*	0.001
	(0.0002)	(0.0006)	(0.0006)	(0.003)	(0.001)	(0.003)
%Δ (Coefficient / Mean)	-0.011	-0.014	0.066	0.005	-0.058	0.015
Mean of outcome	0.009	0.069	0.009	0.069	0.030	0.090
Observations	4,790,035	5,185,139	2,523,483	2,561,570	4,592,401	4,655,163
PANEL F: Conflict, 25+						
Any Conflict Deaths	0.0004	0.003***	-0.003***	0.005***	0.0001	0.001
	(0.0003)	(0.0008)	(0.0007)	(0.002)	(0.0007)	(0.00192)
%Δ (Coefficient / Mean)	0.032	0.037	-0.399	0.104	0.003	0.013
Mean of outcome	0.011	0.070	0.008	0.051	0.029	0.089
Observations	3,346,791	3,924,864	328,588	340,318	1,657,590	1,901,106

Notes: This table shows the impacts of a year with low rainfall, high temperatures, or exposure to conflict on the probability of child marriage within that same year at the individual-year level. Panels A-C show the effects for individuals ages 21 or older at the time of the survey, while Panels D-F show the effects for individuals ages 25 or older at the time of the survey. Data for India come from the DHS, and data for Indonesia and Nepal come from IPUMS. Shock data come from the University of Delaware and UCDP. Regressions include district and observed year fixed effects, as well as controlling for the age of the individual in the relevant year. Standard errors are clustered at the district level. This table uses p<0.1*, p<0.05**, p<0.01***.